
Extending Input to State Stability to Closed Sets: Applications to Robustness of Anytime Constrained Optimization

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Abstract

This document extends the notion of Input to State Stability (ISS) to flows in closed sets with compact sets of equilibria. We call this property Set Input to State Stability (SISS). We give sufficient conditions for the SISS property to hold and underline the key relationships and differences between SISS, local ISS and ISS; highlighting the novel insight given by the SISS framework.

The second part of this document applies the developed SISS framework to study the stability of the Safe Gradient Flow algorithm when implemented in real-world settings. We prove that, under some reasonable assumptions, both the continuous and the discretized Safe Gradient Flow (SGF) showcase the Set Input to State Stability (SISS) property.

Key words: Input-to-State Stability, Control Barrier Functions, Constrained Optimization, Dynamical Systems.

AMS Codes: 93D25, 93D09, 90C26, 37N35, 37N40

Aquest document estén la noció d'Input to State Stability (ISS) a fluxos a sets tancats amb sets de punts d'equilibri compactes. Aquesta propietat l'anomenem Set Input to State Stability (SISS). Donem condicions necessàries perquè es compleixi la propietat SISS i subratllem les principals relacions i diferències entre SISS, local ISS i ISS; emfatitzant la nova informació que ens aporta el marc teòric SISS.

La segona part d'aquest document aplica el marc teòric SISS desenvolupat per estudiar l'estabilitat de l'algorisme Safe Gradient Flow quan s'implementa en situacions reals. Demostrem que, sota certes suposicions raonables, tant el SGF continu com el discretitzat exhibeixen la propietat Set Input to State Stability (SISS).

Paraules Clau: Estabilitat entrada-estat, Funcions Barrera de Control, Optimització Restringida, Sistemes Dinàmics

AMS Codes: 93D25, 93D09, 90C26, 37N35, 37N40

Este documento extiende la noción de Input to State Stability (ISS) a flujos en sets cerrados con sets de puntos de equilibrio compactos. A esta propiedad la llamamos Set Input to State Stability (SISS). Damos condiciones necesarias para que se cumpla la propiedad SISS y subrayamos las principales relaciones y diferencias entre SISS, local ISS y ISS; enfatizando la novedosa información que nos aporta el marco teórico SISS.

La segunda parte de este documento aplica el marco teórico SISS desarrollado para estudiar la estabilidad del algoritmo Safe Gradient Flow cuando se implementa en situaciones reales. Demostramos que, bajo ciertas suposiciones razonables, tanto el SGF continuo como el discretizado exhiben la propiedad Set Input to State Stability (SISS).

Palabras Clave: Estabilidad entrada-estado, Funciones Barrera de Control, Optimización Restringida, Sistemas Dinámicos

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Chapter 1

Introduction

The Input to State Stability (ISS) property was first introduced by Eduardo Sontag in 1989 and quickly became the paradigm to study the effect of the inputs (such as controllers or disturbances) to a system ruled by ordinary differential equations. Since its appearance, ISS property has been studied in detail and also extended to deal with many other types of systems, such as hybrid systems ([1]) or PDEs ([2]).

What is more, ISS property characterizations were able to show close knots to the unperturbed system's own properties.

Thus, it is no surprise that the ISS property is today a standard, taught in many textbooks (see for example [3], or [4] for a specialized book on the topic).

Nevertheless, the recent raise of data-driven control and reinforcement learning methods has highlighted the need to develop yet another extension of the ISS property that enables the study of these new systems, often defined only in open subsets of the whole space. This need was addressed by [5], in which the concept of "ISS on open sets" was developed.

However, once one starts working with systems defined only in open subsets, several difficulties arise. To begin with, such systems often cannot guarantee the forward invariance of the whole open set in which they are defined - a requirement in the "ISS on open sets" theory. Nevertheless, those systems -when unperturbed- do have invariant closed sets: their equilibria, the sub-level sets of their Lyapunov functions or even safe sets that have been specifically designed to never be exited are examples of this.

In this document, we are particularly interested in studying the stability of optimization algorithms. Optimization problems emerge naturally in a wide range of disciplines, and have therefore been widely studied in the literature. An approach that has proven to be incredibly prolific is to model optimization algorithms as continuous time dynamical systems, as this facilitates the study of properties such as stability or convergence and also the synthesis of their discrete time counterparts.

However, the omnipresence of optimization problems in many processes operating in delicate environments -many of which take place in the presence of humans- requires us to go one step further and strive for the development of constrained optimization algorithms that ensure safety guarantees. Recent work has shown that these constrained optimization algorithms can be studied as control systems, an approach that not only allows for the study of properties such as stability or convergence (just as the dynamical

systems approach does) but can also exploit the safety notions developed in the control-theoretic field.

In particular, we are interested in the anytime safety property. A set is forward invariant when any trajectory starting in it remains in it forever. If one is able to render the constraints set forward invariant, then one can rest assured the optimization algorithm will always return a feasible output, no matter when its execution is stopped. This is what we call anytime safety. If the optimization algorithm is connected to a plant and this property transitions to the plant’s operation, we will have safety under interconnection.

The processes we want to create safe, constrained optimization algorithms for might require anytime safety of their output, safety guarantees for the interconnection with a plant or even both simultaneously. Systems that demand the first requirement might include self-driving cars searching for the optimal angle and speed to take a curve (which must pertain to a certain interval to ensure that the car stays in the road and does not overturn) or a power plant choosing the optimal frequency and voltage of its output (which must belong to a certain range to avoid damages). Systems that need to fulfill the second requirement might include robots operating in human environments trying to follow a designated trajectory (through optimal control) without colliding with humans (therefore, the optimal control input to the robot must ensure the safety of the humans surrounding it, so the optimization process that decides that input must have some safety guarantees that are inherited by the robot).

Algorithms such as the Projected Gradient Flow, which is a particular case of the Safe Gradient Flow, are able to guarantee anytime safety. However, we must take into account that any real-life implementation will require them to operate in “imperfect” conditions: our knowledge of the objective function to minimize will be faulty, we might have uncertainty defining the safe set, our sensors will be noisy, our actuators might suffer delays, the controllers will have errors and we will inevitably need to discretize our algorithm. Therefore, we need to guarantee that these safe, constrained optimization algorithms will (at least approximately) retain their properties when deployed in these settings.

That is, we want to see whether these algorithms retain anytime safety of the constraints set under disturbances and, if they do not, whether there exists a “noisy” constraints set that does exhibit such property. We also want to gain insight on their stability under disturbances, this is, how will the trajectories of the perturbed system behave and whether they will converge next to an optimal point of the ideal system.

Literature Review

The first paper to showcase the Input to State Stability property was Sontag’s [6]. In 1995, further work by Sontag & Wang characterized this property, establishing connections with the robust stability of the system ([7]). In that same year, the authors also extended ISS property to deal with compact sets of equilibria in [8]. Later, they showed in [9] that some similar concepts proposed in the literature were, in fact, equivalent to ISS.

In 2021, Della Rossa, Tanwani & Zaccarian introduced a new generalization of the

Input to State Stability property, showing in [10] that the existence of a non-pathological, generalized ISS-Lyapunov function implies ISS of the system.

ISS on open sets was introduced in [5]. To the best of our knowledge, since its appearance in 2022, this concept has been employed in several applications, but there has not been any further work on its characterization.

This paper aims to develop a further extension of the ISS property for ODE systems in finite dimension, so our framework must include previous generalizations of the concept. In other words, our framework must hold sufficient conditions formulated in terms of a non-pathological generalized Lyapunov-like function and it must be able to work with systems with compact sets of equilibria defined only in an open set.

En route to the development of our extended framework, we introduce a natural extension of the Local ISS property (LISS) on open sets. The use of LISS was proposed for the first time in [9] and mainly used to characterize the ISS property, but it has received extensive attention in the infinite dimensional systems literature. One must highlight the work of Dashkovskiy & Mironchenko in [11] and of the latter in [12].

Our proposed framework targets closed forward invariant sets of the system. There are different ways to identify those sets, but it is particularly relevant to mention the Input to State Safety (ISSf) theory. Introduced by Kolathaya & Ames in [13], this concept is tailored to closed sets defined by a series of equalities and inequalities and is able to both identify their forward invariance under disturbances and also design controllers to make them forward invariant under disturbances.

The development of our framework, which we call Set Input to State Stability (SISS), was motivated by the study of the stability of the Safe Gradient Flow ([14]) -a constrained optimization algorithm designed by Allibhoy & Cortés that guarantees anytime safety of the constraints set- and the Safe Monotone Flow ([15]) -its counterpart for variational inequalities.

The Safe Gradient Flow algorithm showcases some very desirable properties (forward invariance of the safe set, exponential stability of the safe set, convergence to KKT points) that are essential for some real-life implementations.

Its development has been possible thanks to the deep ties between dynamical systems and optimization ([16]). It is usually relevant to consider settings in which the optimization problem to be studied as a dynamical system is interconnected with a plant ([17]), as this scenarios are ubiquitous in real life applications ([18], [19], [20]).

In this document we are interested in solving constrained optimization problems. When trying to solve these using the flow of a dynamical system, there have been 3 particularly relevant approaches:

- Geometric techniques ([21])
- Projected Gradient Methods. These were initially developed for solving nonlinear programs in continuous time ([22]) but were extended to solve constrained problems while rendering the constraints set forward invariant in [23]. However, the result is a discontinuous flow which requires strong assumptions to provide guarantees.

- Saddle-Point Dynamics ([24]). They search for saddle points of the associated Lagrangian, but do not leave the constraints set invariant.

The Safe Gradient Flow [14] represents a milestone in the study of optimization algorithms as the flow of a dynamical system. It showcases deep ties to both Projected Gradient Methods and Saddle-Point Dynamics. More importantly, it renders the constraints set forward invariant with a continuous flow that has convergence guarantees in very general settings.

In this document we study the discretization of the Safe Gradient Flow under two of the best-known discretization schemes: the Forward Euler discretization and the Runge-Kutta 4(5) discretization (RK45 from now on). Both Forward Euler discretization and Runge-Kutta methods are addressed in many textbooks, see for example [25].

The Forward Euler discretization is perhaps the most simple discretization scheme there exists, but it is the basis of many other more complex methods such as the RK45. Therefore, when facing a new system it is very convenient to do a first study of its discretization using this scheme.

While every step Forward Euler discretization is direct, Runge-Kutta methods employ interpolation. When the step size is fixed beforehand, we talk about explicit Runge-Kutta methods.

If the step size is adjusted in every step to guarantee the error falls below a certain tolerance, we talk about implicit Runge-Kutta methods ([26]). The RK45 is an implicit Runge-Kutta method.

To study the forward invariance of the constraints set (or noisy, expanded versions of it) we rely on the literature on Discrete Control Barrier Functions (DVCBFs), introduced in [27].

Statement of Contributions

Some results in the last decade have showcased the behavior of the closed forward invariant sets of a system under disturbances ([11], [28], [13]). Therefore, one wonders if it is possible to develop an “ISS on closed subsets of an open set” framework that leverages the existence of both “natural” and “artificial” closed forward invariant sets. That is precisely what we have done in the first half of this document (Chapter 3): introducing a novel extension of the ISS property for ODE systems in finite dimension that is able to give insight into a wider range of systems - from model-free learning to safe control. We have achieved this in 3 steps:

First, we have developed a natural extension of the Local ISS property (LISS) on open sets (Section 3.1).

Second, we have introduced the Set Input to State Stability (SISS) property, a framework which is able to study the input to state stability of the closed forward invariant sets of a system defined on an open set with compact sets of equilibria. We have given sufficient conditions for the SISS property to hold in terms of a non-pathological generalized Lyapunov-like function. This ensures our novel framework encompasses previous extensions of the ISS theory (Section 3.2).

Third, we have characterized the SISS property and its relationship with both LISS and ISS properties (Section 3.3).

The second half of this document (Chapter 4) is devoted to applying the novel framework and concepts introduced in the first half to studying the stability of the Safe Gradient Flow algorithm under disturbances.

In Sections 4.1 and 4.2. We give sufficient conditions for the Safe Gradient Flow algorithm to be stable in the Set Input to State Stability sense under generic disturbances (4.1) and state-dependent disturbances (4.2).

Finally, in Section 4.3 we give results on the discretization of the Safe Gradient Flow algorithm, giving results on the stability of the discretized flow based on the stability of the continuous time system. Section 4.4 illustrates our results.

1.1 Preliminaries

First, we need to establish the notation we are going to work with throughout this document. Once we have done that, in this section we introduce the main results that lay the foundations upon which this thesis has been built. Bachelor's level knowledge of several mathematics topics such as differential equations or topology is assumed. The two last subsections of this section also introduce some slight generalizations of preexisting theory.

Some basic notation

Given a function with its image laying in $\mathbb{R}^m$, $w : A_1 \times \dots \times A_k \rightarrow \mathbb{R}^m$, its infinity norm is $\|w\|_\infty := \sup_{y_1 \in A_1} \dots \sup_{y_k \in A_k} \|w(y_1, \dots, y_k)\|$. Let $\mathcal{M}(A_1 \times \dots \times A_k, \mathbb{R}^m)$ be the set of measurable functions from $A_1 \times \dots \times A_k$ to $\mathbb{R}^m$. Given $c > 0$, let $\mathcal{M}_c(A_1 \times \dots \times A_k, \mathbb{R}^m) := \{w : A_1 \times \dots \times A_k \rightarrow \mathbb{R}^m \mid w \in \mathcal{M}(A_1 \times \dots \times A_k, \mathbb{R}^m) \text{ and } \|w\|_\infty \leq c\}$. If a function is *Locally Lipschitz*, we often abbreviate it as LL.

We say a property holds *almost everywhere* (*a.e.*) if the set of points where it does not hold has Lebesgue measure zero. Equivalently, given a parameter λ , we say a property holds *for almost all* λ ($\forall \lambda$) if the set of values of λ for which it does not hold has Lebesgue measure zero.

If a property is directly related to a constant, we write the corresponding constant in parentheses right before the property name for clarity. For example, the small signal property appears repeatedly along this document. The small-signal property is applied to concepts that only hold when the input/disturbance entering the system is small enough. If a concept, such as stability, is only verified for disturbances $d \in \mathcal{M}_c$, we say we have (c)small-signal stability.

1.1.1 Topology

Given a metric space E , we denote a sequence of points of E as $(x_k)_{k \in \mathbb{N} \cup \{0\}} \subset E$. If the limit when the index k tends to infinity exists, this is, $\lim_{k \rightarrow \infty} x_k = x$, we write $x_k \rightarrow x$ as $k \rightarrow \infty$.

Given an arbitrary set B , we denote its *closure* by $\overline{B}$ and its boundary by ∂B . Let $\|\cdot\|$ be the $\|\cdot\|_2$ norm in $\mathbb{R}^n$ (for any $n > 0$). Given $v_1, v_2 \in \mathbb{R}^n$, $\langle v_1, v_2 \rangle = v_1^t \cdot v_2$ denotes the *euclidean scalar product*.

Given a continuous function $f : D \subset \mathbb{R}^d \rightarrow \mathbb{R}$ (where D is an open set), if f is uniformly continuous on every compact subset of D , then f can be continuously extended to $\overline{D}$. Given a continuous function $g : C \subset \mathbb{R}^d \rightarrow [m, M]$ with C a closed set, then g can be continuously extended to $g^{ext} : \mathbb{R}^d \rightarrow [m, M]$ by *Tietze Extension Theorem*. Note that m can be $-\infty$ and M can be $+\infty$ (in those cases, the interval of the image is open). For further results on topology, we refer the reader to [29].

1.1.2 Non-smooth Analysis Tools

Given a Locally Lipschitz function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we denote by $\Omega_f \subset \mathbb{R}^d$ the set of points at which f fails to be differentiable.

Definition 1 (Generalized Gradient). *The generalized gradient of f , $\partial f : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{R}^d)$, is given at each point $x \in \mathbb{R}^d$ by $\partial f(x) := \text{co}\{\lim_{i \rightarrow \infty} \nabla f(x_i) \mid (x_i)_{i \in \mathbb{N} \cup \{0\}} \subset \mathbb{R}^d \text{ st } x_i \rightarrow x \text{ and } x_i \notin S \cup \Omega_f \forall i \in \mathbb{N} \cup \{0\}\}$ (where S is a null-measure set that can be arbitrarily chosen to help with the computation and co denotes the convex hull of a set).*

Remark 1.1.1 (Generalized Gradient Definition is independent of S). S is a null-measure set that can be arbitrarily chosen to help with the computation of the generalized gradient. Thus, the generalized gradient is well-defined, as it does not depend on the choose of S . co denotes the convex hull of a set. •

Definition 2 (Set-valued Lie Derivative). *Given a set-valued map $\mathcal{F} : \mathbb{R}^n \rightarrow \mathcal{B}(\mathbb{R}^\times)$ and a Locally Lipschitz (LL) function $f : \mathbb{R}^n \rightarrow \mathbb{R}$; the set-valued Lie derivative of f along $\mathcal{F}$ is*

$$\tilde{\mathcal{L}}_{\mathcal{F}} f(x) = \{a \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } \xi^t v = a \forall \xi \in \partial f(x)\}$$

Lemma 1.1.2 (Set-valued Lie Derivative of function sum). *Consider two LL and regular functions $f_1, f_2 : \mathbb{R}^n \rightarrow \mathbb{R}$. Then,*

$$\tilde{\mathcal{L}}_{\mathcal{F}}(f_1 + f_2)(x) = \tilde{\mathcal{L}}_{\mathcal{F}} f_1(x) + \tilde{\mathcal{L}}_{\mathcal{F}} f_2(x)$$

Proof. The sum rule of the generalized gradient [30, Expression (39)] ensures that, if f_1 and f_2 are both regular, then $\partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x)$. Therefore:

$$\begin{aligned} \tilde{\mathcal{L}}_{\mathcal{F}}(f_1 + f_2)(x) &= \{a \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } (\xi_1 + \xi_2)^t v = a \forall \xi = \xi_1 + \xi_2 \in \partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x)\} \\ &= \{a \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } (\xi_1)^t v + (\xi_2)^t v = a \forall \xi_1 \in \partial f_1(x) \text{ and } \forall \xi_2 \in \partial f_2(x)\} \end{aligned}$$

For any $a \in \tilde{\mathcal{L}}_{\mathcal{F}}(f_1 + f_2)(x)$; if we fix some $\bar{\xi}_2 \in \partial f_2(x)$, we will have that $\exists v \in \mathcal{F}(x)$ such that $\xi_1^t v + \bar{\xi}_2^t v = a \forall \xi_1 \in \partial f_1 \Rightarrow \xi_1^t v = a - \bar{\xi}_2^t v =: a_1 \forall \xi_1 \in \partial f_1(x)$.

We can reason symmetrically but fixing $\bar{\xi}_1$ now, and we would get $\xi_2^t v = a - \bar{\xi}_1^t v =: a_2 \forall \xi_2 \in \partial f_2(x)$. Therefore,

$$\{a \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } (\xi_1)^t v + (\xi_2)^t v = a \forall \xi_1 \in \partial f_1(x) \text{ and } \forall \xi_2 \in \partial f_2(x)\}$$

$$\begin{aligned}
&= \{a_1 \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } (\xi_1)^t v = a_1 \ \forall \xi_1 \in \partial f_1(x)\} \\
&\quad + \{a_2 \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } (\xi_2)^t v = a_2 \ \forall \xi_2 \in \partial f_2(x)\} \\
&= \tilde{\mathcal{L}}_{\mathcal{F}} f_1(x) + \tilde{\mathcal{L}}_{\mathcal{F}} f_2(x)
\end{aligned}$$

□

Lemma 1.1.3 (Cardinal of the Set-valued Lie Derivative). *If $\mathcal{F}$ is single-valued, then $\tilde{\mathcal{L}}_{\mathcal{F}} f(x)$ is either $\emptyset$ or $\{a\}$ for a certain $a \in \mathbb{R}$.*

Proof. $\tilde{\mathcal{L}}_{\mathcal{F}} f(x) = \{a \in \mathbb{R} \mid \exists v \in \mathcal{F}(x) \text{ st } \xi^t v = a \ \forall \xi \in \partial f(x)\} = \{a \in \mathbb{R} \mid \xi^t \mathcal{F}(x) = a \ \forall \xi \in \partial f(x)\}$ because $\mathcal{F}$ is single-valued.

Now, assume there exist a_1, a_2 st $a_1 \neq a_2$ and they both pertain to $\tilde{\mathcal{L}}_{\mathcal{F}} f(x)$. Then, we would have that:

$$\begin{aligned}
\xi^t \mathcal{F}(x) &= a_1 \ \forall \xi \in \partial f(x) \\
\xi^t \mathcal{F}(x) &= a_2 \ \forall \xi \in \partial f(x)
\end{aligned}$$

Fixing $\bar{\xi} \in \partial f(x)$ and subtracting, we would get $\bar{\xi}^t \mathcal{F}(x) - \bar{\xi}^t \mathcal{F}(x) = 0 = a_1 - a_2 \Rightarrow a_1 = a_2$.

This contradicts our initial assumption $a_1 \neq a_2$, therefore if it is not empty we must have $\tilde{\mathcal{L}}_{\mathcal{F}} f(x) = \{a\}$ for a certain (unique) $a \in \mathbb{R}$. □

Lemma 1.1.4 (Set-valued Lie Derivative over sum of single-valued maps). *If $\mathcal{F}_1, \mathcal{F}_2 : \mathbb{R}^n \rightarrow \mathcal{B}(\mathbb{R}^n)$ are single-valued, then*

$$\tilde{\mathcal{L}}_{\mathcal{F}_1 + \mathcal{F}_2} f(x) = \tilde{\mathcal{L}}_{\mathcal{F}_1} f(x) + \tilde{\mathcal{L}}_{\mathcal{F}_2} f(x)$$

Proof. The proof is a direct consequence of the previous lemma:

$$\tilde{\mathcal{L}}_{\mathcal{F}_1 + \mathcal{F}_2} f(x) = \{a \in \mathbb{R} \mid \xi^t (\mathcal{F}_1(x) + \mathcal{F}_2(x)) = a \ \forall \xi \in \partial f(x)\}$$

and we know there is at most one element in the set. As we also know that the sets $\tilde{\mathcal{L}}_{\mathcal{F}_1} f(x)$ and $\tilde{\mathcal{L}}_{\mathcal{F}_2} f(x)$ must also be single-valued, this means $\exists a_1, a_2 \in \mathbb{R}$ such that $\xi^t (\mathcal{F}_1(x)) = a_1 \ \forall \xi \in \partial f(x)$ and also $\xi^t (\mathcal{F}_2(x)) = a_2 \ \forall \xi \in \partial f(x)$ (if either of those sets is empty, then necessarily $\tilde{\mathcal{L}}_{\mathcal{F}_1 + \mathcal{F}_2} f(x)$ will be empty). Therefore if a exists we must have $a = a_1 + a_2$ which completes the proof. □

Now, we introduce the Dini derivative. This concept is used to define vector control barrier functions in [14], but it is not necessary for the scope of this document.

Definition 3 (Dini Derivative). *Given a dynamical system $\dot{x} = F(x)$, where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a function $V : \mathbb{R}^n \rightarrow \mathbb{R}$, the upper-right Dini derivative of V along the solutions of the system is $D_F^+ V(\xi) = \limsup_{h \rightarrow 0^+} \frac{1}{h} [V(x(t, \xi)) - V(x)]$.*

If V is differentiable at ξ , then $D_F^+ V(\xi) = \nabla V(x)^t \cdot F(x)$.

Finally, we introduce the concept of tangent cone to a set at a given point. This concept is of uttermost relevance in this document, as Nagumo's Theorem for Set Invariance conditions are related to the tangent cone of the points at the set's boundary.

Definition 4 (Tangent Cone). *Given a set $C \subset \mathbb{R}^n$, the tangent cone to C at $x \in \mathbb{R}^n$ is*

$$T_C(x) = \{d \in \mathbb{R}^n \mid \exists (t_k)_{k \in \mathbb{N} \cup \{0\}} \subset (0, \infty), (x_k)_{k \in \mathbb{N} \cup \{0\}} \subset C \text{ st } t_k \rightarrow 0^+, x_k \rightarrow x \text{ and } \frac{x_k - x}{t_k} \rightarrow d \text{ as } k \rightarrow \infty\}$$

As noted in [14], if C is an embedded sub manifold of $\mathbb{R}^n$ then the tangent cone is just the tangent space.

For further results on variational analysis, we refer the reader to [31]. One can find a tutorial on non smooth analysis and its applications to discontinuous dynamical systems in [30].

1.1.3 Comparison Functions

We say $\alpha : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is:

- (i) a class $\mathcal{K}$ function if it is strictly increasing and $\alpha(0) = 0$.
- (ii) a class $\tilde{\mathcal{K}}$ function if it is strictly increasing and $\alpha(0) \geq 0$
- (iii) a class $\mathcal{K}^b$ function if $\alpha(r) = 0 \forall r \in [0, b]$ and $\alpha(r + b)$ is a class $\mathcal{K}$ function.

If α belongs to any of these classes of functions and additionally $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$, then it is a class $\mathcal{K}_\infty$, $\tilde{\mathcal{K}}_\infty$ or $\mathcal{K}_\infty^b$ function, respectively.

Lemma 1.1.5 (Properties of class $\mathcal{K}_\infty$ functions). *Given α_1, α_2 class $\mathcal{K}_\infty$ functions, their sum, product, inverses and compositions are also class $\mathcal{K}_\infty$ functions.*

1.1.4 Ordinary Differential Equations, Invariance and Stability

We introduce here some basic concepts on dynamical systems using the definitions in [5] and [14].

Consider the system

$$\dot{x} = f(x, d) \tag{1.1}$$

where we assume d can only be a function of the state and time, this is, $d : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}^m$ and $f : \mathcal{X} \times \mathbb{R}^m \rightarrow \mathbb{R}^n$.

From now on, let $\mathcal{M} := \mathcal{M}(\mathcal{X} \times \mathbb{R}, \mathbb{R}^m)$ and, for any $c > 0$, $\mathcal{M}_c := \mathcal{M}_c(\mathcal{X} \times \mathbb{R}, \mathbb{R}^m)$.

We assume $d \in \mathcal{M}$ and also assume f to be Locally Lipschitz. For a discussion on the solutions of the system under this assumptions, see [5, Remark 3.1]. We denote the solutions of the system by $x(t, \xi, d)$, where $\xi \in \mathcal{X}$ is the starting point. When we consider the unperturbed system, we write $x(t, \xi)$.

Consider the unperturbed system. Let $\mathcal{A} \subset \mathcal{X}$ be a compact set of equilibria of the unperturbed system, this is, $f(\mathcal{A}, 0) = 0$. Local Lipschitzness guarantees that the solutions of the system are unique, this is: for every starting point $\xi \in \mathcal{X}$ there exists $T > 0$ and a unique trajectory $x : [0, T) \rightarrow \mathcal{X}$ such that $x(0) = \xi$ and $\dot{x}(t) = F(x(t), 0)$. If $T = +\infty$ for every starting point $\xi \in \mathcal{X}$, we say the system is *forward complete*. If the solutions are defined for all negative times, the system is *backwards complete*. Finally, we say the system is *complete* if for every $\xi \in \mathcal{X}$ solutions are defined for all times $t \in \mathbb{R}$.

We say the set $\mathcal{K} \subseteq \mathcal{X}$ is *forward invariant* if for all $\xi \in \mathcal{K}$, the solution of the system (either perturbed or unperturbed) starting in ξ remains in $\mathcal{K}$ for all $t \geq 0$. The modern theory of Control Barrier functions for set invariance relies in Nagumo's work [32].

Proposition 1.1.6 (Nagumo's Theorem for Set Invariance). *Given a system $\dot{x} = f(x)$, a closed set C and an open set O st $C \subset O$ and the system is forward complete and has unique solutions in O , the set C is forward invariant if and only if $f(x) \in T_C(x)$ for every $x \in \partial C$.*

Consider again the unperturbed system. If $\mathcal{K}$ is forward invariant and $\mathcal{A} \subset \mathcal{K}$, we say $\mathcal{A}$ is *Lyapunov stable* relative to $\mathcal{K}$ if for every open set U st $\mathcal{A} \subset U \subseteq \mathcal{X}$ there exists an open set $\tilde{U} \subseteq U$ such that $\mathcal{A} \subset \tilde{U}$ and for all $x \in \tilde{U} \cap \mathcal{K}$, $x(t, \xi) \in U \cap \mathcal{K}$ for all $t \geq 0$.

We say $\mathcal{A}$ is *asymptotically stable* relative to $\mathcal{K}$ if it is Lyapunov stable relative to $\mathcal{K}$ and there exists an open set V such that $\mathcal{A} \subset V \subseteq \mathcal{X}$ such that $x(t, \xi) \rightarrow \mathcal{A}$ as $t \rightarrow \infty$ for all $\xi \in V \cap \mathcal{K}$.

1.1.5 Uniform Asymptotic Stability

Let $\mathcal{A} \subset \mathcal{X}$ be a compact set of equilibria. One can extend the usual notions of stability to study the stability of $\mathcal{A}$ under the perturbed system.

Definition 5 (Uniform Local Asymptotic Stability). *Given a size function $w_{\mathcal{A}}$ for $(\mathcal{X}, \mathcal{A})$, the system 1.1 is (c)small-gain Uniformly (r)Locally Asymptotically Stable (ULAS) if:*

- *There exists a class $\mathcal{K}_{\infty}$ function δ such that, given $\epsilon > 0$, if $w_{\mathcal{A}}(\xi) \leq \min\{\delta(\epsilon), r\}$ then $w_{\mathcal{A}}(x(t, \xi, d)) \leq \epsilon$ for all $t \geq 0$ and all $d \in \mathcal{M}_c$*
- *For any $0 < s \leq r$ and any $\epsilon > 0$, there exists $T_{s, \epsilon} > 0$ such that given $\xi \in S_s$, $w_{\mathcal{A}}(x(t, \xi, d)) < \epsilon$ for all $t > T$ and all $d \in \mathcal{M}_c$*

There exists a convenient alternative characterization of ULAS in terms of class $\mathcal{KL}$ functions:

Proposition 1.1.7 (Characterization of ULAS). *Given a size function $w_{\mathcal{A}}$ for $(\mathcal{X}, \mathcal{A})$, if the system 1.1 is (c)small-gain Uniformly (r)Locally Asymptotically Stable (ULAS) then there exists $\beta \in \mathcal{KL}$ such that for any $\xi \in S_r$, $w_{\mathcal{A}}(x(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t)$ for all $t \geq 0$ and all $d \in \mathcal{M}_c$.*

1.1.6 Generalizations of the ISS property

Size functions

When one works in an open set $\mathcal{X}$, it is not convenient to use the norm of the whole space $\mathbb{R}^n$. In virtually any mathematical theory in normed spaces, we directly or indirectly need the concept of "balls", which are the sub-level sets of the norm we work with. In our case, for example, we use the norm $\|\cdot\|_2$ of $\mathbb{R}^n$. However, in general these sub-level

sets of the 2-norm in $\mathbb{R}^n$ do not "fit" an arbitrary open set $\mathcal{X}$, this is, $B(\vec{0}, r) \cap \mathcal{X}^c \neq \emptyset$ for some $r > 0$.

This mismatch would cause several complications, and that is the reason why we need to move on from the norm of the whole space to a function that behaves like a norm in $\mathcal{X}$: we want it to be zero only in a compact set $\mathcal{A}$, be greater than zero everywhere else and we want its sub-level sets to fit and fill $\mathcal{X}$. This is what we call Size Functions. We introduce here the notation and results in [5].

Definition 6 (Size Function). *Let $\mathcal{X} \subseteq \mathbb{R}^n$ be an open set. Consider a compact set of equilibrium points for $d = 0$ $\mathcal{A}$. Let $w_{\mathcal{A}} : \mathcal{X} \rightarrow \mathbb{R}$. We will say $w_{\mathcal{A}}$ is a **size function** for $(\mathcal{X}, \mathcal{A})$ if:*

- (i) *it is continuous*
- (ii) $w_{\mathcal{A}}(x) = 0 \iff x \in \mathcal{A}, w_{\mathcal{A}}(x) > 0 \quad \forall x \notin \mathcal{A}$
- (iii) $\forall r \geq 0 \in \mathbb{R}, S_r := \{x \mid \mathcal{A}(x) \leq r\}$ *is a compact subset of $\mathcal{X}$*

Let us have a look at some properties of size functions on an open set with respect to a compact set of equilibria. These were shown to be true in [5] for size functions with respect to a single equilibrium point. We do not show the proof here unless the generalization to a compact set of equilibria requires a proof more elaborate than the original. In general, it suffices to substitute the expression $|x - x^*|$ with $|x|_{\mathcal{A}} = d(x, \mathcal{A})$.

Lemma 1.1.8 (Equivalent Conditions for Size Functions). *The following are equivalent:*

- $\forall r \geq 0 \in \mathbb{R}, S_r := \{x \mid \mathcal{A}(x) \leq r\}$ *is a compact subset of $\mathcal{X}$*
- $\forall \{x_k\}_{k \geq 1}$ *contained in $\mathcal{X}$, if either $x_k \rightarrow \partial\mathcal{X}$ or $|x_k| \rightarrow +\infty$, then $w(x_k) \rightarrow +\infty$*

Lemma 1.1.9 (Reverse Continuity of Size Functions). *If $w_{\mathcal{A}}$ is a size function for $(\mathcal{X}, \mathcal{A})$ then*

$$\forall \epsilon > 0, \exists \delta > 0 \quad \text{st} \quad w_{\mathcal{A}}(x) < \delta \Rightarrow |x|_{\mathcal{A}} < \epsilon$$

Proposition 1.1.10 (Equivalence of Size Functions). *Given two size functions w_1, w_2 for $(\mathcal{X}, \mathcal{A})$;*

$$\exists \alpha \in \mathcal{K}_{\infty} \quad \text{st} \quad w_1(x) \leq \alpha(w_2(x)) \quad \forall x \in \mathcal{X}$$

Proof. Let $\tilde{\alpha}(r) := \max_{\{x \mid w_2(x) \leq r\}} w_1(x)$. Since $\{x \mid w_2(x) \leq r\}$ is compact (because $w_2(x)$ is a size function), $\tilde{\alpha}(r)$ is well-defined and single-valued.

Take $x \in \mathcal{X}$ and let $r = w_2(x)$. Then, $w_1(x) \leq \tilde{\alpha}(r) = \tilde{\alpha}(w_2(x))$ (as x is such that $r = w_2(x)$ and therefore $x \in \{x \mid w_2(x) \leq r\}$)

The set $\{x \mid w_2(x) \leq r\}$ gets bigger as r increases, therefore $\tilde{\alpha}(r)$ is non-decreasing. Also, $\tilde{\alpha}(0) = 0$ by positive definiteness of w_1, w_2

We are now going to prove continuity at 0^+ . This is, we want to see that

$$\forall \epsilon > 0, \exists \delta > 0 \quad \text{st} \quad \forall r < \delta, \tilde{\alpha}(r) < \epsilon$$

Seeing this is equivalent to finding a δ st

$$\forall r < \delta, w_2(x) < r \Rightarrow w_1(x) < \epsilon$$

because if such δ exists then $\forall r < \delta \quad \tilde{\alpha}(r) = \max_{\{x | w_2(x) \leq r\}} w_1(x) < \epsilon$ and continuity is proven.

As $w_1(x)$ is continuous and $w_1(x) = 0 \forall x \in \mathcal{A}$, we know that given an $\epsilon \exists \delta_1$ st $|x|_{\mathcal{A}} < \delta_1 \Rightarrow w_1(x) < \epsilon$. Therefore, we only need to see that there exists a δ such that:

$$w_2(x) \leq r \forall r < \delta \Rightarrow |x|_{\mathcal{A}} < \delta_1$$

This is equivalent to finding a δ st $w_2(x) < \delta \Rightarrow |x|_{\mathcal{A}} < \delta_1$

Now, we apply the above lemma 1.1.9 with its ϵ set as our now δ_1 and its w set as our w_2 , getting that $\exists \delta$ st $w_2(x) < \delta \Rightarrow |x|_{\mathcal{A}} < \epsilon$. This proves continuity of $\tilde{\alpha}$ at 0^+ .

Finally, we can extend our $\tilde{\alpha} \in \mathcal{K}$ to an $\alpha \in \mathcal{K}_\infty$ proceeding just as in Proposition 2.6 in [1]. $\square$

Corollary 1.1.11 (Equivalence of Size Functions II). *Given w_1 a size function for $(\mathcal{X}, \mathcal{A})$ and w_2 a continuous function:*

w_2 is a size function for $(\mathcal{X}, \mathcal{A}) \iff \exists \underline{\alpha}, \bar{\alpha} \in \mathcal{K}_\infty$ st:

$$\underline{\alpha}(w_1(x)) \leq w_2(x) \leq \bar{\alpha}(w_1(x)) \forall x \in \mathcal{X}$$

Finally, some results proved in this document need the employed size function to be Locally Lipschitz. The next Lemma was extracted from [5] and states that we can always find such a size function.

Lemma 1.1.12 (Locally Lipschitz Size Function for $(\mathcal{X}, \mathcal{A})$). *$w_{\mathcal{A}} : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ defined as $w_{\mathcal{A}} = \max\{|x|_{\mathcal{A}}, \frac{1}{d(x, \partial\mathcal{X})} - \frac{a}{d(\mathcal{A}, \partial\mathcal{X})}\}$ is a Locally Lipschitz size function for $(\mathcal{X}, \mathcal{A})$ for any $a \geq 1$.*

Input to State Stability

We can now proceed to define the ISS property on an open set $\mathcal{X}$ combining the work in [5] and [7]:

Definition 7 (Input to State Stability on open sets). *Let $\mathcal{X} \subseteq \mathbb{R}^n$ be an open set. The system 1.1 is **ISS** on the open set $\mathcal{X}$ with respect to (wrt) a compact set of equilibrium points (when $d = 0$) $\mathcal{A}$ if there exists a $\mathcal{KL}$ -function β and a $\mathcal{K}_\infty$ -function γ such that the solutions of the system $x(t, \chi, d)$ are defined $\forall t \geq 0$ and fulfill the inequality:*

$$w_{\mathcal{A}}(x(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|_\infty) \quad \forall t \geq 0, \quad d \in \mathcal{M} \quad \text{and} \quad \xi \in \mathcal{X} \quad (1.2)$$

Remark 1.1.13 (ISS implies forward invariance of $\mathcal{X}$). Observe that this definition ensures that the solutions of the system will never exit the set $\mathcal{X}$, as if a solution approached the boundary of the set then the left-hand term $w_{\mathcal{A}}(x(t, \psi, d))$ would tend to infinity, therefore breaking the inequality (as the right-hand term is bounded) $\bullet$

The concept of Input to State Stability has an ISS-Lyapunov function associated to it: finding a smooth function that verifies some conditions (called the ISS-Lyapunov function) ensures that the system is ISS.

[10] extended the concept of ISS and ISS-Lyapunov functions to systems defined by set-valued maps and non smooth Lyapunov functions in $\mathbf{R}^n$. We will only work with single-valued functions, but are interested in using non smooth ISS-Lyapunov functions for systems defined only on an arbitrary set and wrt a compact set of equilibria. For the reader's convenience, we present here the bridged results in [7] and [10].

Remark 1.1.14 (ISS for systems with multiple solutions). If the system has unique solutions, the definition of ISS in a set $\mathcal{X}$ we have given is also valid even if the system is defined by a set-valued map. If the system had multiple solutions, we would just need to substitute the left-hand term in the ISS inequality by the maximum at a given time and for a given initial condition of the values taken by the size function for each solution starting at that initial condition at the given time. •

Definition 8 (Non-Pathological Functions). A Locally Lipschitz (LL) function $V : \text{dom}(V) \rightarrow \mathbb{R}$ is **non-pathological** if $\forall \phi : \mathbb{R}_+ \rightarrow \text{dom}(V)$ absolutely continuous, we have that

$$\forall \forall t \geq 0, \exists a_t \in \mathbb{R} \text{ st } \langle v, \dot{\phi}(t) \rangle = a_t \forall v \in \partial V(\phi(t))$$

Definition 9 (Generalized ISS-Lyapunov Function). A LL, non-pathological function $V : \mathcal{X} \rightarrow \mathbb{R}$ is called a **generalized ISS-Lyapunov function** for the system 1.1 on $\mathcal{X}$ if:

(i) $V(x)$ is a size function for $(\mathcal{X}, \mathcal{A})$

(ii) $\exists \alpha, \gamma \in \mathcal{K}_\infty \text{ st } \max \tilde{L}_{f(x,d)} V(x) \leq -\alpha(w_{\mathcal{A}}(x)) + \gamma(\|d\|) \quad \forall x \in \mathcal{X}$

Remark 1.1.15 (ISS-Lyapunov function independence of selected size function). Thanks to proposition 1.1.10, the definition does not depend on the $w_{\mathcal{A}}$ selected. •

Theorem 1.1.16 (Sufficient conditions for ISS). If there exists a generalized ISS-Lyapunov function for the system 1.1 on $(\mathcal{X}, \mathcal{A}) \Rightarrow$ the system is ISS on $\mathcal{X}$

Proof. Pick an initial condition $\xi \in \mathcal{X}$. We will abuse the notation by ommiting d in the arguments of the resulting flow, this is, we write $\varphi(t; \xi)$ to denote the solution of the system 1.1 starting at ξ at time t . By Theorem 54, Appendix C.3 in [7], this solution is defined in an interval $[0, t_{max})$. We will also abuse the notation by writing $d(t)$ for the disturbances along the flow of the system, when we should really write $d(x(t), t)$.

Now, let us define $v(t) := V(\varphi(t; \xi))$. As V is non-pathological, we have that $\dot{V}(\varphi(t; \xi)) = \frac{d}{dt} V(\varphi(t; \xi)) = \dot{v}(t) \in \tilde{L}_{f(x(t), d(t))} V(\varphi(t; \xi))$.

Condition (2) in the definition of non-pathological ISS-Lyapunov functions gives us that:

$$\exists \alpha, \gamma \in \mathcal{K}_\infty \text{ st } \max \tilde{L}_{f(x(t), d(t))} V(\varphi(t; \xi)) \leq -\alpha(w_{\mathcal{A}}(\varphi(t; \xi))) + \gamma(\|d(t)\|) \quad \forall t \in [0, t_{max})$$

It is important to keep in mind that condition (2) in the definition of non-pathological ISS-Lyapunov functions ensures that the solution cannot exit $\mathcal{X}$. If a solution went to

the boundary of the set, then $-\alpha(w_{\mathcal{A}}(\varphi(t; \xi))) = -\infty$ and the maximum element in the generalized Lie derivative set could not remain bounded by that value (the only way this would be possible is that the generalized Lie derivative set was empty $\forall x \in \mathcal{X}$, but this is only because it is standard to take the maximum of the empty set to be $-\infty$).

As both $w_{\mathcal{A}}$ and V are size functions, we can apply 1.1.11, getting that

$$\exists \underline{\alpha}, \bar{\alpha} \in \mathcal{K}_{\infty} \text{ st } \underline{\alpha}(V(x)) \leq w_{\mathcal{A}}(x) \leq \bar{\alpha}(V(x)) \quad \forall x \in \mathcal{X}$$

Notice that $\underline{\alpha}(V(x)) \leq w_{\mathcal{A}}(x) \Rightarrow \alpha(\underline{\alpha}(V(x))) \leq \alpha(w_{\mathcal{A}}(x)) \Rightarrow -\alpha(w_{\mathcal{A}}(x)) \leq -\alpha(\underline{\alpha}(V(x)))$.

Let $\tilde{\alpha} := \alpha \circ \underline{\alpha}$. Then:

$$\dot{v}(t) \leq -\tilde{\alpha}(V(\varphi(t; \xi))) + \gamma(\|d(t)\|) = -\tilde{\alpha}(v(t)) + \gamma(\|d(t)\|) \quad \forall t \in [0, t_{max}]$$

Now, consider the IVP

$$\begin{cases} \dot{y} = -\frac{1}{2}\tilde{\alpha}(y) \\ y(0) = s \geq 0 \end{cases}$$

As $\tilde{\alpha} \in \mathcal{K}_{\infty}$, $y = 0$ is the only equilibrium of the system. Moreover, $-\frac{1}{2}\tilde{\alpha}(y) < 0 \quad \forall y > 0$. Therefore, every solution starting at $s \geq 0$ tends to the equilibrium 0.

Let us consider $\beta(t; s)$ the flow of the solutions to the IVP. We can see that it is a $\mathcal{KL}$ -function:

- (i) $\forall s \geq 0; \beta(t; s) \rightarrow 0$ as $t \rightarrow \infty$
- (ii) $\forall t \geq 0; \beta(t; s) \in \mathcal{K}$ as a function of s

The first property is clear as we have seen all solutions starting from $s \geq 0$ tend to the origin. To see the second property, let us reason by contradiction. Assume $\exists 0 < s_1 < s_2$ st for a certain t we have $\beta(t; s_1) > \beta(t; s_2)$. As $\beta(0; s_1) = s_1 < s_2 = \beta(0; s_2)$, for this to happen at some $\tilde{t} < t$ we must have had $\beta(\tilde{t}; s_1) = \beta(\tilde{t}; s_2)$ which would mean that there exist two solution to the IVP

$$\begin{cases} \dot{y} = -\frac{1}{2}\tilde{\alpha}(y) \\ y(0) = \beta(\tilde{t}; s_1) = \beta(\tilde{t}; s_2) \end{cases}$$

As $\tilde{\alpha}$ is continuous, this would contradict the uniqueness of the solutions of the IVP problem (Theorem 54, Appendix C.3 in [7]).

Therefore, for every fixed t , $\beta(t; s) \in \mathcal{K}$ as a function of s and we can conclude that β is a $\mathcal{KL}$ -function.

To match the usual order of the arguments in a $\mathcal{KL}$ -function, let us invert the order of the arguments in our β and write $\beta(s, t)$.

Observe that for any $t \in [0, t_{max}]$, if $\tilde{\alpha}(v(t)) \leq 2\gamma(\|d(t)\|) \Rightarrow v(t) \leq \tilde{\alpha}^{-1}(2\gamma(\|d(t)\|)) \leq \tilde{\alpha}^{-1}(2\gamma(\|d\|))$.

Otherwise, if $\tilde{\alpha}(v(t)) > 2\gamma(\|d(t)\|) \iff \gamma(\|d(t)\|) < -\frac{\tilde{\alpha}(v(t))}{2}$, we see that $\dot{v}(t) \leq -\tilde{\alpha}(v(t)) + \gamma(\|d(t)\|) < -\frac{\alpha(v(t))}{2}$. By the well-known comparison theorem, this implies that $v(t) < \beta(v(0), t)$.

Therefore,

$$v(t) \leq \max\{\beta(v(0), t), \tilde{\alpha}^{-1}(2\gamma(\|d\|))\} \quad \forall t \in [0, t_{max})$$

Now taking into account that $v(0) = V(\xi) \leq \underline{\alpha}^{-1}(w_{\mathcal{A}}(\xi))$:

$$v(t) \leq \max\{\beta(\underline{\alpha}^{-1}(w_{\mathcal{A}}(\xi)), t), \tilde{\alpha}^{-1}(2\gamma(\|d\|))\}$$

Also, $w_{\mathcal{A}}(\varphi(t; \xi)) \leq \bar{\alpha}(V(\varphi(t; \xi))) = \bar{\alpha}(v(t))$ so:

$$w_{\mathcal{A}}(\varphi(t; \xi)) \leq \bar{\alpha}(v(t)) \leq \max\{\bar{\alpha}(\beta(\underline{\alpha}^{-1}(w_{\mathcal{A}}(\xi)), t)), \bar{\alpha}(\tilde{\alpha}^{-1}(2\gamma(\|d\|)))\}$$

Now, define $\tilde{\beta} = \bar{\alpha} \circ \beta(\underline{\alpha}^{-1}(y), t)$ and $\tilde{\gamma} = \bar{\alpha} \circ \tilde{\alpha}^{-1} \circ 2\gamma$. Notice that, as $\underline{\alpha}$, $\bar{\alpha}$, $\tilde{\alpha}$ and their inverses are all class $\mathcal{K}_{\infty}$, the resulting $\tilde{\beta}$ still belongs to the class $\mathcal{KL}$ and $\tilde{\gamma}$ still belongs to the class $\mathcal{K}_{\infty}$.

Let us rewrite the previous inequality using these functions:

$$w_{\mathcal{A}}(\varphi(t; \xi)) \leq \max\{\tilde{\beta}(w_{\mathcal{A}}(\xi), t), \tilde{\gamma}(\|d\|)\} \leq \tilde{\beta}(w_{\mathcal{A}}(\xi), t) + \tilde{\gamma}(\|d\|) \quad \forall t \in [0, t_{max})$$

This shows that the ISS inequality is fulfilled. We are only left to see that the solutions of the system are defined $\forall t \geq 0$.

To do this, fix $r := \max\{\tilde{\beta}(w_{\mathcal{A}}(\xi), t), \tilde{\gamma}(\|d\|)\}$. We have that $w_{\mathcal{A}}(\varphi(t; \xi)) \leq r$ for all times in its interval of existence. As $w_{\mathcal{A}}$ is a size function, the set $\{x \mid w_{\mathcal{A}}(x) \leq r\} =: S_r$ is compact.

Therefore, $\varphi(t; \xi)$ stays in the compact $S_r \forall t \in [0, t_{max})$. By Proposition C.3.6 in [7], this implies that $t_{max} = +\infty$. This completes our proof. $\square$

Remark 1.1.17 (Small-gain ISS). All the results obtained here can be extended to the concept of **small-gain ISS** simply by requiring only that the imposed conditions involving d are satisfied if $\|d\| \leq c$ for a certain (fixed) $c \in \mathbb{R}$. $\bullet$

In general, it is be very hard to find systems that fulfill the ISS property on a set for an arbitrary disturbance input. If we have a constant disturbance with a sufficiently large norm that always points towards the exterior of the set we are working on, our solutions may exit the set. This intuition suggests that, for some systems, we might only be able to prove small-gain ISS but not ISS (and there are, of course, systems that do not fulfill any of those properties).

1.1.7 Generalizations of the Input to State Safety (ISSf) property

We now turn our attention towards a slightly different concept. The ISS property ensures that the solutions of the perturbed system remain within a "tube" revolving around the unperturbed solutions that narrows as we advance along the curves, but we now want to look at the stability of closed sets under disturbances, rather than the stability of the trajectories of the system.

Thus, we now consider a safe set C defined as follows:

$$C = \{x \in \mathbb{R}^n \mid g(x) \leq 0, h(x) = 0\} \quad (1.3)$$

where $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ are both smooth functions.

We want the solutions of 1.1 to remain inside this set, so we design our system in a way such that it renders C forward invariant (this is typically done by using Control Barrier Functions (CBFs) and CBF controllers). When this happens, we say that C is safe.

[13] expands this notion of safety to a notion of Input to State Safety (ISSf), but only does so for a safe set defined by a single inequality. We generalize that concept to a safe set defined as in (1.3).

Definition 10 (Expanded Safe Set). *Consider g, h as in 1.3. Given $d : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ and three functions $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{K}$, we define C_d as the (set of) maximal path connected components of the set*

$$\begin{aligned} \{x \in \mathbb{R}^n \mid \\ g_i(x) - \gamma_1(\|d\|) \leq 0 \ \forall i \in \{1, \dots, m\}, \\ h_j(x) - \gamma_2(\|d\|) \leq 0 \ \forall j \in \{1, \dots, k\}, \\ h_s(x) + \gamma_3(\|d\|) \geq 0 \ \forall s \in \{1, \dots, k\} \} \end{aligned}$$

that contain C .

Remark 1.1.18 (Expanded safe set for all $d \in \mathcal{M}_c$). Given a constant $c \in \mathbb{R}$, we can analogously define C_c simply by substituting $\|d\|$ by c in 10. •

The following Lemma is derived from [33, Definition 4.9]:

Lemma 1.1.19 (Tangent Cone of Expanded Safe Set). *Consider $Act_1(x) = \{i \in \{1, \dots, m\} \mid g_i(x) = \gamma_1(\|d\|)\}$, $Act_2(x) = \{j \in \{1, \dots, m\} \mid h_j(x) = \gamma_2(\|d\|)\}$, $Act_3(x) = \{s \in \{1, \dots, m\} \mid h_s(x) = -\gamma_3(\|d\|)\}$. Then, the tangent cone to C_d in a point $x \in \partial C_d$ is:*

$$\begin{aligned} \mathcal{T}_{C_d} = \{z \in \mathbb{R}^n \mid \\ \nabla g_i(x)^t \cdot z \leq 0 \ \forall i \in Act_1(x) \\ \nabla h_j(x)^t \cdot z \leq 0 \ \forall j \in Act_2(x) \\ \nabla h_s(x)^t \cdot z \geq 0 \ \forall s \in Act_3(x)\} \end{aligned}$$

Proof. See [33, Definition 4.9] and its comments. □

Remark 1.1.20 (Conditions for f to belong to T_C). Given some $f(x, d)$ as in 1.1;

$$f(x, d) \in \mathcal{T}_{C_d} \text{ for some } x \in \partial C_d \iff \begin{cases} \mathcal{L}_f g_i(x) \leq 0 \ \forall i \in Act_1(x) \\ \mathcal{L}_f h_j(x) \leq 0 \ \forall j \in Act_2(x) \\ \mathcal{L}_f h_s(x) \geq 0 \ \forall s \in Act_3(x) \end{cases} \quad \forall t$$

•

Definition 11 (Input to State Safety). *Consider a system as in 1.1 and a set as in 1.3, but now let us assume the system is only defined on an open set D st $C \subset D$. We will say the set C is **Input to State Safe (ISSf)** under the system 1.1 if $\exists c > 0$ such that C_d is forward invariant and included in $D \forall d \in \mathcal{M}_c$.*

Following the work in [13], we give sufficient conditions to guarantee the forward invariance of C_d in the form of ISSf-Barrier Functions (ISSf-BF)

Proposition 1.1.21 (Input to State Safety Barrier Functions). *If $\exists \alpha, \tau_1, \tau_2 \in \mathcal{K}$ and $\tilde{c} > 0 \in \mathbb{R}$ such that $\forall t \geq 0, \forall x \in D \setminus \overset{\circ}{C}$ and $\forall d$ st $\|d\| < \tilde{c}$ we have that:*

$$\begin{cases} \mathcal{L}_{f(x,d)}g_i(x) \leq -\alpha(g_i(x)) + \tau_1(\|d\|) \quad \forall i \in \{1, \dots, m\} \\ -\alpha(h_j(x)) - \tau_2(\|d\|) \leq \mathcal{L}_{f(x,d)}h_j(x) \leq -\alpha(h_j(x)) + \tau_2(\|d\|) \quad \forall j \in \{1, \dots, k\} \end{cases} \quad (1.4)$$

$\Rightarrow \exists \gamma_1, \gamma_2, \gamma_3 \in \mathcal{K}$ and $c^* < \tilde{c}$ such that C_d is forward invariant and contained in D ($C_d \subset D$) for all $d \in \mathcal{M}_{c^*}$.

Proof. Define $\gamma_1 = \alpha^{-1} \circ \tau_1$, $\gamma_2 = \alpha^{-1} \circ \tau_2$ and $\gamma_3 = \beta_{-1} \circ \tau_2$ where $\beta(r) := -\alpha(-r)$ for $r \in \mathbb{R}$. We now the composition of class $\mathcal{K}$ functions results in a class $\mathcal{K}$ function, and the inverse of a class $\mathcal{K}$ function is also a class $\mathcal{K}$ function. Therefore, if we see that β is also a class $\mathcal{K}$ function, we will have seen that $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{K}$.

Clearly, $\beta(0) = 0$ and given $r_1 > r_2 > 0 \in \mathbb{R}$, we have got that $-\alpha(-r_1) > -\alpha(-r_2) > 0$ and our claim is proven. Notice that we need α to be a class $\mathcal{K}$ function in all $\mathbb{R}$, and not only for $\mathbb{R}_{\geq 0}$ (which is a sufficient condition for the other functions considered here). In general, it must hold that the image of τ_1, τ_2 is included inside the image of α .

Now that we have our $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{K}$, we find the biggest c st $c < \tilde{c}$ and also $C_c \subset D$. This will be our c^* .

At this point, we can consider an input disturbance d st $\|d\|_\infty \leq c^* < \tilde{c}$ and its associated set C_d . Then, for any $x \in \partial C_d$, we know $g_i(x) = \gamma_1(\|d\|) \quad \forall i \in Act_1(x)$, $h_j(x) = \gamma_2(\|d\|) \quad \forall j \in Act_2(x)$ and $h_s(x) = -\gamma_3(\|d\|) \quad \forall s \in Act_3(x)$. Therefore, substituting in the equalities in 1.4 we have that

$$\begin{cases} \mathcal{L}_{f(x,d)}g_i(x) \leq -\alpha(\gamma_1(\|d\|)) + \tau_1(\|d\|) \quad \forall i \in Act_1(x) \\ \mathcal{L}_{f(x,d)}h_j(x) \leq -\alpha(\gamma_2(\|d\|)) + \tau_2(\|d\|) \quad \forall j \in Act_2(x) \\ \mathcal{L}_{f(x,d)}h_s(x) \geq -\alpha(-\gamma_3(\|d\|)) - \tau_2(\|d\|) \quad \forall s \in Act_3(x) \end{cases}$$

Decomposing the functions $\gamma_1, \gamma_2, \gamma_3$ according to their definition, we get that:

$$\begin{cases} \mathcal{L}_{f(x,d)}g_i(x) \leq -\tau_1(\|d\|) + \tau_1(\|d\|) = 0 \quad \forall i \in Act_1(x) \\ \mathcal{L}_{f(x,d)}h_j(x) \leq -\tau_2(\|d\|) + \tau_2(\|d\|) = 0 \quad \forall j \in Act_2(x) \\ \mathcal{L}_{f(x,d)}h_s(x) \geq +\tau_2(\|d\|) - \tau_2(\|d\|) = 0 \quad \forall s \in Act_3(x) \end{cases}$$

□

Remark 1.1.22 (Forward Invariance of C_{c^*}). The set C_{c^*} is forward invariant under disturbances d st $\|d\|_\infty \leq c^*$. For $x \in \partial C_{c^*}$, note that $\tau_i(c^*) \geq \tau_i(\|d\|_\infty)$, so $-\tau_i(c^*) \geq \tau_i(\|d\|_\infty)$ for $i = \{1, 2\}$.

$$\begin{cases} \mathcal{L}_{f(x,d)}g_i(x) \leq -\tau_1(c^*) + \tau_1(\|d\|_\infty) \leq -\tau_1(\|d\|_\infty) + \tau_1(\|d\|_\infty) = 0 \quad \forall i \in Act_1(x) \\ \mathcal{L}_{f(x,d)}h_j(x) \leq -\tau_2(c^*) + \tau_2(\|d\|_\infty) \leq -\tau_2(\|d\|_\infty) + \tau_2(\|d\|_\infty) = 0 \quad \forall j \in Act_2(x) \\ \mathcal{L}_{f(x,d)}h_s(x) \geq +\tau_2(c^*) - \tau_2(\|d\|_\infty) \geq \tau_2(\|d\|_\infty) - \tau_2(\|d\|_\infty) = 0 \quad \forall s \in Act_3(x) \end{cases}$$

•

If the pair (g, h) satisfies the conditions in 1.4, we will say the pair is a **Input to State Safety Barrier Function (ISSf-BF)**.

1.1.8 Control Barrier Functions (CBFs)

In this section we introduce the generalization of Control Barrier Functions to (m, k) -Vector Control Barrier Function (VCBF) as found in [14].

Consider a control-affine system

$$\dot{x} = F_0(x) + \sum_{i=1}^r u_i F_i(x)$$

where $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ are LL vector fields for $i \in \{0, \dots, r\}$. Let $U \subset \mathbb{R}^r$ be a set of valid control inputs and consider a LL feedback controller $u : \mathcal{X} \rightarrow U$, with $\mathcal{X} \subset \mathbb{R}^n$ an open set.

Definition 12 ((m, k) -Vector Control Barrier Function (VCBF)). *Consider a closed set C such that $C \subset \mathcal{X}$ and $m, k \in \mathbb{N} \cup \{0\}$. Then, a continuously differentiable function $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^{m+k}$ is a VCBF of C if the two following properties hold:*

- $C = \{x \in \mathbb{R}^n \mid \phi_i(x) \leq 0 \quad \forall i \in \{1, \dots, m\} \text{ and } \phi_j(x) = 0 \quad \forall j \in \{1, \dots, k\}\}$
- *There exists $\alpha \in \mathcal{K}_\infty$ such that the map $K_\alpha : \mathbb{R}^n \rightrightarrows U$ defined as $K_\alpha = \{u \in U \mid D_{F_0}^+ \phi_i(x) + \sum_{l=1}^r u_l D_{F_l}^+ \phi_i(x) + \alpha(\phi_i(x)) \leq 0 \quad \forall i \in \{1, \dots, m\} \text{ and } D_{F_0}^+ \phi_j(x) + \sum_{l=1}^r u_l D_{F_l}^+ \phi_j(x) + \alpha(\phi_j(x)) \leq 0 \quad \forall j \in \{m+1, \dots, m+k\}\}$ takes non-empty values for all $x \in X$.*

Lemma 1.1.23 (Forward Invariance of C under VCBF). *If there exists a VCBF for C on $\mathcal{X}$, then any feedback controller satisfying $u(x) \in K_\alpha(x)$ for all $x \in \mathcal{X}$ and rendering the flow of the control-affine system 1.1.8 LL renders the set C forward invariant and asymptotically stable.*

1.1.9 Discrete Control Barrier Functions

In order to study the discretization of the Safe Gradient Flow, we need to look at the concepts introduced in [27], which we generalize here following the spirit of [14].

Definition 13 (Discrete Vector Control Barrier Functions (DVCBF)). *Given a constraints set C as in 1.3, the map $\phi = (g, h) : C \rightarrow \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k$ is a **Discrete Vector Control Barrier Function (DVCBF)** if there exists a pair of controllers $(u_t, v_t)_{t \in \mathbb{N} \cup 0}$ such that $(u_t, v_t) : C \rightarrow \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k$ and, given $x_0 = \xi \in C$ and $\alpha \in (0, 1)$:*

$$\begin{aligned} \Delta(g_i(x_t); u_t, v_t) + \alpha \cdot g_i(x_t) &\leq 0 \quad \forall i \in \{1, \dots, m\}, \quad \forall t \in \mathbb{N} \cup 0 \\ \Delta(h_j(x_t); u_t, v_t) + \alpha \cdot h_j(x_t) &= 0 \quad \forall j \in \{1, \dots, k\}, \quad \forall t \in \mathbb{N} \cup 0 \end{aligned}$$

Proposition 1.1.24 (Forward Invariance of C with DVCBF). *If the constraints set C has a DVCBF, then it is forward invariant.*

Proof. Note that $\Delta(g_i(x_t); u_t, v_t) = g_i(x_{t+1}) - g_i(x_t)$ and the same goes for h . Therefore, if there exists a DVCBF, for a given $t \in \mathbb{N} \cup 0$ we have the inequalities:

$$\begin{aligned} g_i(x_{t+1}) &\leq (1 - \alpha)g_i(x_t) \quad \forall i \in \{1, \dots, m\} \\ h_j(x_{t+1}) &= (1 - \alpha)h_j(x_t) \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

Recursively, we get:

$$\begin{aligned} g_i(x_{t+1}) &\leq (1 - \alpha)^t g_i(\xi) \leq 0 \quad \forall i \in \{1, \dots, m\} \\ h_j(x_{t+1}) &= (1 - \alpha)^t h_j(\xi) = 0 \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

because $\xi \in C$ and $\alpha \in (0, 1)$. □

Remark 1.1.25 (Forward Invariance of $\tilde{C}$ with DVCBF). Let $\tilde{C} = \{x \in \mathbb{R}^n \mid g(x) < 0, h(x) = 0\}$. If the DVCBF hold, the set $\tilde{C}$ is rendered forward invariant. •

1.1.10 Constrained Optimization

For the reader's convenience and consistence with the Safe Gradient Flow algorithm notation, we introduce here the concepts and notation regarding constrained optimization that appear in [14].

Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} f(x) \quad (P) \\ \text{s.t.} \quad g(x) \leq 0 \\ h(x) = 0 \end{aligned} \tag{1.5}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ are continuously differentiable.

Note that the feasible set is 1.3. At each $x \in \mathbb{R}^n$, we classify the inequality constraints as:

$$\begin{aligned} I_0(x) &= \{i \in \{1, \dots, m\}, \mid g_i(x) = 0\} \\ I_+(x) &= \{i \in \{1, \dots, m\}, \mid g_i(x) > 0\} \\ I_-(x) &= \{i \in \{1, \dots, m\}, \mid g_i(x) < 0\} \end{aligned}$$

We say the optimization problem (1.5) satisfies:

- the Mangasarian-Fromovitz Constraint Qualification (MFCQ) if $\{\nabla h_j(x)\}_{j=1}^k$ are linearly independent and there exists $\xi \in \mathbb{R}^n$ such that $\nabla h_j(x)^t \cdot \xi = 0$ for all $j \in \{1, \dots, k\}$ and $\nabla g_i(x)^t \cdot \xi < 0$ for all $i \in I_0(x)$.
- the Extended Mangasarian-Fromovitz Constraint Qualification (EMFCQ) if $\{\nabla h_j(x)\}_{j=1}^k$ are linearly independent and there exists $\xi \in \mathbb{R}^n$ such that $\nabla h_j(x)^t \cdot \xi = 0$ for all $j \in \{1, \dots, k\}$ and $\nabla g_i(x)^t \cdot \xi < 0$ for all $i \in I_0(x) \cup I_+(x)$.

- the Linear Independence Constraint Qualification (LICQ) if $\{\nabla g_i(x)\}_{i=1}^m \cup \{\nabla h_j(x)\}_{j=1}^k$ are linearly independent.

If $x^* \in C$ is a local minimizer and any of the introduced constraint qualifications hold at x^* , then there exist $u^* \in \mathbb{R}^m$ and $v \in \mathbb{R}^k$ such that the *Karush-Kuhn-Tucker* conditions hold,

$$\begin{cases} \nabla f(x^*) + \frac{\partial g(x^*)}{\partial x} \cdot u^* + \frac{\partial h(x^*)}{\partial x} \cdot v^* = 0 \\ g(x^*) \leq 0 \\ h(x^*) = 0 \\ u^* \geq 0 \\ (u^*)^t \cdot g(x^*) = 0 \end{cases} \quad (1.6)$$

We call (x^*, u^*, v^*) the KKT triple, and (u^*, v^*) the Lagrange multipliers. We denote the set of KKT points of (1.5) as

$$X_{KKT} = \{x^* \in \mathbb{R}^n \mid \exists (u^*, v^*) \in \mathbb{R}^m \times \mathbb{R}^k \text{ st } (x^*, u^*, v^*) \text{ solves (1.6)}\}$$

1.1.11 Safe Gradient Flow (SGF)

We want to solve the constrained optimization problem 1.5 while guaranteeing anytime safety. In other words, we require our algorithm to never exit the safe set $C = \{x \in \mathbb{R}^n \mid g(x) \leq 0, h(x) = 0\}$. The Safe Gradient Flow (SGF) is designed to achieve this objective. We present here a compendium of the main results in Allibhoy and Cortés original paper on the SGF [14].

The SGF transforms the optimization problem into a (controlled) dynamical system of the form

$$\dot{x} = -\nabla f(x) - \frac{\partial g(x)}{\partial x} \cdot u - \frac{\partial h(x)}{\partial x} \cdot v \quad \text{with } (u, v) \in \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k \quad (1.7)$$

Let $\alpha > 0$, then

$$K_\alpha(x) = \left\{ (u, v) \in \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k \left| \begin{aligned} -\frac{\partial g(x)}{\partial x} \cdot \frac{\partial g(x)}{\partial x} \cdot u - \frac{\partial g(x)}{\partial x} \cdot \frac{\partial h(x)}{\partial x} \cdot v &\leq \frac{\partial g(x)}{\partial x} \cdot \nabla f(x) - \alpha g(x) \\ -\frac{\partial h(x)}{\partial x} \cdot \frac{\partial g(x)}{\partial x} \cdot u - \frac{\partial h(x)}{\partial x} \cdot \frac{\partial h(x)}{\partial x} \cdot v &\leq \frac{\partial h(x)}{\partial x} \cdot \nabla f(x) - \alpha h(x) \end{aligned} \right. \right\} \quad (1.8)$$

Remark 1.1.26. We will choose $\begin{bmatrix} u(x) \\ v(x) \end{bmatrix} \in \operatorname{argmin}_{(u,v) \in K_\alpha(x)} \left\{ \left\| \frac{\partial g(x)}{\partial x} \cdot u + \frac{\partial h(x)}{\partial x} \cdot v \right\|^2 \right\}$.

This election minimizes the change introduced to the regular gradient flow drift.

Lemma 1.1.27. ([14, Lemma 4.8]). If MFCQ holds $\forall x \in C$, then $\exists \mathcal{X}$ open set st $C \subset \mathcal{X}$ and $\phi = (g, h) : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k$ is a valid (m, k) -VCBF for the system 1.7 on $\mathcal{X}$ relative to $\mathcal{U} = \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k$

Consider now

$$\begin{aligned} \mathcal{G}_\alpha(x) = \operatorname{argmin}_{\xi \in \mathbb{R}^n} & \frac{1}{2} \|\xi + \nabla f(x)\|^2 \\ \text{s.t.} & \quad \frac{\partial g(x)}{\partial x} \cdot \xi \leq -\alpha g(x) \\ & \quad \frac{\partial h(x)}{\partial x} \cdot \xi = -\alpha h(x) \end{aligned} \quad (1.9)$$

The Lagrangian for the restricted optimization problem 1.9 is

$$\mathcal{L}(\xi, u, v; x) = \frac{1}{2} \|\xi + \nabla f(x)\|^2 + u^t \cdot \left(\frac{\partial g(x)}{\partial x} \cdot \xi + \alpha g(x) \right) + v^t \cdot \left(\frac{\partial h(x)}{\partial x} \cdot \xi + \alpha h(x) \right)$$

Therefore, the KKT conditions on x are:

$$\begin{cases} \xi + \nabla f(x) + \frac{\partial g(x)}{\partial x}^t \cdot u + \frac{\partial h(x)}{\partial x}^t \cdot v = 0 \\ \frac{\partial g(x)}{\partial x} \cdot \xi + \alpha \cdot g(x) \leq 0 \\ \frac{\partial h(x)}{\partial x} \cdot \xi + \alpha \cdot h(x) = 0 \\ u^t \cdot \left(\frac{\partial g(x)}{\partial x} \cdot \xi + \alpha \cdot g(x) \right) = 0 \end{cases} \quad (1.10)$$

Let also $\Lambda_\alpha(x) = \{(u, v) \in \mathbb{R}_{\geq 0}^m \times \mathbb{R}^k \mid \exists \xi \in \mathbb{R}^n \text{ st } (\xi, u, v) \text{ solves 1.10}\}$

Lemma 1.1.28. (*[14, Lemma 4.5]*) If MFCQ conditions are satisfied $\forall x \in C$, then $\exists \mathcal{X}$ open such that $C \subset \mathcal{X}$ and $\Lambda_\alpha(x') \neq \emptyset \forall x' \in \mathcal{X}$

Proposition 1.1.29. (*[14, Proposition 4.6]*) If $\Lambda_\alpha(x) \neq \emptyset$, then:

- (i) $(u, v) \in \Lambda_\alpha(x) \Rightarrow (u, v)$ solves 1.1.26
- (ii) $\mathcal{G}_\alpha(x)$ is the closed-loop dynamics obtained by implementing the design strategy 1.1.26 on 1.7

Proposition 1.1.30. (*[14, Proposition 5.1]*) If MFCQ holds at $x^* \in C$, then:

- (i) $\mathcal{G}_\alpha(x^*) = 0 \iff x^* \in X_{KKT}$
- (ii) if $x^* \in X_{KKT}$, then $\Lambda_\alpha(x^*)$ is the set of Lagrange multipliers of 1.5

Proposition 1.1.31. (*[14, Proposition 5.2]*) If MFCQ conditions hold $\forall x \in C$, f, g and h are all continuously differentiable and their derivatives are LL, then $G_\alpha(x)$ is well defined and LL on $\mathcal{X}$

Theorem 1.1.32. (*[14, Theorem 5.4]*) If the conditions in the previous proposition hold, then C is forward invariant and asymptotically stable under the flow of $G_\alpha(x)$, which is what we call Safe Gradient Flow.

Theorem 1.1.33. (*[14, Theorem 5.7]*) Let x^* be an isolated local minimizer for 1.5. Let U be an open set such that x^* is the only KKT point contained in U . Then,

- (i) if MFCQ holds for all $x \in U \cap C$, then x^* is asymptotically stable relative to C

- (ii) if EMFCQ holds for all $x \in \overline{U}$, then x^* is asymptotically stable relative to $\mathbb{R}^n$
- (iii) If LICQ, strict complementarity and the second-order sufficient condition hold at x^* , then x^* is exponentially stable relative to $\mathbb{R}^n$

Theorem 1.1.34. (*[14, Theorem 5.19]*) If C is bounded, f , g and h are real-analytic functions and LICQ holds everywhere on C , there exists α^* such that $\forall \alpha > \alpha^*$ every trajectory of the safe gradient flow starting in $\mathcal{X}$ converges to some KKT point

We have omitted the results in [14, Section V.C] because in this document we work with the assumption that the set of KKT points of 1.5 is compact, and compact and isolated equilibria behave similarly. To be able to consider arbitrary closed sets of KKT points, we would first need to extend the SISS framework to systems with closed sets of equilibria. This problem has not been addressed in this document for both space and temporal reasons.

1.2 Problem Statement

The fulfillment of the Input to State Stability property comes with a series of desirable consequences. However, intuition tells us that any property with strong implications can only hold on a limited class of systems - as those systems must at the very least verify those strong implications if they are to fulfill such property.

For example, one of the most straightforward implications of Input to State Stability is perhaps that any ISS system is Globally Asymptotically Stable. This gives us an intuition about the strength of the ISS property, but also highlights its limitations. As the ISS property is even more restrictive than Global Asymptotic Stability, it can only hold in a very limited set of systems.

Therefore, it makes sense to ask ourselves whether we can relax the ISS property to deal with systems that may showcase an input to state stable behavior only in closed subsets of the open set $\mathcal{X}$.

The other great limitation of the Input to State Stability on open sets property is that its Lyapunov-like sufficient conditions require the ISS-Lyapunov function V to be a size function on $(\mathcal{X}, \mathcal{A})$. Although it is not difficult to render such functions, they must explode to infinity when $x \rightarrow \partial\mathcal{X}$, this is, they need to have a multiplying term that behaves like $(\frac{1}{d(x, \partial\mathcal{X})})^p$ (where $p \in \mathbb{R}_{>0}$) when $x \rightarrow \partial\mathcal{X}$.

The problem is that one needs to compute the Lie derivative of this expression and prove the inequality (1.2), an extremely complex task when dealing with the distance function to an arbitrary set $\mathcal{X}$ - which might not even be explicitly known (this is the case in [14], for example).

The objectives of the first part of this thesis are therefore twofold:

- (i) To develop a framework that is able to characterize the input to state stability of a system on a closed subset of the space $\mathcal{X}$.
- (ii) To get rid of the complexities associated to operating with the Lie derivative of a size function.

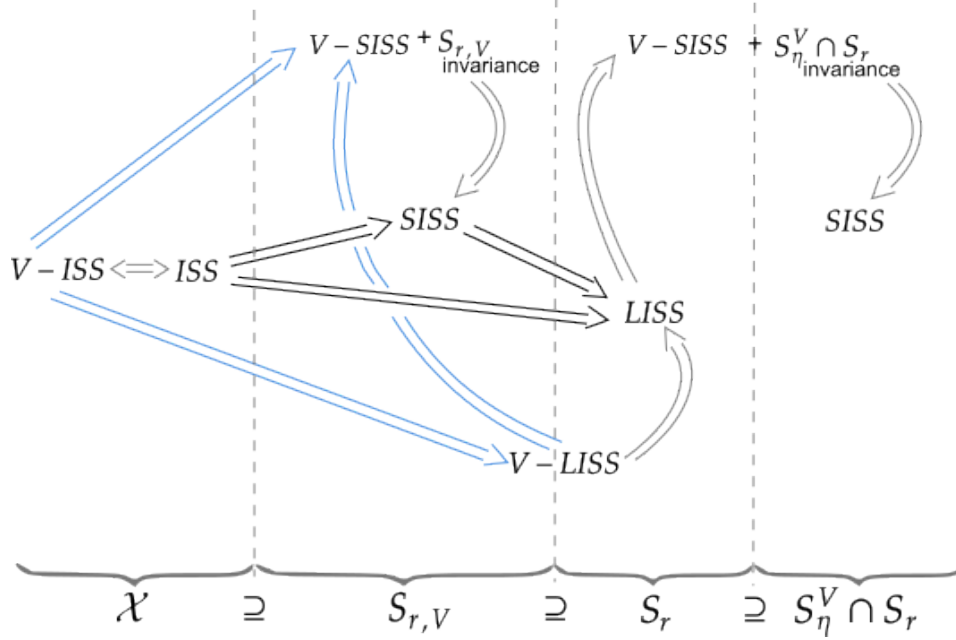


Figure 1.1: The figure displays the relationship between ISS, LISS and SISS in key subsets of $\mathcal{X}$. Given a size function, consider $\rho := \max_{x \in \partial S_r} V(x)$ and $\eta := \min_{x \in \partial S_r} V(x)$. $S_{r,V}$ is the (set of) maximal connected components of S_ρ^V that contain S_r . For a function V to be a LISS-Lyapunov function for S_r , it must satisfy certain conditions on $S_{r,V}$.

We do this in 3 steps:

- **First**, we introduce Local Input to State Stability (LISS) on open sets as a natural extension of the Local Input to State Stability on $\mathbb{R}^n$. This property characterizes input to state stability in a particular class of compact sets: the sub-level sets of a size function. These sets are compact, and therefore we get rid of the complexities of computing the Lie derivative of a term exploding at the boundary of $\mathcal{X}$.
- **Second**, we introduce Set Input to State Stability (SISS). This framework studies input to state stability in arbitrary forward invariant closed sets (which arise naturally but can also be artificially designed) using set size functions, which do not need to explode at the boundary of $\mathcal{X}$.
- **Third** and finally, we fully characterize the relationship between ISS on open sets, LISS on open sets and SISS.

Once we have introduced the novel Set Input to State Stability framework, characterized it and studied its properties, we are interested in applying it to study the stability of the Safe Gradient Flow ([14]). The Safe Gradient Flow (SGF) is a novel algorithm developed by Ahmed Allibhoy and Jorge Cortés which solves constrained optimization problems while ensuring anytime safety of the constraints set. What is more, the SGF renders the constraints set asymptotically stable.

These properties make the SGF particularly fit to solve delicate optimization problems taking place in fast paced environments in which our optimization algorithm might

not be able to keep running until it converges (in other words, it might be stopped before reaching a solution) and it is of paramount importance that, no matter when is it stopped, the output of the algorithm belongs to the constraints set to guarantee the system's safety.

However, it is not enough to know that the SGF as a dynamical system guarantees anytime safety of the constraints set. For real-world applications, we need to ensure that the SGF retains its safety guarantees under both errors and discretization. Thus, our problem can be separated in 2 layers of complexity: first, we need to study the input to state stability of the constraints set under the Safe Gradient Flow. Once that has been done, we need to recall that the (perturbed) Safe Gradient Flow is a dynamical system that needs to be discretized in order to solve it. Therefore, we need to know under which conditions does discretization preserve anytime safety and convergence.

- (i) **Layer I: Errors.** To keep our analysis as general as possible, we assume errors enter the system as additive disturbances. These errors may be caused by a variety of sources. Some of them include:

- Computational errors
- Faulty sensors and actuators
- Imperfect knowledge of the system

These can result in:

- Erroneous objective function
- Erroneous constraints set
- Erroneous CBF controllers

All these errors render an altered flow of the system. In this thesis **we study the input to state stability of the SGF under 2 different settings:**

- (a) A generic additive disturbance
 - (b) Errors exclusively on the objective function and/or the constraints set
- (ii) **Layer II: Discretization.** We study the discretization of both the SGF and the perturbed SGF, establishing conditions for the discretized solution to:
- (a) Retain anytime safety
 - (b) Retain input to state stability

Chapter 2

Methodology and Planning

Methodology

The methodology used in this document follows the standard for research in applied mathematics. In order to solve the problem we are faced with (the study of the stability of the Safe Gradient Flow), we familiarize ourselves with the algorithm, study the pre-existing literature, develop extended concepts to enable the study of our system, apply those novel concepts to analyze the stability of the SGF algorithm and finally perform simulations to showcase our results.

Our results are given in terms of Lemmas, Propositions and Theorems that have been rigorously proven using mathematical logic.

Planning

The planning behind this document has been the following. We set ourselves the objective to study the stability of the Safe Gradient Flow algorithm. In order to do that, we began by familiarizing ourselves with the SGF algorithm and the concept of Input to State Stability, and reviewed the preexisting literature concerning both topics.

Then, we tried to apply the ISS theory on open sets to study the stability of the SGF and ran into an insurmountable obstacle - the ISS on open sets theory could not appropriately capture the input to state stability properties of the SGF. Thus, it was made clear that we needed to develop an extension of the ISS on open sets theory that was able to deal with systems like the SGF.

We set ourselves the objective to develop this concept and apply it to the SGF algorithm by January, and we succeeded in our intent. Once we had an extension of the preexisting ISS theory that was able to study the SGF stability, we realized that this extension was in fact particularly suitable to study the stability of flows in closed sets. Therefore, we focused on characterizing our property (Set Input to State Stability) and establishing connections between ISS, LISS and SISS.

In the meantime, we also studied the stability of the discretized SGF, performed simulations to showcase our results and wrote this document and 2 papers on the matter.

This timeline is illustrated in the Gantt Chart below.

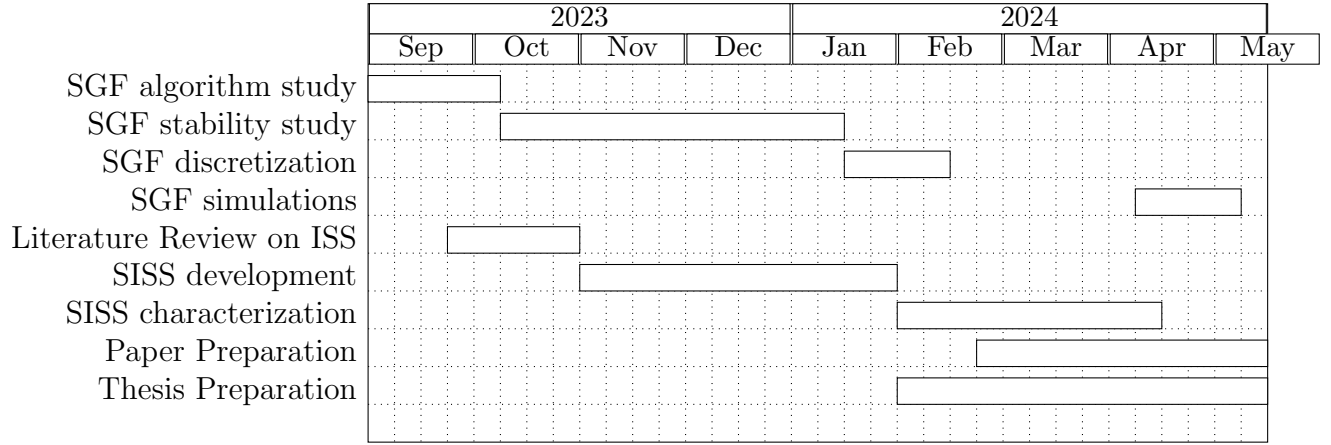


Figure 2.1: Gantt Chart

Chapter 3

Local Input to State Stability (LISS) and Set Input to State Stability (SISS)

3.1 Local ISS on open sets

Just as the class of Locally Asymptotically Stable systems is broader than the class of Globally Asymptotically Stable systems, the concept of Local ISS for systems defined on $\mathbb{R}^n$ extends this stability concept to deal with systems that only fulfill input to state stability properties in a closed ball of the space (see [34]).

In this section, we generalize the Local Input to State Stability (LISS) property on $\mathbb{R}^n$ to deal with systems defined on an open set $\mathcal{X}$. Many of these systems do not fulfill the ISS property on $\mathcal{X}$ but showcase an input to state stable behavior in some sub-level sets of a size function for the space.

Definition 14 (LISS on open sets). *We say the system 1.1 is small signal **Local ISS (LISS)** on $(\mathcal{X}, \mathcal{A})$, where $\mathcal{X}$ is an open set, if there exists $\rho > 0$, $c > 0$, $\mathcal{KL}$ -function β and $\mathcal{K}$ -function γ such that:*

$$w_{\mathcal{A}}(x(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|) \quad (3.1)$$

for all $t \geq 0$ for all $\xi \in \mathcal{X}$ st $w_{\mathcal{A}}(\xi) \leq \rho$, and for all $d \in \mathcal{M}_c$, where $\|d\| = \|d(x(t, \xi, d), t)\|$.
for some $w_{\mathcal{A}}$ a size function for $(\mathcal{X}, \mathcal{A})$.

Definition 15 (LISS-Lyapunov function). *Given $w_{\mathcal{A}}$ a size function on $(\mathcal{X}, \mathcal{A})$ and $r > 0$; consider $S_{r,V}$ the (set of) path-connected component of S_{ρ}^V (where $\rho = \max_{x \in S_r} V(x)$) that contains S_r . A non-pathological, LL function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a generalized **(r)LISS-Lyapunov function** for the set $S_r = \{x \in \mathcal{X} \mid w_{\mathcal{A}}(x) \leq r\}$ if:*

(i) *there exist $\alpha_1, \alpha_2 \in \mathcal{K}_{\infty}$ such that*

$$\alpha_1(w_{\mathcal{A}}(x)) \leq V(x) \leq \alpha_2(w_{\mathcal{A}}(x)) \quad \forall x \in S_{r,V}$$

(ii) there exist $\alpha, \gamma \in \mathcal{K}_\infty$, $c_1 > 0$ such that:

$$\max \tilde{L}_{f(x,d)} V(x) \leq -\alpha(w_{\mathcal{A}}(x)) + \gamma(\|d(x,t)\|)$$

for all $x \in S_{r,V}$, for all $t \geq 0$ and for all $d \in \mathcal{M}_{c_1}$.

Proposition 3.1.1. *[Existence of LISS-Lyapunov function implies LISS] If V is a generalized (r)LISS-Lyapunov function, then there exists $c > 0$ such that the system (1.1) is (c)small-signal (r)LISS on $(\mathcal{X}, \mathcal{A})$.*

Proof. Given V , if we prove forward invariance of the set $S_{r,V}$ then the proof reduces to the one in Theorem 1 in [5]. The proof for the forward invariance of the set $S_{r,V}$ uses the ideas in the proof of Theorem 1 in [11].

To do this, let us first reformulate Lyapunov's inequality. Note that $w_{\mathcal{A}}(x) \geq \alpha^{-1}(2\gamma(\|d\|_\infty)) \iff \alpha(w_{\mathcal{A}}(x)) \geq 2\gamma(\|d\|_\infty) \Rightarrow \gamma(\|d\|_\infty) \leq \frac{1}{2}\alpha(w_{\mathcal{A}}(x)) \Rightarrow \max \tilde{L}_{f(x,d)} V(x) \leq -\frac{1}{2}\alpha(w_{\mathcal{A}}(x))$. Taking $\tilde{\alpha} = \frac{1}{2}\alpha \in \mathcal{K}_\infty$ and $\mu = \alpha^{-1} \circ 2\gamma \in \mathcal{K}_\infty$, we have that:

$$w_{\mathcal{A}}(x) \geq \mu(\|d\|_\infty) \Rightarrow \max \tilde{L}_{f(x,d)} V(x) \leq \tilde{\alpha}(w_{\mathcal{A}}(x))$$

Take $c_\rho := \min\{\mu^{-1}(\alpha_2^{-1}(\rho)), c_1\}$ Consider the set:

$$I = \{x \in S_{r,V} \mid V(x) \leq \alpha_2(\mu(c))\}$$

Consider too $\tilde{I} = \{x \in \mathcal{X} \mid V(x) \leq \alpha_2(\mu(c))\}$ (clearly $I \subseteq \tilde{I}$). Note that any $x \in \tilde{I}$ satisfies: $V(x) \leq \alpha_2(\mu(c)) \Rightarrow \alpha_2^{-1}(V(x)) \leq \mu(c) \leq \mu(\mu^{-1}(\alpha_2^{-1}(\rho))) = \alpha_2^{-1}(\rho) \Rightarrow V(x) \leq \rho$ and thus $\tilde{I} \subseteq S_\rho^V$. As $I = \tilde{I} \cap S_{r,V} \subseteq S_{r,V}$, I is a (collection of) path-connected components of $\tilde{I}$.

Another option would be to take $\tilde{c} := \min\{\mu^{-1}(\alpha_2^{-1}(\alpha_1(r))), c_1\}$. This ensures $I \subseteq S_r \subseteq S_{r,V}$, but we can see that $\alpha_1(r) = \alpha_1(w_{\mathcal{A}}(x)) \forall x \in \partial S_r$ and therefore $\alpha_1(r) \leq V(x) \forall x \in \partial S_r$, so in particular $\alpha_1(r) \leq \max_{x \in \partial S_r} V(x) = \rho$. We conclude $\tilde{c} \leq c$ and therefore we would be overly restrictive if we were to take $\tilde{c}$.

We consider $d \in \mathcal{M}_c$. Take $\xi \in I$ and consider the solution to the perturbed system $x(t, \xi, d)$, with a maximal existence time $w_+(\xi, d)$. Take $v(t) = V(x(t, \xi, d))$. As V is non-pathological, $\dot{v}(t) \in \tilde{L}_{f(x,d)} V(x) \forall t$. Under these conditions, the set I is forward invariant:

Assume the solution only remains in I for a finite time t^* . Then, $\alpha_2(w_{\mathcal{A}}(x(t^*, \xi, d))) \geq V(x(t^*, \xi, d)) = \alpha_2(\mu(c)) \Rightarrow w_{\mathcal{A}}(x(t^*, \xi, d)) \geq \mu(c) \geq \mu(\|d\|_\infty) \Rightarrow \dot{v}(t) \leq -\tilde{\alpha}(w_{\mathcal{A}}(x)) \leq 0$. Therefore, $v(t^*) \in I$ and its value is non-increasing, meaning it cannot exit I , leading to a contradiction: I must be forward invariant for as long as the solutions exist.

Take now $\xi \in S_{r,V} \setminus I$. Again, consider the solution $x(t, \xi, d)$ and take $v(t) = V(x(t, \xi, d))$. Continuity implies that the solution stays inside $S_{r,V} \setminus \overset{\circ}{I}$ for at least some time $t_{r,V,I}^\xi < w_+(\xi, d)$. Therefore, during that time, $w_{\mathcal{A}}(x(t, \xi, d)) \geq \mu(\|d\|_\infty)$ and so:

$$\dot{v}(t) \leq -\tilde{\alpha}(w_{\mathcal{A}}(x(t, \xi, d))) \leq -\tilde{\alpha}(\alpha_2^{-1}(v(t)))$$

Take $\alpha_0 = \tilde{\alpha} \circ \alpha_2^{-1} \in \mathcal{K}_\infty$ and consider the IVP:

$$\begin{cases} \dot{y} = -\alpha_0(y) \\ y(0) = s \end{cases}$$

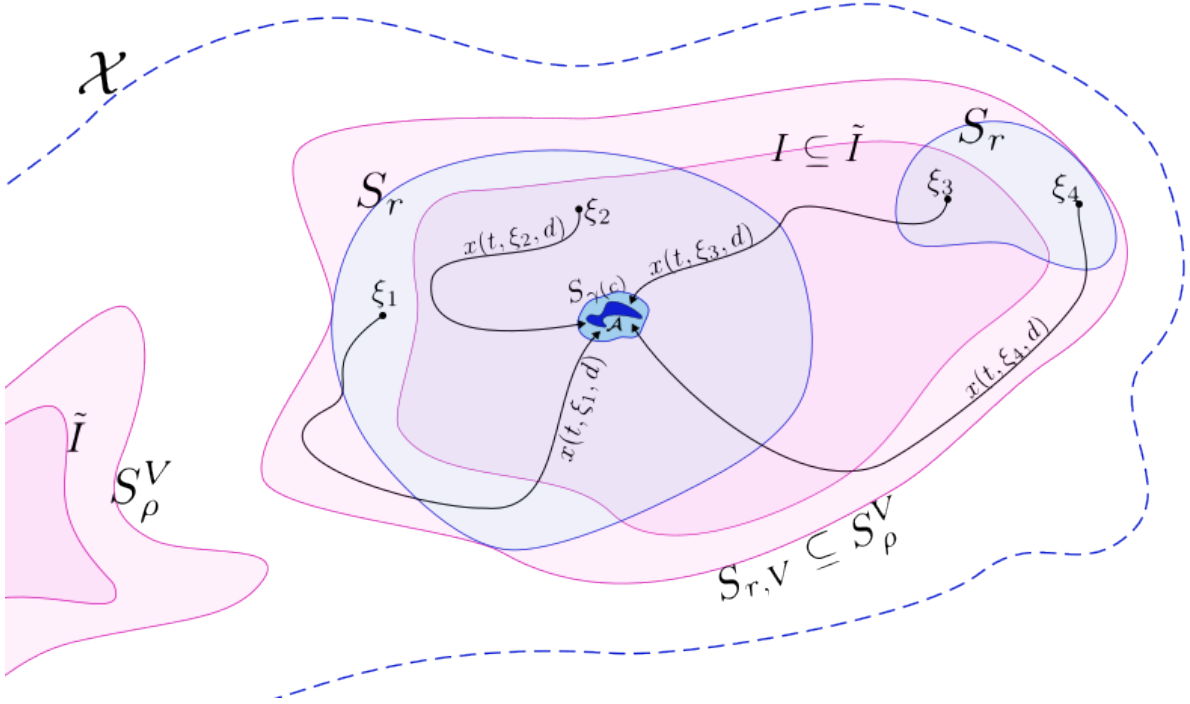


Figure 3.1: The figure showcases the casuistry of the sets S_ρ^V , $\tilde{I}$ and S_r and the trajectories starting inside them. We can see the set S_r is not necessarily invariant, but both $S_{r,V}$ and I are. S_ρ^V and $\tilde{I}$ might have other connected components for which we cannot say anything.

whose solution is a $\mathcal{KL}$ -function $\beta(s, t)$ st $\beta(s, 0) = s$. Therefore, by the comparison principle, $v(t) \leq \beta(V(\xi), t) \leq \beta(V(\xi), 0) = V(\xi) \leq \rho \ \forall t \in [0, t_{r,V,I}^\xi]$. But as $\beta(V(\xi), t)$ is decreasing on t and tends to 0 as $t \rightarrow \infty$, the solution cannot escape $S_{r,V}$ at time $t_{r,V,I}^\xi > 0$ (because at any time $t \in (0, t_{r,V,I}^\xi]$ we have $v(t) < \beta(V(\xi), 0) = V(\xi) \leq \rho$). What is more, this means that at such time $t_{r,V,I}^\xi$ the solution must enter the set I , which we have already proven to be forward invariant.

We conclude the set $S_{r,V}$ is forward invariant for as long as the solutions are defined. This set is compact because it is included in the sub-level set (of the size function $w_{\mathcal{A}}$ we are working with) $S_{\alpha_1^{-1}(\rho)}$, which is compact by definition. This implies solutions are defined for all $t \geq 0$ by Proposition C.3.6 in [9]. $\square$

Remark 3.1.2. Proposition 3.1.1 shows that for every sub-level set of a (r)LISS-Lyapunov function, $S_\nu^V := \{x \in \mathcal{X} \mid V(x) \leq \nu\}$ (where $\nu \leq \rho$), there exists $c_\nu := \min\{\mu^{-1}(\alpha_2^{-1}(\nu)), c_1\}$ st S_ν^V is forward invariant for all $d \in \mathcal{M}_{c_\nu}$. Please note that c_ρ is equal to the c given by Proposition 3.1.1.

An immediate consequence of this fact is that for every sub-level set of an ISS-Lyapunov function (let us abuse the notation and call it V , and let its sub-level sets be S_ν^V) there exists $c_\nu > 0$ such that S_ν^V is forward invariant for all $d \in \mathcal{M}_{c_\nu}$. To see this, proceed as in Proposition 3.1.1 and take $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ given by Corollary 1.1.11 $\bullet$

A weaker converse result also holds, but to prove it we will first need a technical Lemma:

Lemma 3.1.3. *Given a size function $w_{\mathcal{A}}$ and $r > 0$, consider a continuous function $V : U \rightarrow \mathbb{R}_{\geq 0}$ that is continuously differentiable in $S_r \subset U$ and suppose there exist $\alpha_1, \alpha_2 \in \mathcal{K}_{\infty}$ and $c > 0$ such that:*

- $\alpha_1(w_{\mathcal{A}}(x)) \leq V(x) \leq \alpha_2(w_{\mathcal{A}}(x))$ for all $x \in S_r$
- the ISS-Lyapunov inequality 1.2 holds for all $x \in S_r$ and all $d \in \mathcal{M}_c$

Consider $\eta := \min_{x \in \partial S_r} V(x) (> 0)$. Then, $S_{\eta}^V \cap S_r$ is a collection of maximal path-connected components of S_{η}^V and there exists $c_{\eta} > 0$ such that $S_{\eta}^V \cap S_r$ is forward invariant under any disturbance $d \in \mathcal{M}_{c_{\eta}}$.

Proof. It is enough to see this assuming path-connectedness of S_r and $\mathcal{A}$. If they are not path-connected, we repeat the same reasoning for each of their maximal path-connected components.

Proceed as in 3.1.1 to get $\mu := \alpha^{-1} \circ 2\gamma \in \mathcal{K}_{\infty}$ and also $c_{\eta} := \min \{\mu^{-1}(\alpha_2^{-1}(\eta)), c\}$. Define $I := \{x \in U \mid V(x) \leq \alpha_2(\mu(c))\} \cap S_r \subseteq S_{\eta}^V := \{x \in U \mid V(x) \leq \eta\}$. For any $x \in (S_{\eta}^V \cap S_r) \setminus \overset{\circ}{I}$, we see that $w_{\mathcal{A}}(x) \geq \mu(\|d\|_{\infty}) \Rightarrow \mathcal{L}_{f(x,d)}V(x) \leq -\alpha(w_{\mathcal{A}}(x)) < 0$ because $\mathcal{A} \subset \overset{\circ}{I}$.

As V is continuously differentiable, $\mathcal{L}_{f(x,d)}V(x) = (\nabla V(x))^t \cdot f(x, d)$. Therefore, we cannot have $\|\nabla V(x)\| = 0$ for any $x \in (S_{\eta}^V \cap S_r) \setminus \overset{\circ}{I}$ and in particular for any $x \in \partial(S_{\eta}^V \cap S_r)$.

Clearly, $\overset{\circ}{S}_{\eta}^V \cap \partial S_r = \emptyset$ (otherwise, there would exist $y \in \partial S_r$ with $V(y) < \eta$, which is a contradiction).

From now on, suppose $S_r \cap S_{\eta}^V$ is path-connected. If it was not, simply repeat this argument for each of its maximal path-connected components.

Suppose $S_r \cap S_{\eta}^V$ is not a maximal path-connected component of S_{η}^V . Then, there must exist $Y \subset \partial S_r \cap \partial S_{\eta}^V$ such that the maximal path-connected component of S_{η}^V that contains $S_r \cap S_{\eta}^V$ -which we call C_{η}^V - becomes not path-connected when we remove Y . Locally, this means that given $y \in Y$ there exists a path-connected neighborhood $U \subset \mathcal{X}$ of y such that $(U \cap (C_{\eta}^V) \setminus Y)$ is not path-connected. Note that, as $Y \subset \partial S_r \cap \partial S_{\eta}^V \subseteq \partial(S_r \cap S_{\eta}^V)$, we know that $\nabla V(y) \neq \vec{0} \ \forall y \in Y$.

Y must be contained in a manifold in ∂S_{η}^V with dimension strictly smaller than $n-1$, because otherwise we would have $\overset{\circ}{S}_{\eta}^V \cap \partial S_r \neq \emptyset$.

Let us call $D := C_{\eta}^V \cap S_r$ and $F := C_{\eta}^V \setminus (C_{\eta}^V \cap S_r) \cup Y$. It is clear $D \cup F = C_{\eta}^V$, $\overset{\circ}{D} \cap \overset{\circ}{F} = \emptyset$ and $\partial D \cap \partial F = Y$. Thus, Y is closed and thus compact because S_r is compact. This proves the surface ∂S_{η}^V crosses itself at Y .

Suppose that Y is connected, and if it is not simply repeat this argument for every maximal connected component of Y :

Given $y \in Y$, we have $\nabla V(y) \neq \vec{0}$. Therefore there exists $i_y \in \{1, \dots, n\}$ such that $\frac{\partial V}{\partial x_{i_y}}(y) \neq 0$. The function V is continuously differentiable in S_r , this means it is continuously differentiable in a neighborhood of S_r . Therefore, we can apply the Implicit Function Theorem to get δ_y, ϵ_y such that $B(y_{i_y}, \delta_y) \subset \mathbb{R}^{n-1}$ and $\exists g_y : B(y_{i_y}, \delta_y) \rightarrow$

$g_y(B(y_{\hat{i}_y}, \delta_y)) \subseteq B(y_i, \epsilon_i) \subset \mathbb{R}$ a continuously differentiable bijective map such that $V(z, g(z)) = \eta \ \forall z \in B(y_{\hat{i}_y}, \delta_y)$. This means that there is a bijection between the set $\{x \mid x_{\hat{i}_y} \in B(y_{\hat{i}_y}, \delta_y) \text{ and } x_{i_y} \in g_y(B(y_{\hat{i}_y}, \delta_y))\} \subset \partial S_\eta^V$ and the ball $B(y_{\hat{i}_y}, \delta_y)$. Note that $U_y := \{x \mid x_{\hat{i}_y} \in B(y_{\hat{i}_y}, \delta_y) \text{ and } x_{i_y} \in g_y(B(y_{\hat{i}_y}, \delta_y))\}$ is open and connected and cannot be included in Y , cause it is bijective to a manifold of dimension $n - 1$. Notice that $U_y \cap S_r \subset \partial D$ and $(U_y \setminus S_r) \cup Y \subset \partial F$. The surface crosses itself at Y , so it cannot be bijective to a ball, leading to a contradiction.

Thus, we must have $Y = \emptyset$ and therefore $S_r \cap S_\eta^V = C_\eta^V$, this is, $S_r \cap S_\eta^V$ is a (collection of) maximal path-connected component of S_η^V . Proceed as in 3.1.1 to see that there exists $c_\eta > 0$ such that $S_\eta^V \cap S_r$ is forward invariant under any disturbance $d \in \mathcal{M}_{c_\eta}$. $\square$

Proposition 3.1.4. *[LISS implies existence of LISS-Lyapunov function] If the system 1.1 is (c)small-signal (r)LISS in $(\mathcal{X}, \mathcal{A})$ for a size function w which is Locally Lipschitz in $S_r \setminus \mathcal{A}$, then there exist $c_V > 0$, $\alpha_1^V, \alpha_2^V, \alpha_V, \gamma_V \in \mathcal{K}_\infty$ and a continuously differentiable function $V : S_r \rightarrow \mathbb{R}_{\geq 0}$ such that:*

- $\alpha_1^V(w(x)) \leq V(x) \leq \alpha_2^V(w(x)) \ \forall x \in S_r$

-

$$\mathcal{L}_{f(x,d)}V(x) \leq \frac{\alpha_V(w(x))}{2} + \gamma_V(\|d(x,t)\|)$$

for all $x \in S_r$, all $t \geq 0$ and all $d \in \mathcal{M}_{c_V}$

What is more, the set $S_\eta^V \cap S_r$ is forward invariant.

Proof. The proof closely follows the steps in Lemma 2.12 and Lemma 2.13 in [8] and Theorem 2.8 in [35]. However, some adjustments need to be made to deal with the LISS property on an open set $\mathcal{X}$, so we include the proof here for clarity and completeness.

Recall that (c)small-signal (r)LISS implies that there exist $\beta_0 \in \mathcal{KL}$ and $\gamma_0 \in \mathcal{K}_\infty$ st for every $\xi \in S_r$ and every d with $\|d\|_\infty \leq c$,

$$w(x(t, \xi, d)) \leq \beta_0(w(\xi), t) + \gamma_0(\|d\|) \quad \forall t \geq 0$$

Also, notice that $\beta_0(w(\xi), t) + \gamma_0(\|d(t, \xi)\|) \leq \max\{2\beta_0(w(\xi), t), 2\gamma_0(\|d(t, \xi)\|)\}$. Simply take $\beta_1 := 2\beta_0 \in \mathcal{KL}$ and $\gamma_1 = 2\gamma_0 \in \mathcal{K}_\infty$ and we get that for every $\xi \in S_r$ and every $d \in \mathcal{M}_c$, the system 1.1 also fulfills:

$$w(x(t, \xi, d)) \leq \max\{\beta_1(w(\xi), t), \gamma_1(\|d(t, \xi)\|)\} \quad \forall t \geq 0 \quad (3.2)$$

Define $\tilde{\alpha}_{aux}(s) := \beta_1(s, 0) \in \mathcal{K}_\infty$ and take $\alpha_{aux}(s) := \max\{\alpha_{aux}(s), 2s\} \in \mathcal{K}_\infty$. Note that $\alpha_{aux} \in \mathcal{K}_\infty$ and that $\alpha_{aux}^{-1}(s) < s$. Take an arbitrarily small ϵ_c , any $\sigma(s) \in \mathcal{K}_\infty$ st $\sigma(s) < \frac{1}{c^*} \gamma_1^{-1}(\frac{1}{4} \alpha_{aux}^1(s))$ and also $\rho(s), \psi(r) \in \mathcal{K}_\infty$ st $(1 - \epsilon_c) \frac{1}{c^*} \gamma_1^{-1}(\frac{1}{4} \alpha_{aux}^1(s)) \leq \psi(s) < \rho(s) \leq \sigma(s)$. Define $\varphi(\xi) := \rho(w(\xi))$. We can see that:

$$\psi(w(\xi)) \leq \varphi(\xi) \leq \sigma(w(\xi)) \quad \forall \xi \in \mathcal{X} \quad (3.3)$$

Consider now the system

$$\dot{x} = f(x, \varphi(x) \cdot d(x, t)) =: f_{wr}(x, d) \quad (3.4)$$

and for any given $d \in \mathcal{M}_{c^*}$ (where $c^* = \min\{1, c\}$) and $\xi \in \mathcal{X}$ let $x(t, \xi, d)$ be the solution to this system. Let us abuse the notation and write d along the solutions of the system as $d(t, \xi) := d(x(t, \xi, d), t)$. Observe that $\gamma_0(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi, d)\|) \leq \gamma_0(\varphi(x(t, \xi, d)) \cdot \|d\|_\infty) \forall t \geq 0$.

Claim 3.1.5. $\forall \xi \in S_r$ and $\forall d \in \mathcal{M}_{c^*}$,

$$w(x(t, \xi, d)) \leq \alpha_{aux}(w(\xi)) \quad \forall t \geq 0$$

where $x(t, \xi, d)$ is a solution of the system (3.4).

Proof. Take $\xi \in \mathcal{X}$ and $d \in \mathcal{M}_{c^*}$. $\gamma_1(\|\varphi(x(0, \xi, d)) \cdot d(t, \xi)\|) \leq \gamma_1(\varphi(x(0, \xi, d)) \cdot \|d\|_\infty) \leq \gamma_1(\sigma(w(\xi)) \cdot \|d\|_\infty) < \gamma_1(\frac{1}{c}\gamma_1^{-1}(\frac{1}{4}\alpha_{aux}^{-1}(w(\xi))) \cdot \|d\|_\infty) \leq \gamma_1(\gamma_1^{-1}(\frac{1}{4}\alpha_{aux}^{-1}(w(\xi))))$ as $\frac{\|d\|_\infty}{c} \leq 1$.

Now, $\gamma_1(\gamma_1^{-1}(\frac{1}{4}\alpha_{aux}^{-1}(w(\xi)))) = \frac{1}{4}\alpha_{aux}^{-1}(w(\xi)) < \frac{1}{4}w(\xi)$ because $\alpha^{-1}(s) < s$.

By continuity, we conclude that there exists $t_0 > 0$ such that $\gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|) \leq \frac{1}{4}w(\xi) \quad \forall t \leq t_0$.

Now, take

$$t_1 := \inf\{t > 0 \mid \gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|) > \frac{1}{2}w(\xi)\}$$

We know $t_1 > 0$ because we have seen there exists $t_0 > 0$ st $\gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|) \leq \frac{1}{4}w(\xi) \quad \forall t \leq t_0$.

Assume $t_1 < \infty$. Then, $\gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|) \leq \frac{1}{2}w(\xi) < \frac{1}{2}\alpha_{aux}(w(\xi)) \quad \forall t \in [0, t_1)$ (because $\alpha_{aux}(s) > s$). This means $w(x(t, \xi, d)) \leq \max\{\beta_1(w(\xi), 0), \gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|)\} \leq \max\{\alpha_{aux}(w(\xi)), \frac{1}{2}\alpha_{aux}(w(\xi))\} = \alpha_{aux}(w(\xi)) \quad \forall t \in [0, t_1]$.

Therefore, for $t = t_1$ we have $\gamma_1(\|\varphi(x(t_1, \xi, d)) \cdot d(t_1, \xi)\|) \leq \gamma_1(\varphi(x(t_1, \xi, d)) \cdot \|d\|_\infty) \leq \gamma_1(\sigma(w(x(t_1, \xi, d))) \cdot \|d\|_\infty) \leq \gamma_1(\frac{1}{c^*}\gamma_1^{-1}(\frac{1}{4}\alpha_{aux}^{-1}(w(x(t_1, \xi, d)))) \cdot \|d\|_\infty) \leq \frac{1}{4}\alpha_{aux}^{-1}(w(x(t_1, \xi, d))) \leq \frac{1}{4}w(\xi)$, which contradicts the fact that for t_1 we should have $\gamma_1(\|\varphi(x(t_1, \xi, d)) \cdot d(t_1, \xi)\|) \geq \frac{1}{2}w(\xi)$. This means $t_1 = +\infty$.

Given $\xi \in S_r$ and $d \in \mathcal{M}_{c^*}$ we conclude that $w(x(t, \xi, d)) \leq \max\{\alpha_{aux}(w(\xi)), \gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|)\} \leq \max\{\alpha_{aux}(w(\xi)), \frac{1}{2}w(\xi)\} \leq \max\{\alpha_{aux}(w(\xi)), \frac{1}{2}\alpha_{aux}(w(\xi))\} = \alpha_{aux}(w(\xi))$ for all $\xi \in S_r$ and all $d \in \mathcal{M}_{c^*}$. This concludes the proof for the claim. $\square$

We have thus proved stability of the system (3.4), the first condition in definition of Uniform Local Asymptotic Stability. The following claim proves that the second condition in Definition 5 is also fulfilled.

Claim 3.1.6. $\forall \epsilon > 0 \exists t_\epsilon > 0$ and $s > 0$ st $\forall t \geq t_\epsilon, \forall \xi$ with $w(\xi) \leq s$ and $\forall d \in \mathcal{M}_{c^*}$ it holds true that $w(x(t, \xi, d)) < \epsilon$

Proof. Recall that $\gamma_1(\|\varphi(x(t, \xi, d)) \cdot d(t, \xi)\|) \leq \frac{1}{2}w(\xi)$ by the previous claim. Therefore, we have that for every $\xi \in S_r$ and $d \in \mathcal{M}_{c^*}$:

$$w(x(t, \xi, d)) \leq \max\{\beta_1(w(\xi), t), \frac{1}{2}w(\xi)\}$$

By the properties of class $\mathcal{KL}$ functions, $\forall s \geq 0 \exists T_s \geq 0$ st $\beta_1(s, t) \leq \frac{1}{2}s \quad \forall t \geq T_s$.

Therefore, we have that given $s \geq 0 \exists T_s \geq 0$ st $\forall \xi$ with $w(\xi) \leq s$:

$$w(x(t, \xi, d)) < \frac{1}{2}s \quad \forall t \geq T_s \quad (3.5)$$

Now, given $\epsilon > 0$, choose $k \in \mathbb{N}$ st $2^{-k} \cdot s < \epsilon$. Consider the series $\{s_i\}_{i \in \{1, \dots, k\}}$ defined following the rule $s_1 = s, s_i = \frac{1}{2}s_{i-1}$. For every s_i , take T_{s_i} . Let $t_\epsilon = \sum_{i=1}^k T_{s_i}$. Then, for $t \geq t_\epsilon$, ξ st $w(\xi) \leq s$ and $d \in \mathcal{M}_{c^*}$,

$$w(x, t, \xi, d) \leq \frac{s}{2^k} < \epsilon$$

where we have applied (3.5) recursively. $\square$

This completes the proof that the conditions in Definition 5 hold, and therefore (3.4) is Locally Asymptotically Stable in S_r and therefore by Proposition 1.1.7 $\exists \beta_2 \in \mathcal{KL}$ st for all $\xi \in S_r$, $w(x(t, \xi, d)) \leq \beta_2(w(\xi, t)) \forall t \geq 0$.

As the solutions of (3.4) are unique, consider now the time transformation

$$\tau = \int_0^t 1 + \|f_{wr}(x(s), d(x(s), s))\| + w(x(s)) ds$$

Note that $1 + \|f_{wr}(x, d)\| + w(x) \geq 1 \forall x \in \mathcal{X}$. $\frac{d\tau}{dt} = 1 + \|f_{wr}(x(t), d(x(t), t))\| + w(x(t))$ and by the inverse function theorem $\frac{d\tau}{dt}^{-1} = \frac{1}{1 + \|f_{wr}(x, d)\| + w(x)}$.

As $\frac{dx}{dt} = \frac{dx}{d\tau} \cdot \frac{d\tau}{dt}$, we get

$$\frac{dx}{d\tau} = \frac{d\tau}{dt}^{-1} \cdot f_{wr}(x, d) = \frac{f_{wr}(x, d)}{1 + \|f_{wr}(x, d)\| + w(x)}$$

Observe that for each $i \in \{1, \dots, n\}$, $\lim_{x \rightarrow \partial \mathcal{X}} \frac{|f_{wr_i}(x)|}{1 + \|f_{wr}(x, d)\| + w(x)} \leq \lim_{x \rightarrow \partial \mathcal{X}} \frac{|f_{wr_i}(x)|}{|f_{wr_i}(x)| + w(x)} = 0$ because $w(x) \rightarrow \infty$ as $x \rightarrow 0$. If $\mathcal{X}$ is unbounded, then the limit when $\lim_{\|x\| \rightarrow \infty} \frac{|f_{wr_i}(x)|}{1 + \|f_{wr}(x, d)\| + w(x)} = 0$ too (cause $w(x)$ will also tend to infinity).

Following the ideas in Appendix C.8 in [3], define now the system:

$$\dot{x} = \begin{cases} \frac{f_{wr}(x, d)}{1 + \|f_{wr}(x, d)\| + w(x)} & \text{if } x \in \mathcal{X} \\ 0 & \text{if } x \in \partial \mathcal{X} \end{cases} \quad (3.6)$$

which is well-defined thanks to the limit properties showcased above and has unique solutions because the original system does. We now once again abuse the notation and:

- (i) denote by $x(t, \xi, d)$ the solution to the system under a disturbance $d \in \mathcal{M}_{c^*}$ with initial condition $\xi \in \overline{\mathcal{X}}$
- (ii) let $d(t, \xi) := d(x(t, \xi, d), t)$

Note that $\partial \mathcal{X}$ is a set of equilibria for system (3.6) and $\mathcal{A}$ is still a compact set of equilibria for (3.6). Note also that given $\xi \in \mathcal{X}$, $\|x(t, \xi, d) - \xi\| = \left\| \int_0^t \frac{f_{wr}(x, d)}{1 + \|f_{wr}(x, d)\| + w(x)} ds \right\| \leq$

$\int_0^t \left\| \frac{f_{wr}(x,d)}{1+\|f_{wr}(x,d)\|+w(x)} \right\| ds \leq t$. Therefore, if $\|x(t, \xi, d)\| \rightarrow \infty$ it always does so at a slower rate than τ and therefore such solutions are defined for all $t \geq 0$. Meanwhile, if $x(t, \xi, d) \rightarrow \partial\mathcal{X}$ then the solution tends towards an equilibrium of the system and is therefore also defined $\forall t \geq 0$. Finally, every solution starting at $\partial\mathcal{X}$ remains still $\forall t \geq 0$. From Theorem 2 in Chapter 2 in [36] we know that, if the forward time of existence of a solution is finite, then that solution must tend to either $\partial\mathcal{X}$ or ∞ when t approaches the forward time of existence. However, we have seen that every solution that approaches $\partial\mathcal{X}$ or ∞ has infinite forward time of existence, meaning the system is forward complete. Note that this reasoning is symmetrical for time-reversed solutions, so we can conclude **the system (3.6) is complete**.

Note that for any starting point $\xi \in S_r$, the trajectories of system (3.6) and system (3.4) will be the same: they will just be followed at different paces, because we have only performed a time transformation (and extended the system, but that does not affect trajectories starting in S_r). Another way to look at it is to observe that, at every point, the vectors $f_{wr}(x, d)$ and $\frac{f_{wr}(x,d)}{1+\|f_{wr}(x,d)\|+w(x)}$ have different modulus but the same direction (for a technical proof of this fact, see Lemma 7.2 in [35]). Therefore, $\exists \beta \in \mathcal{KL}$ st for all $\xi \in S_r$:

$$x(t, \xi, d) \leq \beta(w(\xi), t) \quad \forall t \geq 0$$

We will work with system (3.6) from now and try to find a Local Lyapunov function for it, because if we have a continuously differentiable function W st W is a Local Lyapunov function for (3.6) on S_r then it is also a Local Lyapunov function for (3.4) on S_r :

$$\begin{aligned} \mathcal{L}_{\frac{f_{wr}(x,d)}{1+\|f_{wr}(x,d)\|+w(x)}} W(x) = \\ \nabla W(x)^t \cdot \frac{f_{wr}(x,d)}{1+\|f_{wr}(x,d)\|+w(x)} \leq -\alpha_W(x) \Rightarrow \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{f_{wr}(x,d)} W(x) = \\ \nabla W(x)^t \cdot f_{wr}(x,d) \leq -(1+\|f_{wr}(x,d)\|+w(x))\alpha_W(x) \\ \leq -\alpha_W(x) \end{aligned}$$

where $\alpha_W(x) \in \mathcal{K}_\infty$.

Let $\alpha(r) := \beta(r, 0) \in \mathcal{K}_\infty$.

For every $\xi \in S_r$, define $t_\xi^e := \inf\{t \leq 0 \mid w(x(t, \xi, d)) \leq r\}$ and $E_\xi = \{t \in [t_\xi^e, 0] \mid w(x(t, \xi, d)) > r\}$, which is an open set. Note that for every $\xi \in S_r$ and every $\tilde{t} \in [t_\xi^e, 0] \setminus E_\xi$, it holds that:

$$w(x(t - \tilde{t}, x(\tilde{t}, \xi, d), d_{\tilde{t}})) \leq \alpha(w(x(\tilde{t}, \xi, d))) \quad (3.7)$$

where $d_{\tilde{t}}(\cdot, t) = d(\cdot, t + \tilde{t})$. Finally, for $\xi \in S_r$, define:

$$g(\xi) = \inf\{w(x(t, \xi, d)) \mid t \leq 0, d \in \mathcal{M}_{c^*}\} \quad (3.8)$$

Notice that from equation (3.7) with $t = 0$ we get that $w(\xi) \leq \alpha(w(x(\tilde{t}, \xi, d))) \Rightarrow w(x(\tilde{t}, \xi, d)) \geq \alpha^{-1}(w(\xi)) \quad \forall \tilde{t} \in [t_\xi^e, 0] \setminus E_\xi$.

As the set S_r is not re-entered by the solution for times smaller than t_ξ^e , we can rewrite g as $g(\xi) = \inf\{w(x(t, \xi, d)) \mid t \in [t_\xi^e, 0] \setminus E_\xi, d \in \mathcal{M}_{c^*}\} \geq \alpha^{-1}(w(\xi))$.

Note also that $\inf\{w(x(t, \xi, d)) \mid t \in [t_\xi^e, 0] \setminus E_\xi, d \in \mathcal{M}_{c^*}\} \leq r \leq \inf\{w(x(t, \xi, d)) \mid t < t_\xi^e \text{ or } t \in E_\xi\}$ and also that $w(x(0, \xi, d)) = w(\xi) \Rightarrow \inf\{w(x(t, \xi, d)) \mid t \in [t_\xi^e, 0] \setminus E_\xi, d \in \mathcal{M}_{c^*}\} \leq w(\xi)$.

We conclude that

$$\alpha^{-1}(w(\xi)) \leq g(\xi) \leq w(\xi) \quad \forall \xi \in S_r$$

The next step is to prove that g is Lipschitz in every ring $K_{s_1, s_2} = \{x \in \mathcal{X} \mid s_1 \leq w(x) \leq s_2\} \subset S_r \setminus \mathcal{A}$. To see this, we first need a technical result.

Claim 3.1.7. *Given $s \in (0, r]$, $\forall \epsilon > 0 \exists q_{\epsilon, s} \leq 0$ st $\forall \xi \in K_{\epsilon, s}$ and any $d \in \mathcal{M}_{c^*}$,*

$$w(x(t, \xi, d)) > s \quad \forall t < q_{\epsilon, s}$$

Proof. Suppose the statement is false, this is, suppose $\exists \xi \in K_{\epsilon, s}$ and $\exists d \in \mathcal{M}_{c^*}$ such that $w(x(t, \xi, d)) \leq s \forall t < 0$.

By the properties of class $\mathcal{KL}$ functions there exists $\tau_{\epsilon, s} \geq 0$ st $w(x(t, \xi, d)) < \epsilon \forall t > \tau_{\epsilon, s}$, $\forall \xi \in S_s$.

Take $t_{aux} \in (-\tau_{\epsilon, s}, 0)$ and consider $x(t_{aux}, \xi, d)$, which must be in S_s . Thus, $x(t - t_{aux}, x(t_{aux}, \xi, d), d_{t_{aux}})$ must satisfy $w_{\mathcal{A}}(x(t - t_{aux}, x(t_{aux}, \xi, d), d_{t_{aux}})) < \epsilon \forall t - t_{aux} > \tau_{\epsilon, s}$.

Taking $t = 0$, we have $-t_{aux} > \tau_{\epsilon, s}$ (cause we have chosen $t_{aux} \in (-\tau_{\epsilon, s}, 0)$). Therefore:

$$w(x(-t_{aux}, x(t_{aux}, \xi, d), d_{t_{aux}})) = w(\xi) < \epsilon$$

which is a contradiction if $\xi \in K_{\epsilon, s}$. This concludes the proof of the claim. $\square$

We can now prove Lipschitzness of the function g on every ring of S_r .

Claim 3.1.8. *For every pair s_1, s_2 such that $0 < s_1 < s_2 \leq r$, $g(\xi)$ is Lipschitz on K_{s_1, s_2} .*

Proof. Take $\xi_1, \xi_2 \in K_{s_1, s_2}$, and consider WLOG that $g(\xi_1) > g(\xi_2)$. Consider the constant q_{s_1, s_2} associated to K_{s_1, s_2} given by Claim 3.1.7.

Note that $\forall \epsilon > 0 \exists t_{2, \epsilon} \in [q_{s_1, s_2}, 0]$ and $d_{2, \epsilon} \in \mathcal{M}_{c^*}$ such that $g(\xi_2) > w_{\mathcal{A}}(x(t_{2, \epsilon}, \xi_2, d_{2, \epsilon})) - \epsilon$. Then:

$$|g(\xi_1) - g(\xi_2)| = g(\xi_1) - g(\xi_2) < w(x(t_{2, \epsilon}, \xi_1, d_{2, \epsilon})) - w(x(t_{2, \epsilon}, \xi_2, d_{2, \epsilon})) + \epsilon.$$

$$\text{Making } \epsilon \text{ tend to } 0, \text{ we get } |g(\xi_1) - g(\xi_2)| \leq w(x(t_{2, \epsilon}, \xi_1, d_{2, \epsilon})) - w(x(t_{2, \epsilon}, \xi_2, d_{2, \epsilon})).$$

As w is locally Lipschitz in $S_r \setminus \mathcal{A}$, it is Lipschitz in the compact K_{s_1, s_2} with a Lipschitz constant the we will call l_{w, s_1, s_2} . Note also that, as our system (3.6) is complete, we can apply Proposition 5.5 in [13] with $K = K_{s_1, s_2}$ and $T = |q_{s_1, s_2}|$ to get that there exists L_{s_1, s_2} st $\|x(t, \xi_1, d) - x(t, \xi_2, d)\| \leq L_{s_1, s_2} \|\xi_1 - \xi_2\|$ for all t st $|t| \leq |q_{s_1, s_2}|$, all $\xi_1, \xi_2 \in K_{s_1, s_2}$ and all $d \in \mathcal{M}_{c^*}$. Then:

$$\begin{aligned} |g(\xi_1) - g(\xi_2)| &\leq l_{w, s_1, s_2} \cdot \|x(t_{2, \epsilon}, \xi_1, d_{2, \epsilon}) - x(t_{2, \epsilon}, \xi_2, d_{2, \epsilon})\| \\ &\leq l_{w, s_1, s_2} \cdot L_{s_1, s_2} \cdot \|\xi_1 - \xi_2\| \end{aligned}$$

for all $\xi_1, \xi_2 \in K_{s_1, s_2}$. $\square$

Note also that g is continuous and equal to zero on $\mathcal{A}$, as it is sandwiched between $\alpha^{-1}(w(\xi))$ and $w(\xi)$ which are both continuous and tend to 0 as ξ tends to $\mathcal{A}$.

Now, define $U : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ as:

$$U(\xi) := \sup_{t \geq 0, d \in \mathcal{M}_{c^*}} \{g(x(t, \xi, d)) \cdot \frac{1+2t}{1+t}\} \quad (3.9)$$

We can see that $U(\xi) \leq 2 \cdot \sup_{t \geq 0, d \in \mathcal{M}_{c^*}} \inf_{\tilde{t} \in [t_{x(t, \xi, d)}^e, 0], \tilde{d} \in \mathcal{M}_{c^*}} w(x(\tilde{t}, x(t, \xi, d), \tilde{d}))$. By (3.7), we know that for every fixed $t \geq 0$ and $d \in \mathcal{M}_{c^*}$ taking $d_t \in \mathcal{M}_{c^*}$ and $\tilde{t} = -t$ gives us $x(-t, x(t, \xi, d), d_t) = \xi$ (and this proves $-t \geq t_{x(t, \xi, d)}^e$). Note also that $\inf_{\tilde{t} \in [t_{x(t, \xi, d)}^e, 0], \tilde{d} \in \mathcal{M}_{c^*}} w(x(\tilde{t}, x(t, \xi, d), \tilde{d})) \leq w(x(-t, x(t, \xi, d), d_t)) = w(\xi)$, so we end up getting:

$$U(\xi) \leq 2 \cdot \sup_{t \geq 0, d \in \mathcal{M}_{c^*}} w(\xi) = 2w(\xi)$$

On the other side, $U(\xi) \geq (w(x(t, \xi, d)) \cdot \frac{1+2t}{1+t})|_{t=0} = g(\xi) \geq \alpha^{-1}(w(\xi))$. Therefore:

$$\alpha^{-1}(w(\xi)) \leq U(\xi) \leq 2w(\xi) \quad \forall \xi \in S_r$$

U is continuous and equal to 0 at $\mathcal{A}$ for the same reasons g is: it is sandwiched between continuous functions of ξ that tend to 0 when ξ tends to $\mathcal{A}$.

U , just as g , is Lipschitz in every ring of S_r . To prove that, we first need a technical result. For convinience, from now on let $k(t) := \frac{1+2t}{1+t}$.

Given $0 < s_1 < s_2 \leq r$, by the properties of $\mathcal{KL}$ functions, there always exists $t^* \geq 0$ st $\beta(s_s, t) < \frac{\alpha^{-1}(s_1)}{4} \forall t > t^*$. Let $\sigma_{s_1, s_2} := t^*$.

Claim 3.1.9. $U(\xi) = \sup\{g(x(t, \xi, d)) \cdot k(t) \mid t \in [0, \sigma_{s_1, s_2}], d \in \mathcal{M}_{c^*}\} \quad \forall \xi \in K_{s_1, s_2}$

Proof. Suppose the claim is false. Then, $\forall \epsilon > 0 \exists t_\epsilon > \sigma_{s_1, s_2}$ and $d^\epsilon \in \mathcal{M}_{c^*}$ st $\alpha^{-1}(s_1) \leq U(\xi) < g(x(t_\epsilon, \xi, d^\epsilon)) \cdot k(t_\epsilon) + \epsilon \leq 2w(x(t_\epsilon, \xi, d^\epsilon)) + \epsilon$. Make $\epsilon \rightarrow 0$ and we get $\alpha^{-1}(s_1) \leq U(\xi) \leq 2w(x(t_\epsilon, \xi, d^\epsilon))$.

As $t_\epsilon > \sigma_{s_1, s_2}$, $w(x(t_\epsilon, \xi, d^\epsilon)) \leq \beta(w(\xi), t) \leq \beta(s_2, t) < \frac{\alpha^{-1}}{4}$. Therefore, we are left with: $\alpha^{-1}(s_1) \leq \frac{\alpha^{-1}}{2}$, which is a contradiction. This proves the claim. $\square$

Claim 3.1.10. U is Lipschitz in every ring K_{s_1, s_2}

Proof. Take $\xi_1, \xi_2 \in K_{s_1, s_2}$ and suppose WLOG $U(\xi_1) \geq U(\xi_2)$. Remember that $U(\xi) = \sup\{g(x(t, \xi, d)) \cdot k(t) \mid t \in [0, \sigma_{s_1, s_2}], d \in \mathcal{M}_{c^*}\} \quad \forall \xi \in K_{s_1, s_2}$. Then,

$$|U(\xi_1) - U(\xi_2)| = \sup_{t \in [0, \sigma_{s_1, s_2}], d \in \mathcal{M}_{c^*}} g(x(t, \xi_1, d)) \cdot k(t) - \sup_{t \in [0, \sigma_{s_1, s_2}], d \in \mathcal{M}_{c^*}} g(x(t, \xi_2, d)) \cdot k(t).$$

We know that $\forall \epsilon > 0 \exists t_{1, \epsilon} \in [0, \sigma_{s_1, s_2}]$ and $d_{1, \epsilon} \in \mathcal{M}_{c^*}$ st $U(\xi_1) < g((x(t_{1, \epsilon}, \xi_1, d_{1, \epsilon}))) \cdot k(t_{1, \epsilon}) + \epsilon$. By definition, $U(\xi_2) \geq g((x(t_{1, \epsilon}, \xi_2, d_{1, \epsilon}))) \cdot k(t_{1, \epsilon})$. Therefore:

$$|U(\xi_1) - U(\xi_2)| < (g((x(t_{1, \epsilon}, \xi_1, d_{1, \epsilon}))) - g((x(t_{1, \epsilon}, \xi_2, d_{1, \epsilon})))) \cdot k(t_{1, \epsilon}) + \epsilon \leq 2(g((x(t_{1, \epsilon}, \xi_1, d_{1, \epsilon}))) - g((x(t_{1, \epsilon}, \xi_2, d_{1, \epsilon})))) + \epsilon \leq L_{s_1, s_2}^g \|x(t_{1, \epsilon}, \xi_1, d_{1, \epsilon}) - x(t_{1, \epsilon}, \xi_2, d_{1, \epsilon})\| + \epsilon$$

where L_{s_1, s_2}^g is the Lipschitz constant for g in K_{s_1, s_2} (we have already seen g Lipschitz in every ring).

As our system (3.6) is complete, we can again apply Proposition 5.5 in [35] with $K = K_{s_1, s_2}$ and $T = \sigma_{s_1, s_2}$ to get that there exists L_{s_1, s_2} st $\|x(t, \xi_1, d) - x(t, \xi_2, d)\| \leq L_{s_1, s_2} \|\xi_1 - \xi_2\|$ for all t st $|t| \leq |\sigma_{s_1, s_2}|$, all $\xi_1, \xi_2 \in K_{s_1, s_2}$ and all $d \in \mathcal{M}_{c^*}$.

Taking the limit $\epsilon \rightarrow 0$, we finally get:

$$|U(\xi_1) - U(\xi_2)| \leq L_{s_1, s_2}^g \cdot L_{s_1, s_2} \cdot \|\xi_1 - \xi_2\|$$

This proves the claim. $\square$

Define now $\sigma_s := \min\{t \mid \beta(r, t) \leq \frac{\alpha^{-1}(\frac{s}{2})}{4}\}$, which is clearly unique because β is decreasing on t . Let $\alpha_r(t) = \beta(r, t)$, which is continuous, decreasing on t and tends to 0 as t tends to infinity. Its inverse is $\alpha_r^{-1} : (0, \alpha_r(0)] \rightarrow \mathbb{R}_{\geq 0}$, which follows $\lim_{v \rightarrow 0^+} \alpha_r^{-1}(v) = \infty$ and $\alpha_r^{-1}(\alpha_r(0)) = 0$. It is clear that its inverse is continuous in its domain. Now, define:

$$\overline{\alpha_r^{-1}}(v) = \begin{cases} \alpha_r^{-1}(v) & \text{if } v \leq \alpha_r(0) = \beta(r, 0) = \alpha(r) \\ 0 & \text{if } v > \alpha_r(0) = \alpha(r) \end{cases}$$

It is also clear that $\overline{\alpha_r^{-1}} : (0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ is continuous in its domain. Note that we can write $\sigma_s = \overline{\alpha_r^{-1}}(\frac{\alpha^{-1}(\frac{s}{2})}{4})$. This implies that σ_s is continuous in s , as it is the composition of two continuous functions. From now on, for convenience, we will write:

$$\sigma_\xi := \min\{t \mid \beta(r, t) \leq \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{4}\} = \overline{\alpha_r^{-1}}(\frac{\alpha^{-1}(\frac{w(\xi)}{2})}{4})$$

where σ_ξ is continuous with respect to $w(\xi)$. It is clear that, given $\xi \in S_r$ and $d \in \mathcal{M}_{c^*}$, we have that $w(x(t, \xi, d)) \leq \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{4} \forall t > \sigma_\xi$.

Take now $\xi \in S_r$. By continuity, there exists $h_\xi > 0$ st $w(x(t, \xi, \mathbf{d})) \geq \frac{w(\xi)}{2} \forall t \in [0, h_0]$ and $\mathbf{d} \in C_{c^*}$.

Take $h \in [0, h_\xi]$ and let $\mu_{\mathbf{d}} = x(h, \xi, \mathbf{d})$. Then, $\forall \epsilon > 0$ exists $t_{\mathbf{d}, \epsilon} > 0$ and $d^{\mathbf{d}, \epsilon} \in \mathcal{M}_{d^*}$ such that $U(\mu_{\mathbf{d}}) < g(x(t_{\mathbf{d}, \epsilon}, \mu_{\mathbf{d}}, d^{\mathbf{d}, \epsilon})) \cdot k(t_{\mathbf{d}, \epsilon}) + \epsilon$.

From now on, let:

$$\tilde{d}^{\mathbf{d}, \epsilon} = \begin{cases} \mathbf{d} & \text{if } 0 \leq t \leq h \\ d_h^{\mathbf{d}, \epsilon} & \text{if } h < t \leq t_{\mathbf{d}, \epsilon} \end{cases}$$

Note that $g(x(t_{\mathbf{d}, \epsilon}, \mu_{\mathbf{d}}, d^{\mathbf{d}, \epsilon})) = g(x(t_{\mathbf{d}, \epsilon} + h, \xi, \tilde{d}^{\mathbf{d}, \epsilon}))$. Then:

$$\begin{aligned} U(\mu_{\mathbf{d}}) &< g(x(t_{\mathbf{d}, \epsilon} + h, \xi, \tilde{d}^{\mathbf{d}, \epsilon})) \cdot k(t_{\mathbf{d}, \epsilon}) + \epsilon = g(x(t_{\mathbf{d}, \epsilon} + h, \xi, \tilde{d}^{\mathbf{d}, \epsilon})) \cdot (k(t_{\mathbf{d}, \epsilon} + h) - k(t_{\mathbf{d}, \epsilon} + h) + k(t_{\mathbf{d}, \epsilon})) + \epsilon \\ &= g(x(t_{\mathbf{d}, \epsilon} + h, \xi, \tilde{d}^{\mathbf{d}, \epsilon})) \cdot k(t_{\mathbf{d}, \epsilon}) \cdot (1 - \frac{k(t_{\mathbf{d}, \epsilon} + h) - k(t_{\mathbf{d}, \epsilon})}{k(t_{\mathbf{d}, \epsilon} + h)}) + \epsilon. \end{aligned}$$

Observe that $g(x(t_{\mathbf{d}, \epsilon} + h, \xi, \tilde{d}^{\mathbf{d}, \epsilon})) \cdot k(t_{\mathbf{d}, \epsilon}) \leq U(\xi)$. Thus, we have the inequality: $U(\mu_{\mathbf{d}}) \leq U(\xi) \cdot (1 - \frac{k(t_{\mathbf{d}, \epsilon} + h) - k(t_{\mathbf{d}, \epsilon})}{k(t_{\mathbf{d}, \epsilon} + h)}) + \epsilon$ which we can rewrite as:

$$U(x(h, \xi, \mathbf{d})) - U(\xi) < -U(\xi) \cdot \frac{k(t_{\mathbf{d}, \epsilon} + h) - k(t_{\mathbf{d}, \epsilon})}{k(t_{\mathbf{d}, \epsilon} + h)} + \epsilon$$

As $k(t_{\mathbf{d}, \epsilon} + h) \leq 2$, $U(x(h, \xi, \mathbf{d})) - U(\xi) < -U(\xi) \cdot \frac{k(t_{\mathbf{d}, \epsilon} + h) - k(t_{\mathbf{d}, \epsilon})}{2} + \epsilon = -\frac{U(\xi)}{2} \cdot k'(t_{\mathbf{d}, \epsilon} + \theta h) \cdot h + \epsilon$ with $\theta \in [0, 1]$ by the Mean Value Theorem.

The derivative of k is simply $\frac{1}{(1+t)^2}$, which is strictly decreasing in t and tends to 0 as t tends to infinity. To continue with the proof, we need one last claim:

Claim 3.1.11. *Given $\xi \in S_r$ and $h \in [0, h_\xi]$, $t_{d,\epsilon} + h \leq \sigma_\xi \ \forall d \in C_{c^*}$ and $\forall \epsilon \in (0, \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{4})$.*

Proof. Suppose the claim was not true. Then, there would exist $\mathbf{d}^* \in C_{c^*}$ and $\epsilon^* \in (0, \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{2})$ st $t_{\mathbf{d}^*, \epsilon^*} + h > \sigma_\xi^h$.

Recall that $t_{\mathbf{d}^*, \epsilon^*}$ is st $U(\mu_{\mathbf{d}^*}) < g(x(t_{\mathbf{d}^*, \epsilon^*}, \mu_{\mathbf{d}^*}, d^{\mathbf{d}^*, \epsilon^*})) \cdot k(t_{\mathbf{d}^*, \epsilon^*}) + \epsilon^*$ where $\mu_{\mathbf{d}^*} = x(h, \xi, \mathbf{d}^*)$ and also that $w(\mu_{\mathbf{d}^*}) \geq \frac{w(\xi)}{2}$. Then, by the properties of class $\mathcal{K}_\infty$ functions and functions U and g we get that: $\alpha^{-1}(\frac{w(\xi)}{2}) \leq \alpha^{-1}(\mu_{\mathbf{d}^*}) \leq U(\mu_{bd^*}) < g(x(t_{\mathbf{d}^*, \epsilon^*}, \mu_{\mathbf{d}^*}, d^{\mathbf{d}^*, \epsilon^*})) \cdot k(t_{\mathbf{d}^*, \epsilon^*}) + \epsilon^* \leq 2w(x(t_{\mathbf{d}^*, \epsilon^*}, \mu_{bd^*}, d^{\mathbf{d}^*, \epsilon^*})) + \epsilon^* = 2w(x(t_{\mathbf{d}^*, \epsilon^*} + h, \xi, \tilde{d}^{\mathbf{d}^*, \epsilon^*})) + \epsilon^*$.

Finally, observe that $w(x(t_{\mathbf{d}^*, \epsilon^*} + h, \xi, \tilde{d}^{\mathbf{d}^*, \epsilon^*})) < \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{4}$ because $t_{\mathbf{d}^*, \epsilon^*} + h > \sigma_\xi^h$ and notice that $\epsilon^* < \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{2}$. Then:

$$\alpha^{-1}(\frac{w(\xi)}{2}) < \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{2} + \frac{\alpha^{-1}(\frac{w(\xi)}{2})}{2} = \alpha^{-1}(\frac{w(\xi)}{2})$$

which is a contradiction. This proves the claim. $\square$

We know $t_{d,\epsilon} + \theta h \leq t_{d,\epsilon} + h \leq \sigma_\xi$, and therefore $-k'(t_{d,\epsilon} + \theta h) \leq -k'(\sigma_\xi)$ as k' is decreasing.

Recall $U(x(h, \xi, \mathbf{d})) - U(\xi) < -\frac{U(\xi)}{2} \cdot k'(t_{d,\epsilon} + \theta h) \cdot h + \epsilon \leq -\frac{U(\xi)}{2} \cdot k'(\sigma_\xi) \cdot h + \epsilon$. Thus, when we make ϵ tend to zero we get $U(x(h, \xi, \mathbf{d})) - U(\xi) \leq -\frac{U(\xi)}{2} \cdot k'(\sigma_\xi) \cdot h$ and therefore

$$\frac{U(x(h, \xi, \mathbf{d})) - U(\xi)}{h} \leq -\frac{U(\xi)}{2} \cdot k'(\sigma_\xi)$$

Taking the definition of the Lie derivative: $\mathcal{L}_{f(\xi, \mathbf{d})} = \lim_{h \rightarrow 0^+} \frac{U(x(h, \xi, \mathbf{d})) - U(\xi)}{h} \leq \lim_{h \rightarrow 0^+} -\frac{U(\xi)}{2} \cdot k'(\sigma_\xi) = -\frac{U(\xi)}{2} \cdot k'(\sigma_\xi) \leq -\frac{\alpha^{-1}(w(\xi))}{2} \cdot k'(\sigma_\xi) = -\frac{\alpha^{-1}(w(\xi))}{2} \cdot \frac{1}{(1+\sigma_\xi)^2}$.

Notice that, as $w(\xi) \rightarrow \infty$, σ_ξ becomes equal to zero (in fact, this happens for all $w(\xi) \geq 2\alpha(4\alpha(r))$) and thus $k'(\sigma_\xi) = 1$ when $w(\xi) \geq 2\alpha(4\alpha(r))$. As α^{-1} is a class $\mathcal{K}_\infty$ function, we conclude $\frac{\alpha^{-1}(w(\xi))}{2} \cdot \frac{1}{(1+\sigma_\xi)^2}$ is class $\mathcal{K}_\infty$ as a function of $w(\xi)$. Let $\tilde{\alpha}(s) = \frac{\alpha^{-1}(s)}{2} \cdot \frac{1}{(1+\sigma_s)^2} \in \mathcal{K}_\infty$. Then:

$$\mathcal{L}_{f(\xi, \mathbf{d})} \leq -\tilde{\alpha}(w(\xi))$$

As the function U is LL in $\overset{\circ}{S}_r \setminus \mathcal{A}$ (because it is Lipschitz in every ring) and ∂S_r has measure zero, this holds almost everywhere on $S_r \setminus \mathcal{A}$.

Now, apply Theorem 6.0.1 to U in $S_r \setminus \mathcal{A}$ and $B(0, c^*)$ with $\nu(\xi) = \frac{\tilde{\alpha}(w(\xi))}{2}$ and $\mu(\xi) = \frac{U(\xi)}{2}$ (which are both strictly positive functions in $S_r \setminus \mathcal{A}$). This gives us a smooth function $\tilde{V} : S_r \setminus \mathcal{A} \rightarrow \mathbb{R}$ such that:

- $|\tilde{V}(\xi) - U(\xi)| < \frac{U(\xi)}{2}$
- $\mathcal{L}_{f(\xi, \mathbf{d})} \tilde{V}(\xi) \leq -\frac{\tilde{\alpha}(w(\xi))}{2}$

for all $\xi \in S_r \setminus \mathcal{A}$ and all $\mathbf{d} \in C_{c^*}$ (note that there exists a continuous bijection from $\overline{B}(0, c^*) \subset \mathbb{R}^m$ to $C_{c^*} \subset \mathcal{M}_{c^*}$ and therefore C_{c^*} is compact).

From the first condition on $\tilde{V}$, we get that $\frac{U(\xi)}{2} < \tilde{V}(\xi) < \frac{3}{2}U(\xi)$, so $\frac{\alpha^{-1}(w(\xi))}{2} < \tilde{V}(\xi) < 3w(\xi) \quad \forall \xi \in S_r$.

Observe that both $\frac{\alpha^{-1}(w(\xi))}{2}$ and $3w(\xi)$ are size functions for $(\mathcal{X}, \mathcal{A})$ because $\alpha^{-1}, \mathbf{3}(s) := 3s \in \mathcal{K}_\infty$ and w is a size function for $(\mathcal{X}, \mathcal{A})$. As $\xi \rightarrow \mathcal{A}$, both $\frac{\alpha^{-1}(w(\xi))}{2}$ and $3w(\xi)$ tend to zero. Therefore, $\tilde{V}(\xi) \rightarrow 0$ when $\xi \rightarrow \mathcal{A}$. We can continuously extend $\tilde{V}$ by defining $\tilde{V}(\xi) = 0$ for all $\xi \in \mathcal{A}$. Therefore:

$$\frac{\alpha^{-1}(w(\xi))}{2} \leq \tilde{V}(\xi) \leq 3w(\xi) \quad \forall \xi \in S_r$$

Now, as $\tilde{V}$ is smooth and defined in all S_r , positive in $S_r \setminus \mathcal{A}$ and 0 in $\mathcal{A}$, we can apply Lemma 4.3 in [35] to get $\beta_V \in \mathcal{K}_\infty$ st $V = \beta_V \circ \tilde{V}$ smooth on all S_r (it is easy to check this lemma also works for any domain of $\mathbb{R}^n$ that includes the set $\mathcal{A}$). We will have:

$$\beta_V\left(\frac{\alpha^{-1}(w(\xi))}{2}\right) \leq V(\xi) \leq \beta_V(3w(\xi)) \quad \forall \xi \in S_r$$

and

$$\mathcal{L}_{f(\xi, \mathbf{d})}V(\xi) \leq -\alpha_V(w(\xi)) \quad \forall \xi \in S_r$$

where $\alpha_V(s) = \max_{t \in [\beta_V(\frac{\alpha^{-1}(s)}{2}), \beta_V(3w(\xi))]} \beta_V(t) \cdot \frac{\tilde{\alpha}(s)}{2}$.

We conclude V is a Local Uniform Lyapunov Function for the system (3.6) and therefore for system (3.4) too. This means that $\mathcal{L}_{f(\xi, \mathbf{d} \cdot \varphi(\xi))}V(\xi) \leq -\alpha_V(w(\xi))$. For each $\xi \in S_r$, by choosing different $\mathbf{d}$'s in C_{c^*} we can cover the hole set $\overline{B}(0, c^* \cdot \varphi(\xi)) \subset \mathbb{R}^m$.

Therefore, we have $\mathcal{L}_{f(\xi, \lambda)}V(\xi) \leq -\alpha_V(w(\xi)) \quad \forall \lambda \in \overline{B}(0, c^* \cdot \varphi(\xi))$. From (3.3) we get that $\psi(w(\xi)) \leq \varphi(\xi) \quad \forall \xi \in S_r$. Therefore, the condition on the Lie derivative holds whenever $\|\lambda\| \leq c^* \cdot \psi(w(\xi)) (\leq c^* \cdot \varphi(\xi))$. Let $\chi(s) = \psi^{-1}(s)$. We can rewrite our conditions on the Lie derivative as:

$$\begin{aligned} \mathcal{L}_{f(\xi, \lambda)}V(\xi) &\leq -\alpha_V(w(\xi)) \\ \forall \xi \in S_r, \lambda \in \mathbb{R}^m \text{ st } w(\xi) &\geq \chi\left(\frac{\|\lambda\|}{c^*}\right) \end{aligned}$$

Define

$$\gamma_d(s) := \max \left\{ 0, \frac{\tilde{\alpha}\left(\chi\left(\frac{s}{c^*}\right)\right)}{2} + \max_{\lambda \in \overline{B}(0, \frac{s}{c^*}), \xi \in S_{\chi(\frac{s}{c^*})}} \mathcal{L}_{f(\xi, \lambda)}V(\xi) \right\}$$

Notice we need $\chi(\frac{s}{c^*}) \leq r \iff \frac{s}{c^*} \leq \psi(r)$. It is enough to have $\frac{s}{c^*} \leq (1 - \epsilon_c) \frac{1}{c^*} \gamma_1^{-1}(\frac{1}{4} \alpha_{aux}^1(r)) \leq \psi(r)$. Therefore, we can ensure this holds for $\|\lambda\| \leq (1 - \epsilon_c) \gamma_1^{-1}(\frac{1}{4} \alpha_{aux}^1(r)) =: c_V$. Please note this is not an optimal bound on the disturbances.

γ_d is continuous, non-decreasing and 0 when $s = 0$ (as the set over which we maximize the Lie derivative in the definition of γ_d becomes $\{\lambda = 0 \in \mathbb{R}^m\} \times \mathcal{A}$, and $\mathcal{L}_{f(\xi, 0)}V(\xi)$ is zero). We can majorize to an increasing function in $[0, c_V]$ and then extend this function so that the result is a class $\mathcal{K}_\infty$ function. Call this function γ_V .

It is clear that we have

$$\mathcal{L}_{f(\xi,\lambda)}V(\xi) \leq \frac{\alpha_V(w(\xi))}{2} + \gamma_V(\lambda)$$

for all $\xi \in S_r$ and all λ st $\|\lambda\| \leq c_V$, and therefore for all $d \in \mathcal{M}_{c_V}$.

Take $\eta := \min_{x \in \partial S_r} V(x)$. Observe that the conditions in Lemma 3.1.3 hold in S_r for V , and therefore $S_\eta^V \cap S_r$ is forward invariant. $\square$

3.2 Set Input to State Stability

In the previous section, we have looked at the (Local) ISS property on open sets, which can capture input to state stability in the sub-level sets of a size function and can also avoid working with the Lie derivatives of a term $(\frac{1}{d(x,\partial\mathcal{X})})^p$.

However, it is relevant to study the input to state stable behavior of arbitrary closed sets because we often deal with systems defined on an open set $\mathcal{X}$ naturally or artificially showcasing some positive invariance properties under disturbances on a smaller closed set $C \subset \mathcal{X}$ (which need not be compact nor the sub-level set of a size function). Examples of such systems include systems with Input to State Safe Sets (see [13]).

Therefore, the following questions arise: What can we say about the trajectories of the perturbed system inside a closed forward invariant set under the disturbance? Can we establish a more general ISS concept that encompasses arbitrary closed sets while avoiding the complexities of computing the Lie derivative of a size function for a set $\mathcal{X}$?

In this section we introduce Set Input to State Stability (SISS), an extended input to state stability property which gives a positive answer to these questions.

3.2.1 Set Size Functions

We begin by relaxing the definition of size function in an open set.

Definition 16 (Set Size Function). *Consider a compact set $\mathcal{A}$ of equilibrium points of interest for the system 1.1 with $d = 0$. Let $w_{\mathcal{A}} : \text{dom}(w_{\mathcal{A}}) \rightarrow \mathbf{R}$ where $C \subseteq \text{dom}(w_{\mathcal{A}})$. We will say $w_{\mathcal{A}}$ is a **set size function** for $(C, \mathcal{A})$ if:*

- (i) *it is continuous*
- (ii) *If $x \in C$, then $w_{\mathcal{A}}(x) = 0 \iff x \in C \cap \mathcal{A}$, also $w_{\mathcal{A}}(x) > 0 \quad \forall x \in C \setminus (C \cap \mathcal{A})$*
- (iii) *$\forall r \geq 0 \in \mathbf{R}$, $S_r := \{x \in C \mid |w_{\mathcal{A}}(x)| \leq r\}$ is a compact subset of $\mathbf{R}^n$*

Lemma 3.2.1 (Necessary and Sufficient conditions for Set Size Functions). *If conditions i) and ii) in Definition 16 hold, then $w_{\mathcal{A}}$ is a set size function if and only if $\lim_{n \rightarrow \infty} w_{\mathcal{A}}(x_n) = \infty$ for all $(x_n)_{n \in \mathbb{N} \cup \{0\}} \subseteq C$ such that $\lim_{n \rightarrow \infty} \|x_n\| = \infty$*

Proof. Suppose the conditions for the reverse implication hold, then we apply Tietze Extension Theorem to continuously $w_{\mathcal{A}}$ from C to $\mathbf{R}^n$, and add the term $d(x, C)$, which is continuous. Let us call the result $w_{\mathcal{A}}^{ext}$. By the properties of Tietze extension theorem

and taking into account that $d(x, C) = 0 \ \forall x \in C$ and $d(x, C) > 0 \ \forall x \in \mathbb{R}^n \setminus C$; we can see that $w_{\mathcal{A}}^{ext}|_C = w_{\mathcal{A}}$, $w_{\mathcal{A}}^{ext}(x) > 0 \ \forall x \in \mathbb{R}^n \setminus C$. What is more, $\lim_{n \rightarrow \infty} w_{\mathcal{A}}^{ext}(x_n) = \infty$ for all $(x_n)_{n \in \mathbb{N} \cup \{0\}} \subseteq \mathbb{R}^n$ such that $\lim_{n \rightarrow \infty} \|x_n\| = \infty$. Therefore, $S_r^{w_{\mathcal{A}}^{ext}}$ is compact for all $r \geq 0$. Notice that $S_r^{w_{\mathcal{A}}^{ext}} \cap C = S_r$, and therefore S_r is compact $\forall r \geq 0$. We conclude $w_{\mathcal{A}}$ is a set size function.

For the direct implication, imagine the sets S_r were compact $\forall r \geq 0$ but $\exists (x_n)_{n \in \mathbb{N} \cup \{0\}} \subseteq C$ st $\lim_{n \rightarrow \infty} \|x_n\| = \infty$ and $\lim_{n \rightarrow \infty} w_{\mathcal{A}}(x_n) < \infty$. This means that $\exists M > 0$ st $w_{\mathcal{A}}(x_n) \leq M \ \forall n \in \mathbb{N} \cup \{0\}$ and thus $x_n \in S_M \ \forall n \in \mathbb{N} \cup \{0\}$. However, $\lim_{n \rightarrow \infty} \|x_n\| = \infty$, so $(x_n)_{n \in \mathbb{N} \cup \{0\}}$ cannot be contained in any compact. As S_M is compact, this leads to a contradiction. $\square$

It is relevant to note that many systems have two intertwined sets of equilibria with different properties such that one includes the other. For example, in constrained gradient systems we are interested in the minima of the potential function, which are included in the set of KKT points. In general, we are not able to formulate a Lyapunov inequality in terms of a size function for the KKT points but rather two size functions, one for the set of minima and other for the set of KKT points. The definition of "weak" set size functions enables us to deal with these kinds of settings.

Definition 17 (Weak Set Size Function). *Consider a compact set $\tilde{\mathcal{A}}$ of equilibrium points for the system 1.1 with $d = 0$ such that $\mathcal{A} \subseteq \tilde{\mathcal{A}}$. Let $w_{\tilde{\mathcal{A}}} : \text{dom}(w_{\tilde{\mathcal{A}}}) \rightarrow \mathbf{R}$ where $C \subseteq \text{dom}(w_{\tilde{\mathcal{A}}})$. We will say $w_{\tilde{\mathcal{A}}}$ is a **weak set size function** for $(C, \mathcal{A})$ if it is a set size function for $(C, \tilde{\mathcal{A}})$.*

We can show that Set Size Functions share properties with their Size Function counterparts:

Lemma 3.2.2 (Set Size Function Inverse Continuity). *If $w_{\mathcal{A}}$ is a set size function for $(C, \mathcal{A})$ then given an $x \in C$:*

$$\forall \epsilon > 0, \exists \delta > 0 \quad \text{st} \quad w_{\mathcal{A}}(x) < \delta \Rightarrow |x|_{\mathcal{A}} < \epsilon$$

Proof. The proof is equivalent to the one in Lemma 2.5 in [5]. We just need to take into account that $B := \{x \in C \mid |x|_{\mathcal{A}} \geq \bar{\epsilon} \text{ and } w_{\mathcal{A}}(x) \leq 1\} = C \cap \{x \mid |x|_{\mathcal{A}} \geq \bar{\epsilon}\} \cap \{x \mid |w_{\mathcal{A}}(x)| \leq 1\}$ because $w_{\mathcal{A}}(x) \geq 0 \in \mathcal{A}$, and therefore B is compact (it is the intersection of two closed sets -the first two ones- with a compact -the last one, because it is a sub-level set of a set size function). $\square$

Proposition 3.2.3 (Equivalence of Set Size Functions). *Given two set size functions w_1, w_2 for $(C, \mathcal{A})$; $\exists \alpha \in \mathcal{K}_{\infty}$ such that:*

$$w_1(x) \leq \alpha(w_2(x)) \ \forall x \in C$$

Proof. The proof is equivalent to the one in Proposition 2.6 in [5]. We just have to be careful and take $\tilde{\alpha}(r) := \max_{\{x \in C \mid w_2(x) \leq r\}} w_1(x)$. Since $\{x \in C \mid w_2(x) \leq r\}$ is compact (because $w_2(x)$ is a size function and we are intersecting one of its sub-level sets with the closed set C where $w_2(x) \geq 0$), we can proceed equivalently, keeping in mind that every time we write $w(x)$ to be smaller than some quantity, we do not need to write absolute value because we are only considering $x \in C$. $\square$

Corollary 3.2.4. *Given w_1 a set size function for $(C, \mathcal{A})$ and w_2 a continuous function: w_2 is a set size function for $(C, \mathcal{A}) \iff \exists \underline{\alpha}, \bar{\alpha} \in \mathcal{K}_\infty$ st:*

$$\underline{\alpha}(w_1(x)) \leq w_2(x) \leq \bar{\alpha}(w_1(x)) \quad \forall x \in C$$

When we are dealing with a weak set size function and a set size function, we cannot establish an equivalence between the two of them. However, we can see there is a weaker relationship between both that proves to be very useful.

Lemma 3.2.5 (Weak Equivalence Set Size Function - Weak Set Size Function). *If $w_{\tilde{\mathcal{A}}}$ is a weak set size function for $(C, \mathcal{A})$ and $w_{\mathcal{A}}$ is a set size function for $(C, \mathcal{A})$, then $\exists \alpha \in \mathcal{K}_\infty$ such that*

$$w_{\mathcal{A}}(x) \leq \alpha(w_{\tilde{\mathcal{A}}}(x)) \quad \forall x \in C$$

Proof. Let $\tilde{\alpha}(r) := \max_{\{x \in C \mid w_{\tilde{\mathcal{A}}}(x) \leq r\}} w_{\mathcal{A}}$. As the set $\{x \in C \mid w_{\tilde{\mathcal{A}}}(x) \leq r\}$ is compact (it is the sub-level set of a set size function intersected with the closed set C), the function $\tilde{\alpha}(r)$ is well defined, single-valued and non-decreasing (as r increases we evaluate over bigger sets).

Take $x \in C$, and take $r = w_{\tilde{\mathcal{A}}}(x)$. Then, $w_{\mathcal{A}}(x) \leq \tilde{\alpha}(r) = \tilde{\alpha}(w_{\tilde{\mathcal{A}}}(x))$.

We see that, if we see continuity and then we majorize, $\tilde{\alpha} \in \mathcal{K}_\infty$ and not $\mathcal{K}_\infty$ because $\tilde{\alpha}(0) := \max_{\{x \in C \mid w_{\tilde{\mathcal{A}}}(x) \leq 0\}} w_{\mathcal{A}} = \max_{\tilde{\mathcal{A}} \cap C} w_{\mathcal{A}}$.

Taking into account that $w_{\mathcal{A}}(x) = 0 \quad \forall x \in \mathcal{A}$, we can rewrite $\tilde{\alpha}(0) = \max_{(\tilde{\mathcal{A}} \setminus \mathcal{A}) \cap C} w_{\mathcal{A}} =:$

a. Clearly, $a > 0$ and thus our previous claim is proven.

Now, we need to see continuity at 0^+ . This is, we need to see that

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \text{st} \quad \forall r < \delta, \quad \tilde{\alpha}(r) - a < \epsilon$$

As $w_{\mathcal{A}}$ is not always equal to zero in $\tilde{\mathcal{A}}$, we cannot proceed as we did in proposition ???. But because $\tilde{\mathcal{A}}$ is a compact, we can define $M := \max_{z \in \tilde{\mathcal{A}}} |z|$. Now take an arbitrary $\delta_1 > 0$ and consider $\overline{B}(0; M + \delta_1)$.

$|x|_{\tilde{\mathcal{A}}} < \delta_1 \iff \min_{z \in \tilde{\mathcal{A}}} |x - z| < \delta_1$. Therefore, $|x|_{\tilde{\mathcal{A}}} < \delta_1 \Rightarrow x \in \overline{B}(0; M + \delta_1)$ because $|x| \leq |x - z| + |z|$ and, taking $z^* \in \operatorname{argmin}_{z \in \tilde{\mathcal{A}}} |x - z|$, we get that $|x| \leq |x - z^*| + |z^*| < \delta_1 + M \Rightarrow |x| \in \overline{B}(0; M + \delta_1)$. If additionally we have that $x \in C$, then clearly we will have $x \in \overline{B}(0; M + \delta_1) \cap C$, which is a compact set (because it is the intersection of a compact with a closed set).

Now, consider the set size function $w_{\mathcal{A}}$ defined in $\overline{B}(0; M + \delta_1) \cap C \subseteq C \subseteq \operatorname{dom}(w_{\mathcal{A}})$. As $w_{\mathcal{A}}$ is continuous, by the Heine-Cantor Theorem it is uniformly continuous on $\overline{B}(0; M + \delta_1) \cap C$. This means that:

$$\begin{aligned} \forall \epsilon > 0 \quad \exists \delta_2 > 0 \quad \text{st} \quad \forall y_1, y_2 \in \overline{B}(0; M + \delta_1) \cap C \quad \text{st} \\ |y_1 - y_2| < \delta_2, \quad |w_{\mathcal{A}}(y_1) - w_{\mathcal{A}}(y_2)| < \epsilon \end{aligned}$$

Given $\epsilon > 0$, take $\delta_3 < \min\{\delta_2, \delta_1\}$. As $w_{\tilde{\mathcal{A}}}$ is a set size function for $(C, \tilde{\mathcal{A}})$, we can apply Lemma 3.2.2 to get that

$$\text{Given } \delta_3 > 0 \quad \exists \delta > 0 \quad \text{st} \quad w_{\tilde{\mathcal{A}}}(x) < \delta \Rightarrow |x|_{\tilde{\mathcal{A}}} < \delta_3$$

We see that, for every $x \in C$ satisfying $|x|_{\tilde{\mathcal{A}}} < \delta_3$, we have that $|x|_{\tilde{\mathcal{A}}} < \delta_3 < \delta_1 \Rightarrow x \in \overline{B}(0; M + \delta_1) \cap C$. Also, taking $z^* \in \operatorname{argmin}_{z \in \tilde{\mathcal{A}}} |x - z|$ we get that $|x - z^*| < \delta_3 < \delta_2$. Therefore, by uniform continuity with $y_1 = x, y_2 = z^*$ we get that $|w_{\mathcal{A}}(x) - w_{\mathcal{A}}(z^*)| < \epsilon$.

As $z^* \in \tilde{\mathcal{A}}$, we have that $w_{\mathcal{A}}(z^*) \leq a$. Thus, we can conclude that $w_{\mathcal{A}}(x) < a + \epsilon \iff w_{\mathcal{A}}(x) - a < \epsilon$.

Summing up, we have seen that

$$\forall \epsilon > 0 \exists \delta > 0 \text{ st } \forall x \in C \text{ st } w_{\tilde{\mathcal{A}}}(x) < \delta, w_{\mathcal{A}}(x) - a < \epsilon$$

Therefore, when evaluating $\tilde{\alpha}(r)$ for $r < \delta$, we are taking the maximum of $w_{\mathcal{A}}(x)$ for $x \in \{x \in C \mid w_{\tilde{\mathcal{A}}}(x) \leq r < \delta\}$. Clearly, the maximum of $w_{\mathcal{A}}(x)$ over that set will be strictly smaller than $a + \epsilon$.

We conclude that

$$\forall \epsilon > 0 \exists \delta > 0 \text{ st } \forall r < \delta, \tilde{\alpha}(r) - a < \epsilon$$

We have therefore seen that $\exists \tilde{\alpha}$ non-decreasing with $\tilde{\alpha}(0) \geq 0$ such that $\forall x \in C, w_{\mathcal{A}} \leq \tilde{\alpha}(w_{\tilde{\mathcal{A}}}(x))$.

Following the same procedure as Proposition 2.6 in [?], we can majorize $\tilde{\alpha}$ to $\alpha \in \tilde{\mathcal{K}}_{\infty}$. $\square$

3.2.2 The SISS property

We can now move on to define an extended version of the Input to State Stability property tailored for systems evolving inside a closed set. We include in our definition the concept of practical stability introduced by Sontag and Wang in [9].

Definition 18 (Set Input to State Stability (SISS)). *The system 1.1 defined on a set $C \subseteq \mathbf{R}^n$ is small gain **Set Input to State practically Stable (SISpS)** if there exists $c > 0 \in \mathbf{R}$ st $\forall d \in \mathcal{M}_c$ the solutions of the system are defined $\forall t \geq 0$ and we have that:*

- The set C is forward invariant
- There exists a $\mathcal{KL}$ -function β and $\tilde{\mathcal{K}}$ -function γ such that:

$$w_{\mathcal{A}}(x(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|) \\ \text{for all } t \geq 0 \text{ and for all } \xi \in C$$

for some $w_{\mathcal{A}}$ a set size function for $(C, \mathcal{A})$.

Remark 3.2.6 (Connection between ISS and SISS). If $C = \mathcal{X} = \mathbf{R}^n$, then the concepts of ISS and SISS are equivalent. $\bullet$

Remark 3.2.7. If $\gamma \in \mathcal{K}$, we say the system is (small gain) **Set Input to State Stable (SISS)**

If $c = +\infty$, we say that the system is **Set Input to State (practically) Stable (SIS(p)S)**. $\bullet$

Definition 19 (Approximate SISS). *We say system 1.1 is approximately (small signal) SIS(p)S on $(C, \mathcal{A})$ if there exists $0 < \epsilon^* \leq \infty$ such that for all $\epsilon \leq \epsilon^*$, the following conditions on a noisy open set C^ϵ hold:*

- (i) *If $\epsilon_1 < \epsilon_2$ then $C^{\epsilon_1} \subseteq C^{\epsilon_2}$, $d(C, C^\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ and $\overline{C^{\epsilon^*}} \subset \mathcal{X}$*
- (ii) *There exists an adjustable parameter λ such that for all $0 < \epsilon < \epsilon^*$, there exists $\delta_\lambda > 0$ st $\forall \lambda < \delta_\lambda$ we have forward invariance of C^ϵ and the (small signal) SIS(p)S inequality holds for all trajectories starting in C^ϵ with a set size function for $(\overline{C^\epsilon}, \mathcal{A})$.*

Just as ISS-Lyapunov functions grant the fulfillment of the ISS property, we can define SISS-Lyapunov functions whose existence, together with the forward invariance under disturbances of the set, guarantees the SISS property holds.

Definition 20 (SISS-Lyapunov function). *A LL, non-pathological function $V : \mathbf{R}^n \rightarrow \mathbf{R}$ is called a weak generalized **SISS-Lyapunov function** on $(C, \mathcal{A})$ if:*

- (i) *$V(x)$ is a set size function for $(C, \mathcal{A})$*
- (ii) *$\exists \alpha_0, \gamma \in \mathcal{K}_\infty$ st $\max \tilde{L}_{f(x,d)} V(x) \leq -\alpha_0(w_{\tilde{\mathcal{A}}}(x)) + \gamma(\|d\|) \quad \forall x \in C, \forall t \geq 0$ and $\forall d \in \mathcal{M}_c$.*

For sufficiently small disturbances, the sub-level sets of a SISS-Lyapunov function are forward invariant. This is a crucial result for the characterization of the SISS property carried on in Section 3.3.

Lemma 3.2.8. *[Forward Invariance of the sub-level sets of a SISS-Lyapunov function] If V is a non-pathological generalized SISS-Lyapunov function on $(\tilde{S}_\nu^V, \mathcal{A})$ (where $\tilde{S}_\nu^V \subset \mathcal{X}$ is a (set of) path-connected components of its own subset S_ν^V , $\nu > 0$), then there exists $c_\nu > 0$ such that $\tilde{S}_\nu^V$ is forward invariant for all $d \in \mathcal{M}_{c_\nu}$.*

Proof. Use Corollary 3.2.4 to get $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ such that $\alpha_1(w_{\mathcal{A}}(x)) \leq V(x) \leq \alpha_2(w_{\mathcal{A}}(x)) \quad \forall x \in S_\nu^V$ and follow the steps in Proposition 3.1.1. $\square$

The following proposition establishes sufficient conditions for the fulfillment of the SISpS property, showcasing a direct link between the concept of weak (set) size functions and practical stability.

Proposition 3.2.9 (Sufficient conditions for SISS). *If $V : \mathbf{R}^n \rightarrow \mathbf{R}$ is a (c_1) small-signal weak generalized SISS-Lyapunov function on $(C, \mathcal{A})$ and C is forward invariant under all disturbances d st $d \in \mathcal{M}_{c_2}$ for a certain $c_2 > 0$, then the system is (c) small gain SISpS on $(C, \mathcal{A})$, where $c := \min\{c_1, c_2\}$*

Proof. As V is a (c_1) small-signal weak generalized SISS-Lyapunov function, we know that $\exists \alpha_0, \gamma \in \mathcal{K}_\infty$ st $\max \tilde{L}_{f(x,d)} V(x) \leq -\alpha_0(w_{\tilde{\mathcal{A}}}(x)) + \gamma(\|d\|) \quad \forall x \in C, \forall t \geq 0$ and $\forall d$ st $d \in \mathcal{M}_{c_1}$ (and thus also for all $d \in \mathcal{M}_c$)

$\alpha_0 \in \mathcal{K}_\infty$ implies that, given $r_2 \geq r_1 \geq 0$, we have $0 \geq -\alpha_0(r_1) \geq -\alpha_0(r_2)$.

Now take some $w_{\mathcal{A}}$ a set size function for $(C, \mathcal{A})$ and α_w as in Proposition 3.2.5. Then, $\forall x \in C$ we have that $w_{\mathcal{A}}(x) \leq \alpha_w(w_{\tilde{\mathcal{A}}}(x))$.

We cannot directly work with $\alpha_w^{-1}(s)$ cause $\alpha_w^{-1} : [\alpha_w(0), \infty) \rightarrow [0, \infty)$ but $w_{\mathcal{A}}(x)$ can take values smaller than $\alpha_w(0)$ (in fact, if $x \in \mathcal{A}$ then $w_{\mathcal{A}}(x) = 0 \leq \alpha_w(0)$). To work our way around this problem, we define $\overline{\alpha_w^{-1}} : [0, \infty) \rightarrow [0, \infty)$ in the following way:

$$\overline{\alpha_w^{-1}}(s) = \begin{cases} \alpha_w^{-1}(s) & \text{if } s \geq \alpha_w(0) \\ 0 & \text{if } 0 \leq s < \alpha_w(0) \end{cases}$$

Clearly, $\overline{\alpha_w^{-1}}(s)$ is non-decreasing: if we take $s_1 > s_2 \geq \alpha_w(0)$ then $\alpha_w^{-1}(s_1) > \alpha_w^{-1}(s_2)$ because the original $\alpha_w(r)$ is strictly increasing. If we take $s_1 \geq \alpha_w(0) > s_2$, then it is also clear that $\alpha_w^{-1}(s_1) \geq \alpha_w^{-1}(s_2) = 0$ and if $\alpha_w(0) > s_1 > s_2$ then simply $\alpha_w^{-1}(s_1) = \alpha_w^{-1}(s_2) = 0$. It is also continuous because $\alpha_w^{-1}(\alpha_w(0)) = 0$, and it goes to infinity as $s \rightarrow \infty$ because the original $\alpha_w(r)$ tends to infinity when r does too.

Now, we check that, for $x \in C$, if $w_{\mathcal{A}}(x) \geq \alpha_w(0)$, then $\alpha_w^{-1}(w_{\mathcal{A}}(x)) = \overline{\alpha_w^{-1}}(w_{\mathcal{A}}(x)) \leq \overline{\alpha_w^{-1}}(\alpha_w(w_{\mathcal{A}}(x))) = w_{\mathcal{A}}(x)$. If $w_{\mathcal{A}}(x) < \alpha_w(0)$, then $\alpha_w^{-1}(w_{\mathcal{A}}(x)) = 0 \leq \overline{\alpha_w^{-1}}(\alpha_w(w_{\mathcal{A}}(x))) = w_{\mathcal{A}}(x)$.

We conclude that, $\forall x \in C$, $\overline{\alpha_w^{-1}}(w_{\mathcal{A}}(x)) \leq w_{\mathcal{A}}(x) \Rightarrow -\alpha_0(w_{\mathcal{A}}(x)) \leq -\alpha_0(\overline{\alpha_w^{-1}}(w_{\mathcal{A}}(x)))$.

Now, let $\alpha_{00} := \alpha_0 \circ \overline{\alpha_w^{-1}}$. Clearly, $\alpha_{00} \in \mathcal{K}_{\infty}^{\alpha_w(0)}$. We have that:

$$\max \tilde{L}_{f(x,d)} V(x) \leq -\alpha_{00}(w_{\mathcal{A}}(x)) + \gamma(|d|) \quad \forall x \in C, \quad \forall t \geq 0 \text{ and } \forall d \text{ st } d \in \mathcal{M}_c$$

We proceed as in Theorem 1 in [5], picking an $x_0 \in C$ and defining $v(t) := V(\varphi(t; \xi))$. As V is weak, we know that $\dot{V}(\varphi(t; \xi)) = \frac{d}{dt} V(\varphi(t; \xi)) = \dot{v}(t) \in \tilde{L}_{f(x(t), d(t))} V(\varphi(t; \xi))$ and we can rest assured that the solutions remain inside the set thanks to the required forward invariance under disturbances with infinity norm smaller than c_2 (therefore, this holds for disturbances with infinity norm smaller than c).

As V and $w_{\mathcal{A}}$ are both set size functions, we can apply Proposition 3.2.4 to see that $\exists \underline{\alpha}, \bar{\alpha} \in \mathcal{K}_{\infty}$ such that:

$$\underline{\alpha}(V(x)) \leq w_{\mathcal{A}}(x) \leq \bar{\alpha}(V(x)) \quad \forall x \in C$$

Notice that $\underline{\alpha}(V(x)) \leq w_{\mathcal{A}}(x) \Rightarrow -\alpha_{00}(w_{\mathcal{A}}(x)) \leq -\alpha_{00}(\underline{\alpha}(V(x))) \quad \forall x \in C$.

Let $b = \underline{\alpha}^{-1}(\alpha_w(0))$ and let $\tilde{\alpha} := \alpha_{00} \circ \underline{\alpha} \in \mathcal{K}_{\infty}^b$. As $v(t)$ stays in C , we have that:

$$\dot{v}(t) \leq -\tilde{\alpha}(w_{\mathcal{A}}(\varphi(t; x_0))) + \gamma(\|d(t)\|) \quad \forall t \in [0, t_{\max})$$

Let us define

$$\tilde{\alpha}^{-1}(s) = \begin{cases} b & \text{if } s = 0 \\ \tilde{\alpha}^{-1}(s) & \text{if } s > 0 \end{cases}$$

If $s > 0$, then we have that $\exists! y > b$ such that $\tilde{\alpha}(y) = s$ and given $s_1 > s_2 > 0$, we have that the corresponding y_1, y_2 can be written as $y_1 = b + z_1, y_2 = b + z_2$. As $\tilde{\alpha}(b + z)$ is a class $\mathcal{K}_{\infty}$ function (by definition of $\mathcal{K}_{\infty}^b$ functions), this means that $s_1 > s_2 > 0 \Rightarrow z_1 > z_2 > 0 \Rightarrow y_1 > y_2 > b$ and therefore $\tilde{\alpha}^{-1}(s)$ is strictly increasing.

Also, $\tilde{\alpha}(b + z) \rightarrow 0^+$ when $z \rightarrow 0^+$, which means $\tilde{\alpha}^{-1}(s) \rightarrow b^+$ when $s \rightarrow 0^+$. This ensures continuity of $\tilde{\alpha}^{-1}(s)$.

We can therefore conclude that $\tilde{\alpha}^{-1}(s) \in \tilde{\mathcal{K}}_{\infty}$

What is more, if $\tilde{\alpha}(v(t)) \leq a$, we have that:

- (i) if $v(t) \leq b$, then $\bar{\alpha}^{-1}(a) \geq b \geq v(t)$.
- (ii) if $v(t) > b$, then $\tilde{\alpha}(v(t)) > 0$ and therefore $\bar{\alpha}^{-1}(\tilde{\alpha}(v(t))) = \tilde{\alpha}^{-1}(\tilde{\alpha}(v(t))) = v(t)$. As $\bar{\alpha}^{-1}(s)$ is strictly increasing, we have that $\bar{\alpha}^{-1}(\tilde{\alpha}(v(t))) \leq \bar{\alpha}^{-1}(a) \Rightarrow v(t) \leq \bar{\alpha}^{-1}(a)$.

We can therefore conclude that $\tilde{\alpha}(v(t)) \leq a \Rightarrow v(t) \leq \bar{\alpha}^{-1}(a)$

Now, take an arbitrarily small $\epsilon > 0$. For any $t \in [0, t_{\max})$, either:

- (i) $\tilde{\alpha}(v(t)) \leq 2\gamma(\|d(t)\|) + \epsilon \Rightarrow v(t) \leq \bar{\alpha}^{-1}(2\gamma(\|d\|_{\infty}) + \epsilon)$.

As $\gamma \in \mathcal{K}_{\infty}$, $2\gamma(\cdot) + \epsilon \in \tilde{\mathcal{K}}_{\infty}$. Take $\tilde{\gamma} := \bar{\alpha}^{-1} \circ (2\gamma(\cdot) + \epsilon)$. Then,

$$v(t) \leq \tilde{\gamma}(\|d\|_{\infty})$$

- (ii) $\tilde{\alpha}(v(t)) > 2\gamma(\|d(t)\|) + \epsilon \Rightarrow \gamma(\|d(t)\|) < \frac{\tilde{\alpha}(v(t))}{2} - \frac{\epsilon}{2} \Rightarrow \dot{v}(t) < -\frac{\tilde{\alpha}(v(t))}{2} - \frac{\epsilon}{2}$.

Take

$$\alpha(r) = \begin{cases} \frac{r}{b} \cdot \frac{\epsilon}{2} & \text{if } r \leq b \\ \frac{\tilde{\alpha}(r)}{2} + \frac{\epsilon}{2} & \text{if } r > b \end{cases}$$

It is clear that α is continuous, strictly increasing (it is the continuous concatenation of two strictly increasing functions, as $\tilde{\alpha}(b+z) \in \mathcal{K}_{\infty}$) and tends to infinity when r does (because $\tilde{\alpha}$ does too). This is, $\alpha(r) \in \mathcal{K}_{\infty}$.

We can also see that $\alpha(r) \leq \frac{\tilde{\alpha}(r)}{2} + \frac{\epsilon}{2}$: equality holds when $r > b$ by definition, while clearly $\frac{r}{b} \cdot \frac{\epsilon}{2} \leq \frac{\epsilon}{2} = \frac{\tilde{\alpha}(r)}{2} + \frac{\epsilon}{2}$ for $r \leq b$. Reversing the inequality, we get $-\frac{\tilde{\alpha}(r)}{2} - \frac{\epsilon}{2} \leq -\alpha(r)$. Therefore, we have:

$$\dot{v}(t) < -\alpha(v(t))$$

As $\alpha \in \mathcal{K}_{\infty}$, we can proceed as in the proof of Theorem 1 in [5]: we know the solution to the IVP

$$\begin{cases} \dot{y} = -\alpha(y) \\ y(0) = s \end{cases}$$

is a class $\mathcal{KL}$ function $\beta(s, t)$. In this case, we fix $s = v(0) = x_0$ and, by the comparison theorem, we get

$$v(t) < \beta(v(0), t)$$

Observe that $v(0) = V(x_0) \leq \underline{\alpha}^{-1}(w_{\mathcal{A}}(x_0))$, so as β is a class $\mathcal{K}_{\infty}$ function in the first argument and $\underline{\alpha} \in \mathcal{K}_{\infty}$ too, we are left with

$$v(t) < \beta(\underline{\alpha}^{-1}(w_{\mathcal{A}}(x_0)), t)$$

Putting cases 1 and 2 together, we get that

$$v(t) \leq \max\{\beta(\underline{\alpha}^{-1}(w_{\mathcal{A}}(x_0)), t), \tilde{\gamma}(\|d\|_{\infty})\} \quad \forall t \in [0, t_{\max})$$

Now, recall that $w_{\mathcal{A}}(\varphi(t; x_0)) \leq \bar{\alpha}(V(\varphi(t; x_0))) = \bar{\alpha}(v(t))$. Now, we define $\tilde{\beta} := \bar{\alpha} \circ \beta(\underline{\alpha}^{-1}(\cdot, t)) \in \mathcal{KL}$ and $\tilde{\gamma} := \bar{\alpha} \circ \tilde{\gamma} \in \tilde{\mathcal{K}}_{\infty}$ and we see that:

$$w_{\mathcal{A}}(\varphi(t; \xi)) \leq \max\{\tilde{\beta}(w_{\mathcal{A}}(x_0), t), \tilde{\gamma}(\|d\|_{\infty})\}$$

$$\leq \tilde{\beta}(w_{\mathcal{A}}(x_0), t) + \tilde{\gamma}(\|d\|_{\infty}) \quad \forall t \in [0, t_{\max})$$

Take $\max\{\tilde{\beta}(w_{\mathcal{A}}(x_0), t), \tilde{\gamma}(\|d\|_{\infty})\} =: r$ for any $\xi \in C$ and any $t \in [0, t_{\max})$. The sub-level sets of $w_{\mathcal{A}}$ are compact in $\mathbf{R}^n$ because it is a set size function. Therefore, the solutions stay inside $C \cap \{x \mid w_{\mathcal{A}}(x) \leq r\}$ which is still a compact (intersection of closed and compact set). This ensures that the solutions are defined for all times $t \geq 0$ (this is, $tw_+(x, d) = +\infty \quad \forall (x, d) \in C \times \mathcal{M}_c$) by Proposition C.3.6 in [37]. We can therefore conclude that the system is small gain weak SISS. $\square$

Corollary 3.2.10 (ISSf and SISS-Lyapunov imply SISS). *If the set C is ISSf for $d \in \mathcal{M}_c$ (where $c > 0$) and there exists a weak generalized SISS-Lyapunov function $V : \mathbf{R}^n \rightarrow \mathbf{R}$ for $(C_c, \mathcal{A}) \Rightarrow$ the system is small gain SISS on the set C_c .*

Proof. If C is ISSf with constant c , this means $C_c \subset X$ (so the system stays well-defined) and C_c is forward invariant (Recall that, by construction, $C_d^* \subseteq C_c \quad \forall d$ st $\|d\|_{\infty} \leq c$ with equality if $\|d\|_{\infty} = c$, and all those C_d^* are forward invariant). Therefore, Proposition 3.2.9 ensures that C_c is (c) small gain SISpS. $\square$

What is more, using Lemma 3.2.8, we can see that generalized SISS-Lyapunov functions are SISS on their sub-level sets, so we do not need to rely on "inflated" ISSf sets when studying the stability of SISS-Lyapunov sub-level sets.

Corollary 3.2.11 (SISS-Lyapunov function is SISS in its own subsets). *If there exists $r > 0$ such that $V : \mathbf{R}^n \rightarrow \mathbf{R}$ is a (c) small-signal generalized SISS-Lyapunov function for $(\tilde{S}_r^V, \mathcal{A})$ (where $\tilde{S}_r^V$ is a (set of) path-connected components of its own subset S_r^V), then there exists $c_r > 0$ such that the system is (c_r) small gain SISS on $(S_r^V, \mathcal{A})$.*

Proof. Apply Lemma 3.2.8 to get $c_r \leq c$ such that S_r^V is forward invariant. Now, simply apply Proposition 3.2.9. $\square$

Remark 3.2.12. Under the hypotheses of Proposition 3.2.9, if additionally

- (i) $\mathcal{A} = \tilde{\mathcal{A}}$, then $\alpha_w \in \mathcal{K}_{\infty}$ and therefore $\tilde{\gamma} \in \mathcal{K}_{\infty}$, so **the system is (small gain) SISS**.
- (ii) $c = \infty$, then **the system is (weak) SISS**

•

Finally, note that a controller u that is both an ISSf controller for the set C and an input to state stabilizing controller for the extended set C_c would render the system SISS on $(C_c, \mathcal{A})$, but the study of the feasibility of such a controller is not trivial.

3.3 Characterization of SISS and its relationship with ISS

We have presented the SISS property. However, it has happened before that some of the proposed "extensions" to the ISS property in the literature have in fact been proven to be equivalent to it ([9]).

We prove here that the SISS property on an arbitrary closed set is in fact not equivalent to neither Local ISS (LISS) nor ISS. We also showcase the different insights given by the SISS property on some classes of systems where the ISS property might not be applicable. Please notice this does not in any way diminish the interest of the ISS property on open sets, but rather identifies situations in which it might be useful to use the SISS property instead.

3.3.1 SISS property characterization

In [9], the relationship between Asymptotically Stable Equilibria and ISS is established (ISS implies Global Asymptotic Stability and LISS implies Local Asymptotic Stability. Further, Global Asymptotic Stability with an asymptotic gain property implies ISS). We begin here by showing an analogous relationship between the SISS property and the stability of the unperturbed system's equilibria.

First, it is easy to see that SISS implies Local Asymptotic Stability:

Proposition 3.3.1 (SISS implies Local Asymptotic Stability). *If the system 1.1 is SISS on $(C, \mathcal{A})$ where C is a closed set and $\mathcal{A}$ is a compact set of equilibria of the unperturbed system, then the set of equilibria $\mathcal{A} \cap \overset{\circ}{C}$ is locally asymptotically stable*

Proof. The proof is straightforward from the SISS definition. We need to intersect $\mathcal{A}$ with $\overset{\circ}{C}$ because for the equilibria $\mathcal{A} \cap \partial C$ there might not exist an open set containing them such that every trajectory of the unperturbed system starting there converges to them (because there does not exist such set contained in C and we only have SISS conditions over C). $\square$

Under stronger conditions, we can prove a reciprocal implication:

Proposition 3.3.2 (Local Lyapunov function implies SISS under additive disturbance). *Consider a size function $w_{\mathcal{A}}$ for $(\mathcal{X}, \mathcal{A})$ and let $V : \mathcal{X} \rightarrow \mathbb{R}$ be a continuously differentiable (r)Local Lyapunov function for the system $\dot{x} = f(x) + d(x, t)$ defined in $\mathcal{X}$ (V need only be differentiable in a neighborhood of S_r). If there exist $\alpha \in \mathcal{K}_{\infty}$ and W a positive definite function with respect to $\mathcal{A}$ in S_r such that:*

$$\mathcal{L}_f V(x) \leq -\alpha(W(x)) \quad \forall x \in S_r \quad (3.10)$$

Then, taking $\eta := \min_{x \in S_r} V(x)$ there exists $c_{\eta} > 0$ st for all $d \in \mathcal{M}_{c_{\eta}}$ the system is (c_{η}) small-gain SISS on $(S_{\eta}^V \cap S_r, \mathcal{A})$

Proof. As $S_{\eta}^V \cap S_r$ is compact and both V and W are greater or equal than zero on it (with equality only on $\mathcal{A}$), we can conclude both V and W are set size functions for $(S_{\eta}^V \cap S_r)$.

We can see that

$$\|\nabla V(x)\| \leq B_{V,r} \in \mathbb{R}_{\geq 0} \quad \forall x \in S_r$$

because S_r is compact.

Notice that:

$$\begin{aligned}\mathcal{L}_{f(x)+d(x,t)}V(x) &= \mathcal{L}_{f(x)}V(x) + d(x,t)^t \cdot \nabla V(x) \\ &\leq -\alpha(W(x)) + B_{V,r} \|d\|_\infty \quad \forall x \in S_r\end{aligned}$$

We conclude V is a SISS-Lyapunov function on $(S_\eta^V \cap S_r, \mathcal{A})$. Note that $S_\eta^V \cap S_r$ is a (collection of) maximal path-connected components of S_η^V by Lemma 3.1.3. Then, Lemma 3.2.8 states that there exists $c_\eta > 0$ such that $S_\eta^V \cap S_r$ is forward invariant under any disturbance $d \in \mathcal{M}_{c_\eta}$, and therefore by Proposition 3.2.9 the perturbed system is (c_η) small-gain SISS on $(S_\eta^V \cap S_r, \mathcal{A})$. $\square$

Corollary 3.3.3 (Local Asymptotic Stability implies SISS under additive disturbance). *If $\mathcal{A}$ is a locally stable set of equilibria for the system $\dot{x} = f(x) + d(x, t)$ with a region of attraction R_A , then there exists a continuously differentiable function $V : R_A \rightarrow \mathbb{R}$ such that for all $r > 0$ exists $c > 0$ such that the pair $(S_r, \mathcal{A})$ is (c) small-gain SISS.*

Proof. Theorem 4.17 in [3] grants the existence continuously differentiable function Lyapunov function V for the whole region of attraction R_A st the inequality (3.10) in Proposition ?? is fulfilled $\forall x \in R_A$ and $V(x) \rightarrow \infty$ as $x \rightarrow \partial R_A$. V is then a set size function for every closed subset of R_A and its sub-level sets are compact. Thus, both V and W are set size functions on every sub-level set of V .

In fact, V is a SISS-Lyapunov function on its subsets, as for every $r > 0$ we once again can see that there exists $B_{V,r} > 0$ st $\|\nabla V(x)\| \leq B_{V,r} \quad \forall x \in S_r^V$ and therefore $\mathcal{L}_{f(x)+d(x,t)}V(x) \leq -\alpha(W(x)) + B_{V,r} \|d\|_\infty$. Lemma 3.2.8 states that for every r there exists c_r such that the system is (c_r) small-signal on $(S_r^V, \mathcal{A})$. $\square$

3.3.2 SISS in Gradient Systems

We will further attempt to justify the use of the SISS notion looking at its guarantees in the problem that motivated the introduction of ISS in open sets.

Proposition 3.3.4 (Theorem 2 in Sontag's Local ISS). *Consider the system $\dot{x} = -\beta \cdot \nabla V(x) + d(x, t)$ where $\beta > 0 \in \mathbb{R}$, $V : \mathcal{X} \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable, $\nabla V(x)$ is LL and $\|d\|_\infty < \infty$. Consider the set of equilibria of the system $\mathcal{A} \subset \mathcal{X}$, and assume $V(x) = V^* \quad \forall x \in \mathcal{A}$.*

Then, if both $V - V^$ and ∇V are size functions in $(\mathcal{X}, \mathcal{A})$, the system is ISS on $(\mathcal{X}, \mathcal{A})$.*

Notice that, for the conditions of the Theorem to be fulfilled, we need V and ∇V to tend towards infinity as they approach the boundary of $\mathcal{X}$, which greatly restricts the class of perturbed gradient flow problems we can study with the ISS notion.

We will see how the SISS notion is able to give a different insight on the stability of a broader class of perturbed gradient flows.

Proposition 3.3.5 (Gradient Flow is SISS). *Let w_A be a size function for $(\mathcal{X}, \mathcal{A})$. Consider the system $\dot{x} = -\beta \cdot \nabla V(x) + d(x, t)$ defined on $\mathcal{X}$ where $\beta > 0 \in \mathbb{R}$, $V : \mathcal{X} \rightarrow \mathbb{R}$ is continuously differentiable, $\nabla V(x)$ is LL and:*

(i) $V(x) = V^* \forall x \in \mathcal{A}$ and $V(x) > V^* \forall x \in \mathcal{X} \setminus \mathcal{A}$

(ii) $\|\nabla V(x)\| = 0 \iff x \in \mathcal{A}$

Then, for every $r > 0$, taking $\eta_r := \min_{x \in \partial S_r} V(x) - V^* > 0$ there exists $c_{\eta_r} > 0$ st for all $d \in \mathcal{M}_{c_{\eta_r}}$ the system is (c_{η_r}) small gain SISS in $(S_{\eta_r}^{V-V^*} \cap S_r, \mathcal{A})$.

Proof. Under the conditions in the Proposition both $V - V^*$ and $\|\nabla V\|$ are set size functions on $S_{\eta}^{V-V^*} \cap S_r$, which is also compact.

Now, we can operate with the Lie derivative to see that $\mathcal{L}_{-\beta \cdot \nabla V(x) + d(x,t)}(V(x) - V^*) = -\beta \cdot \|\nabla V(x)\|^2 + d(x,t) \cdot \nabla V(x) \leq -\alpha(\|\nabla V(x)\|) + \gamma(\|d(x,t)\|) \forall x \in \mathcal{X}$, where $\alpha(r) = \frac{\beta}{2} \cdot r^2 \in \mathcal{K}_{\infty}$ and $\gamma(r) = \frac{1}{2\beta} \cdot r \in \mathcal{K}_{\infty}$.

We conclude that $V(x) - V^*$ is a SISS-Lyapunov function for $(S_{\eta}^{V-V^*} \cap S_r, \mathcal{A})$. By Lemma 3.1.3 $S_{\eta}^{V-V^*} \cap S_r$ is a (collection of) connected components of $S_{\eta}^{(V-V^*)^{ext}}$ and thus Lemma 3.2.8 ensures that there $\exists c_{\eta_r} > 0$ such that $S_{\eta}^{V-V^*}$ is forward invariant $\forall d \in \mathcal{M}_{c_{\eta_r}}$. Therefore, by Proposition 3.1.1 the system is (c_{η_r}) small-gain SISS on $(S_{\eta_r}^{V-V^*}, \mathcal{A})$. $\square$

Corollary 3.3.6 (Stronger results on Gradient Flow SISS properties). *If the conditions in Proposition 3.3.5 hold and additionally:*

(iii) *for all $(x_k)_{k \in \mathbb{N} \cup \{0\}} \subset \mathcal{X}$ such that $\|x_k\| \rightarrow \infty$ when $k \rightarrow \infty$, $V(x_k) - V^* \rightarrow \infty$ as $k \rightarrow \infty$*

Then, for every $r > 0$ st $\partial S_r^{V-V^*} \cap \partial \mathcal{X} = \emptyset$ exists $c_r > 0$ such that the system is (c_r) small-gain on $(S_r^{V-V^*}, \mathcal{A})$.

Proof. If $\partial S_r^{V-V^*} \cap \partial \mathcal{X} = \emptyset$, then $S_r^{V-V^*}$ is a closed subset of $\mathcal{X}$. It is clear that $V - V^*$ is a set size function for all closed subsets of $\mathcal{X}$. What is more, recall that $\mathcal{L}_{-\beta \cdot \nabla V(x) + d(x,t)}(V(x) - V^*) \leq -\alpha(\|\nabla V(x)\|) + \gamma(\|d(x,t)\|) \forall x \in \mathcal{X}$ and thus $V - V^*$ is a SISS-Lyapunov function for $(S_r^{V-V^*}, \mathcal{A})$. Then, Lemma 3.2.8 ensures that there $\exists c_{\eta_r} > 0$ such that $S_r^{V-V^*}$ is forward invariant $\forall d \in \mathcal{M}_{c_{\eta_r}}$, so the system is (c_r) small-signal SISS on $(S_r^{V-V^*}, \mathcal{A})$. $\square$

Example 3.3.7. (*SISS does not imply ISS*) Consider the system

$$\dot{x} = x^3 - x + d(x, t)$$

where $d: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is LL.

The unperturbed system is a gradient descent with $V = -\frac{x^4}{4} + \frac{x^2}{2}$:

$$\dot{x} = -\nabla\left(-\frac{x^4}{4} + \frac{x^2}{2}\right)$$

For this system, the origin is an local asymptotically stable point with basin of attraction $(-1, 1)$. This will be our set $\mathcal{X}$. We have a single equilibrium of the system in $\mathcal{X}$: $\mathcal{A} = \{0\}$.

Let us check that the conditions in Corollary 3.3.6 are fulfilled. The function V is defined and continuously differentiable in all $\mathbb{R}$. Further:

(i) $V(0) = 0$, $V(x) = -\frac{x^4}{4} + \frac{x^2}{2} > 0 \forall x \in (-1, 1) \setminus \{0\}$

(ii) Given $x \in (-1, 1)$, $\nabla V(x) = -x^3 + x = 0 \iff x = 0$

(iii) $(-1, 1)$ is bounded so no sequence can tend to infinity

What is more, $\forall x \in (-1, 1)$, $S_{V(x)} = \{y \in \mathbb{R} \mid |-\frac{y^4}{4} + \frac{y^2}{2}| \leq -\frac{x^4}{4} + \frac{x^2}{2}\}$ does not intersect with $\partial\mathcal{X} = \{-1, 1\}$ (this is easy to check: $V(1) = V(-1) = \frac{1}{4} > V(x) \forall x \in (-1, 1)$).

Therefore, by Corollary 3.3.6, given x_0 in $(-1, 1)$, the system is small gain SISS in $(S_{V(x_0)}, \{0\})$.

However, $|-\frac{x^4}{4} + \frac{x^2}{2}| \rightarrow \infty$ only when $|x| \rightarrow \infty$, so we can only study local ISS with V either in $\mathbb{R}$ or in open sets unbounded in $-\infty$ and $+\infty$. However, the system clearly is not ISS in those sets:

Consider $d(x, t) = \epsilon > 0$, so $\|d\|_\infty = \epsilon$ can be chosen to be arbitrarily small. Take $x_0 \geq 1$. Clearly, $\dot{x}_0 = x_0^3 - x_0 + \epsilon > 0$ and the trajectory will tend to infinity. Therefore, the system is in fact not even small signal ISS in those sets.

In fact, the system is not small-signal ISS on $(-1, 1)$ either. Consider the negation of the small-signal ISS property: $\forall c > 0, \exists d$ st $d \in \mathcal{M}_c$ and the system is not ISS under the disturbance d on $((-1, 1), \{0\})$.

Given $c < 1$, as $\lim_{x \rightarrow 1} x^3 - x = 0$ and $x^3 - x$ is a negative function in the interval $(0, 1)$, we know that $\forall \epsilon > 0 \exists \delta > 0$ st $\forall 1 - \delta < y \leq 1, -\epsilon < y^3 - y \leq 0$. Take $\epsilon^* < c$ and its corresponding δ^* . Take $d(x, t) = \epsilon$, so that $\|d\|_\infty = \epsilon < c$. Then, $\forall x_0 > 1 - \delta^*, -\epsilon < x_0^3 - x_0 \rightarrow x_0^3 - x_0 + \epsilon > 0$. This means the set $(1 - \delta^*, \infty)$ is forward invariant and its trajectories tend to $+\infty$.

If $c \geq 1$ simply take $\tilde{c} < 1$ and repeat the previous argument, obtaining d st $\|d\|_\infty < \tilde{c} < c$ that makes the trajectories starting from certain points within $\mathcal{X}$ to escape to infinity.

In conclusion, $\forall c > 0 \exists d$ st $d \in \mathcal{M}_c$ and the ISS-Lyapunov inequality cannot be fulfilled for some initial conditions in $\mathcal{X}$ (which explode to infinity under the disturbance d). Therefore, the system is not small-gain ISS on $((-1, 1), \{0\})$ even though it is small-gain SISS on all the ISSf-extended sub-level sets of V in $\mathcal{X}$. •

The results obtained in this section show that the SISS property can give us insight into the stability of some systems for which the ISS is not be able to do so. However, are there systems in which the ISS property is fulfilled but the SISS property is not? We can intuitively state that the fulfillment of the ISS property implies the LISS property also holds. Thus, our question boils down to the following one: what is the relationship between SISS and LISS when we work with the sub-level set of a SISS/LISS-Lyapunov function?. In the following subsection, we draw the connections between the SISS, LISS and ISS properties.

3.3.3 ISS-LISS-SISS implication map

To be able to draw the map that relates the SISS property to the ISS property on open sets and understand the role played by LISS on open sets in this landscapes, it is first necessary to establish a connection between size functions and set size functions.

Lemma 3.3.8 (Size Functions are Set Size functions). *If $w_{\mathcal{A}} : \mathbb{R}^n \rightarrow \mathbb{R}$ is a size function on $(\mathcal{X}, \mathcal{A})$, then it is a set size function on $(C, \mathcal{A})$ where C is any closed subset of $\mathcal{X}$*

It is clear that the Lemma holds. We can see that set size functions can also be extended to size functions:

Lemma 3.3.9 (Set Size functions can be extended to Size Functions). *Given an open set $\mathcal{X}$, a closed set $C \subset \mathcal{X}$, a compact set of equilibria $\mathcal{A} \subseteq C$ and set size function for $(C, \mathcal{A})$, we can continuously extend $w_{\mathcal{A}}$ so that $w_{\mathcal{A}}^{ext}$ is a size function for $(\mathcal{X}, \mathcal{A})$.*

Proof. Take $w_{\mathcal{A}}|_C$ and apply Tietze's extension to the whole space, $\mathbb{R}^n$. Add the term $\frac{d(x, C)}{d(x, \partial \mathcal{X})}$, which is continuous $\forall x \in \mathbb{R}^n \setminus \partial \mathcal{X}$, and consider the resulting function restricted to the set $\mathcal{X}$, which we will call $w_{\mathcal{A}}^{ext} : \mathcal{X} \rightarrow \mathbb{R}$.

Clearly $w_{\mathcal{A}}^{ext}$ is continuous in its domain. We know Tietze's extension preserves the original infimum of $w_{\mathcal{A}}|_C$, which is zero, therefore $w_{\mathcal{A}}^{ext} > 0 \forall x \in \mathcal{X} \setminus \mathcal{A}$ because if $x \in \mathcal{X} \setminus C$ then $\frac{d(x, C)}{d(x, \partial \mathcal{X})} > 0$ and if $x \in C$ then $w_{\mathcal{A}}^{ext}(x) = w_{\mathcal{A}}(x)$. Also, $w_{\mathcal{A}}^{ext}(x) = 0 \iff x \in \mathcal{A}$.

Finally, observe that $w_{\mathcal{A}}(x) \rightarrow \infty$ when $x \rightarrow \partial \mathcal{X}$, so condition *iii*) in ?? is also fulfilled. $\square$

Proposition 3.3.10 (ISS implies SISS). *Consider a system $\dot{x} = f(x, d)$, where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is LL and $d : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^m$ is LL. Consider an open set $\mathcal{X}$, and a compact set of equilibria of the unperturbed system, $\mathcal{A}$. If the system is (small-signal) ISS on $\mathcal{X}$, then there exists $V : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $r > 0$, $\exists c_r > 0$ such that the system is (c_r) small-signal SISS on $(S_r^V, \mathcal{A})$ where V is an ISS-Lyapunov function.*

The reciprocal is false (see previous example).

Proof. If the system is small signal ISS with constant c^1 on $\mathcal{X}$, then by Theorem ?? there exists a continuously differentiable ISS-Lyapunov V for the system on $(\mathcal{X}, \mathcal{A})$. Note that V is a size function in $(\mathcal{X}, \mathcal{A})$ and this implies that it is a set size function in every closed subset of $\mathcal{X}$. We also know that $\exists \alpha, \gamma \in \mathcal{K}_{\infty}$ st:

$$\mathcal{L}_{f(x, d)} V(x) \leq -\alpha(w_{\mathcal{A}}(x)) + \gamma(\|d\|_{\infty}) \quad \forall x \in \mathcal{X}, \quad \forall d \in \mathcal{M}_{c^1}$$

where $w_{\mathcal{A}}$ is a size function in $(\mathcal{X}, \mathcal{A})$ and therefore also a set size function on every closed subset of $\mathcal{X}$. We conclude V is a SISS-Lyapunov function on every closed subset of $\mathcal{X}$.

Thus, Lemma 3.2.8 tells us that for every $r \geq 0$ $\exists c_r \leq c_1$ such that the set S_r^V is forward invariant for any disturbance $d \in \mathcal{M}_{c_r}$. Then, system is (c_r) small signal SISS on $(S_{c_r}^V, \mathcal{A})$ by Proposition 3.2.9. $\square$

There is one possible link left to explore: might the SISS property be equivalent to the LISS on open sets property?

It is clear that a sub-level set of a size function which is SISS will also be LISS on open sets. In fact, approximate SISS is enough to guarantee LISS in the sub-level sets of size functions:

Proposition 3.3.11 (Approximate SISS implies LISS). *Fix If the system 1.1 is approximately SISS on $(C, \mathcal{A})$ over the upper bound of the disturbances infinity norms, where $\mathcal{A} \subset C \subset \mathcal{X}$, then there exists $c > 0$ such that the system is (c) small signal (r) LISS (wrt to a fixed size function $w_{\mathcal{A}}$ for $(\mathcal{X}, \mathcal{A})$) for all r st $S_r \subseteq C$.*

Proof. Take any $\epsilon > 0$ st $C^\epsilon \subset \mathcal{X}$. The definition of approximate SISS tells us that there $\exists \delta_\epsilon > 0$ st $\forall d \in \mathcal{M}_\delta$, every perturbed trajectory starting in C remains within C^ϵ and for any set size function w for $(C^\epsilon, \mathcal{A})$ there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}_\infty$ st $w(x(t, \xi, d)) \leq \beta(w(\xi), t) + \gamma(\|d\|_\infty) \forall t \geq 0, \forall \xi \in C$ and $\forall d \in \mathcal{M}_{\delta_\epsilon}$.

Take any $\epsilon > 0$ such that $C^\epsilon \subset \mathcal{X}$. Take $c := \delta_\epsilon$. Note that, by Lemma 3.3.8, $w_\mathcal{A}$ is a set size function for every closed subset of $\mathcal{X}$, so there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}_\infty$ such that the previous inequality holds with $w = w_\mathcal{A}$.

Therefore, given r st $S_r \subset C$, for any $\xi \in S_r$ and any $d \in \mathcal{M}_c$ the trajectories of the system stay inside C^ϵ and:

$$w_\mathcal{A}(x(t, \xi, d)) \leq \beta(w_\mathcal{A}(\xi), t) + \gamma(\|d\|_\infty) \quad \forall t \geq 0$$

We conclude the system is (c)small gain (r)LISS. $\square$

Lemma 3.3.12 (V-LISS is also V-SISS). *If $V : S_r \rightarrow \mathbb{R}^n$ is a (c)small-signal generalized LISS-Lyapunov function for S_r (for a certain $r > 0$, $c > 0$ and a size function $w_\mathcal{A}$ on $(\mathcal{X}, \mathcal{A})$), then it is also a (c)small-signal generalized SISS-Lyapunov function for $(S_{r,V}, \mathcal{A})$.*

Proof. Note that $w_\mathcal{A}$ is also a set size function on $(S_r, \mathcal{A})$ by Lemma 3.3.8. By the definition of generalized LISS-Lyapunov functions, there exist $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ st $\alpha_1(w_\mathcal{A}(x)) \leq V(x) \leq \alpha_2(w_\mathcal{A}(x)) \forall x \in S_r$.

This implies that $V(x) = 0 \iff x \in S_{r,V} \cap \mathcal{A} = \mathcal{A}$ and also that its sub-level sets are compact: they must be closed because V is continuous and they must be bounded because they must be included in the sub-level sets of $\alpha_1(w_\mathcal{A}(x))$ (given $x \in S_{r,V}$, $V(x) \leq s$, then $\alpha_1(w_\mathcal{A}(x)) \leq s$) which are compact because $w_\mathcal{A}$ is a size function and α_1 is a class $\mathcal{K}_\infty$ function.

It is clear that the ISS-Lyapunov inequality holds. This completes the proof. $\square$

Proposition 3.3.13 (LISS implies SISS). *If the system 1.1 is (c)small-signal (r)LISS (for a certain $r > 0$, $c > 0$ and a size function $w_\mathcal{A}$ on $(\mathcal{X}, \mathcal{A})$), then there exists $V : S_r \Rightarrow \mathbb{R}_{\geq 0}$, $\eta := \min_{x \in \partial S_r} V(x)$ and $c_\eta > 0$ such that the system is (c_η) small-signal SISS on $(S_\eta^V, \mathcal{A})$.*

Proof. If the system is (c)small-signal (r)LISS, Proposition 3.1.4 tells us there exists a continuously differentiable function $V : S_r \rightarrow \mathbb{R}$ and a constant $c_V > 0$ such that:

- $\alpha_1^V(x) \leq V(x) \leq \alpha_2^V \quad \forall x \in S_r$
- $\mathcal{L}_{f(x,d)} V(x) \leq \frac{\alpha_V(w(x))}{2} + \gamma_V(\|d(x, t)\|)$ for all $x \in S_r$, all $t \geq 0$ and all $d \in \mathcal{M}_{c_V}$

What is more, the set $S_\eta^V \cap S_r$ is forward invariant for disturbances $d \in \mathcal{M}_{c_\eta}$ where $c_\eta \leq c_V$.

Then, V is a (c_V) small-signal SISS-Lyapunov function for $(S_\eta^V \cap S_r, \mathcal{A})$. If we require $d \in \mathcal{M}_{c_\eta}$, then $S_\eta^V \cap S_r$ is forward invariant and Proposition 3.2.9 guarantees the system is (c_η) small-signal SISS on $(S_r, \mathcal{A})$. $\square$

Corollary 3.3.14 (SISS in sub-level set implies existence of V-SISS function). *If the system 1.1 is (c)small-signal SISS on $(S_r, \mathcal{A})$ (for a certain $r > 0$, $c > 0$ and a size function $w_\mathcal{A}$ on $(\mathcal{X}, \mathcal{A})$), then there exists a continuously differentiable function $V : S_r \rightarrow \mathbb{R}$ and a constant c_V such that V is a (c_V) small-signal SISS-Lyapunov function for $(S_r, \mathcal{A})$.*

Proof. As $\mathcal{A} \subset S_r$, applying Lemma 3.3.11 the problem is reduced to the one in Proposition 3.3.13. $\square$

Example 3.3.15 (In general, SISS does not imply LISS). Consider the system $\dot{x} = f(x, d)$ with the following form:

$$\begin{cases} \dot{x}_1 = x_2 - x_1 \\ \dot{x}_2 = x_2^3 - x_2 + d(x, t) \end{cases}$$

For the unperturbed system, it is clear that the origin is an equilibrium point. In fact,

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -1 & 1 \\ 0 & 3x^2 - 1 \end{bmatrix}$$

and therefore $\frac{\partial f}{\partial x}(0)$ has eigenvalues at -1 , which proves the origin is locally asymptotically stable.

We can check that the system is not a gradient system, so the general results obtained in Section 3.3.2 do not apply.

It is clear the region of attraction of the origin has to be included or equal to the set $F = \{x \in \mathbb{R}^2 \mid |x_2| < 1\}$, because the second equation of the system is decoupled and corresponds to the system studied in Example 3.3.7

Consider the continuously differentiable function $V(x_1, x_2) = \frac{1}{2}(x_1 - x_2)^2 + \frac{x_2^2}{2} - \frac{x_2^4}{4} \geq 0 \forall (x_1, x_2) \in \overline{F} \subset \mathbb{R}^2$.

Its Lie derivative along the trajectories of the unperturbed system is $\mathcal{L}_f V(x) = -\frac{1}{4}(x_2^3 - 3x_2 + 2x_1)^2 - \frac{3}{4}x_2^2(x_2^4 - x_2^2 + 1)$.

The expression $x_2^4 - x_2^2 + 1$ is positive $\forall x_2 \in \mathbb{R}$, so $\frac{1}{4}(x_2^3 - 3x_2 + 2x_1)^2 + \frac{3}{4}x_2^2(x_2^4 - x_2^2 + 1) \geq 0$ with equality if and only if $(x_1, x_2) = (0, 0)$. Also, whenever $\|x\| \rightarrow \infty$, $\frac{1}{4}(x_2^3 - 3x_2 + 2x_1)^2 + \frac{3}{4}x_2^2(x_2^4 - x_2^2 + 1) \rightarrow \infty$. Continuity is clear, so $w_0((x_1, x_2)) := \frac{1}{4}(x_2^3 - 3x_2 + 2x_1)^2 + \frac{3}{4}x_2^2(x_2^4 - x_2^2 + 1)$ is a set size function for any pair $(C, \{0\})$ where C is a closed set. Therefore, $\mathcal{L}_f V(x) \leq -w_0(x) \forall x \in \mathbb{R}^2$. Notice that $V(x) \rightarrow \infty$ when $|x_1| \rightarrow \infty$ so it is a set size function for any closed subset of F by Lemma 3.2.1

The system cannot be ISS on F because the decoupled second equation of the system is not ISS on $(-1, 1)$, but $\forall r \in (0, 1)$, $\exists c_r > 0$ such that the system is (c_r) small-signal SISS on $([-r, r], 0)$ as seen in Example 3.3.7. Given r , consider the set $F_r = \{x \in \mathbb{R}^2 \mid |x_2| \leq r\}$. This set is forward invariant under disturbances with infinity norm smaller than c_r (the x_1 coordinate cannot escape it because it encompasses all of its span). Therefore, by Proposition 3.2.9, $\forall r \in (0, 1)$ $\exists c_r$ st the system is (c_r) small-signal SISS on $(F_r, \{0\})$.

Notice that no set F_r is included inside a sub-level set of a LISS, ISS or SISS Lyapunov function. The SISS property thus gives us insight on sets that cannot be treated with the LISS or ISS properties. $\bullet$

Chapter 4

SISS properties of the perturbed Safe Gradient Flow (SGF)

4.1 SISS property of the SGF under generic additive disturbances

We now consider a generic disturbance entering the SGF algorithm, which can be written as:

$$\dot{x} = \tilde{\mathcal{G}}_\alpha(x, t) = \mathcal{G}_\alpha(x) + d(x, t) \quad (4.1)$$

We work on the open set $\mathcal{X}$ in which the unperturbed SGF is defined, and assume that both the compact set of global minimizers of the optimization problem 1.5 (which we call $\mathcal{A}$) and the set of KKT points of 1.5 (which we call $\tilde{\mathcal{A}}$) are both compact. Notice $\mathcal{A} \subseteq \tilde{\mathcal{A}}$.

Our first objective is to study how does this general disturbance affect the forward invariance of the constraints set C . In order to do this, we use the Input to State Safety notions introduced in 1.1.7.

Proposition 4.1.1. *If $|\nabla g_i(x)|$ and $|\nabla h_j(x)|$ are bounded along the boundary of the safe set $C \ \forall i \in \{1, \dots, m\}$ and $\forall j \in \{1, \dots, k\}$, then $\forall \epsilon > 0 \ \exists c > 0 \in \mathbb{R}$ such that the pair (g, h) is an ISSf-BF of the perturbed SGF $\dot{x} = \mathcal{G}_\alpha(x) + d(x, t)$ in a set $D = D_\epsilon \cap \mathcal{X}$ for all $d \in \mathcal{M}_c$.*

Proof. We know that

$$\begin{aligned} |\nabla g_i(x)| &< B_g^i \in \mathbb{R} \ \forall x \in \partial C, \ \forall i \in \{1, \dots, m\} \\ |\nabla h_j(x)| &< B_h^j \in \mathbb{R} \ \forall x \in \partial C, \ \forall j \in \{1, \dots, k\} \end{aligned}$$

Take $B_g := \max_{i \in \{1, \dots, m\}} B_g^i$ and $B_h := \max_{j \in \{1, \dots, k\}} B_h^j$.

By continuity of $|\nabla g_i(x)|$ and $|\nabla h_j(x)|$, given $x \in \partial C$ we have that $\exists \delta_1^x, \delta_2^x$ st $\forall y \in B(x; \delta_1^x)$ it follows that $|\nabla g_i(x)| < B_g + \epsilon \ \forall i \in \{1, \dots, m\}$ and also $\forall y \in B(x; \delta_2^x)$ it follows that $|\nabla h_j(x)| < B_h + \epsilon \ \forall j \in \{1, \dots, k\}$.

Now, define $D_\epsilon = C \cup (\cup_{x \in \partial C} B(x; \delta_1^x) \cap B(x; \delta_2^x))$ and take $D := D_\epsilon \cap \mathcal{X}$. This is the set D we will use to look at the input to state safety property.

We can see that $\forall x \in D \setminus \mathring{C}$:

$$|\nabla g_i(x)| < B_g + \epsilon \quad \forall i \in \{1, \dots, m\} \text{ and } |\nabla h_j(x)| < B_h + \epsilon \quad \forall j \in \{1, \dots, k\}$$

Taking this into account together with the fact that when working with smooth functions $\mathcal{L}_{f_1+f_2} = \mathcal{L}_{f_1} + \mathcal{L}_{f_2}$, we have that:

$$\mathcal{L}_{\mathcal{G}_\alpha} g_i(x) + \mathcal{L}_{d(x,t)} g_i(x) \leq -\alpha \cdot g_i(x) + \|d(x,t)\| \cdot |\nabla g_i(x)| \leq -\alpha \cdot g_i(x) + (B_g + \epsilon) \cdot \|d\| \quad \forall i \in \{1, \dots, m\} \quad \forall t \geq 0$$

And also:

$$\begin{aligned} \mathcal{L}_{\mathcal{G}_\alpha} h_j(x) + \mathcal{L}_{d(x,t)} h_j(x) &= -\alpha \cdot h_j(x) + d(x,t)^t \cdot \nabla h_j(x) \Rightarrow \\ -\alpha \cdot h_j(x) - \|d(x,t)\| \cdot |\nabla h_j(x)| &\leq \mathcal{L}_{\mathcal{G}_\alpha} h_j(x) + \mathcal{L}_{d(x,t)} h_j(x) \leq -\alpha \cdot h_j(x) + \|d(x,t)\| \cdot |\nabla h_j(x)| \Rightarrow \\ -\alpha \cdot h_j(x) - (B_h + \epsilon) \cdot \|d\| &\leq \mathcal{L}_{\mathcal{G}_\alpha} h_j(x) + \mathcal{L}_{d(x,t)} h_j(x) \leq -\alpha \cdot h_j(x) + (B_h + \epsilon) \cdot \|d\| \\ &\quad \forall j \in \{1, \dots, k\} \quad \forall t \geq 0 \end{aligned}$$

The conditions in Proposition 1.1.21 are fulfilled with $\tilde{c} = +\infty$. We can see that, for SGF, C_c is the (set of) maximal path connected component of

$$\{x \in \mathbb{R}^n \mid \quad (4.2)$$

$$g_i(x) - \frac{(B_g + \epsilon) \cdot c}{\alpha} \leq 0 \quad \forall i \in \{1, \dots, m\}, \quad (4.3)$$

$$h_j(x) - \frac{(B_h + \epsilon) \cdot c}{\alpha} \leq 0 \quad \forall j \in \{1, \dots, k\}, \quad (4.4)$$

$$h_s(x) + \frac{(B_h + \epsilon) \cdot c}{\alpha} \geq 0 \quad \forall s \in \{1, \dots, k\} \} \quad (4.5)$$

that contains C . We take the biggest c such that $C_c \subset D \subseteq \mathcal{X}$ and the proposition ensures that C_c is forward invariant (or, in other words, that C_d is forward invariant $\forall d \in \mathcal{M}_c$). This is equivalent to saying that (g, h) are an ISSf-BF. $\square$

Remark 4.1.2. *If C is bounded, then we can choose c such that C_c is bounded too because g and h are continuously differentiable.*

Once we have established conditions under which the expanded constraints set remains forward invariant, we can study the Set Input to State Stability of the Safe Gradient Flow on $(C_c, \mathcal{A})$ (recall that the definition of Set Input to State Stability, 18, requires the set to be forward invariant).

We saw in 3.3.1 that Set Input to State Stability implies Local Asymptotic Stability, so for the Safe Gradient Flow to be Set Input to State Stable on the expanded set C_c , the system must be asymptotically stable in C_c . We know that if the LICQ conditions hold in C , C is bounded and f , g and h are real-analytic then every trajectory starting in $\mathcal{X}$ converges to a KKT point (Proposition 1.1.34). However, we are not studying the input to state stability of the whole set $\mathcal{X}$ here (doing that would require guaranteeing its forward invariance, which we have not been able prove), but rather the input to state stability of the expanded constraints set C_c . What is more, we are interested in studying

SISS of C_c with respect to the set of global equilibria of the constrained optimization problem, $\mathcal{A}$. As the KKT points showcase the same stability properties than the global minimizers, it makes sense that we can only prove Set Input to State practical Stability, as we need to account for the fact that trajectories might converge to KKT points that are not global equilibria of the system. When the optimization problem is convex, we are able to move past the practical stability condition because $\mathcal{A} = \tilde{\mathcal{A}}$.

Even though by proving SISpS of C_c we are not requiring the system to be GAS in $\mathcal{X}$, it is to be expected that the conditions needed to ensure SISpS of $(C_c, \mathcal{A})$ resemble those of Proposition 1.1.34. However, working in C_c instead of $\mathcal{X}$ actually allow us to relax those conditions in order to show practical stability.

In [14], 3 different Lyapunov functions are used. The first option considered is $f - f^*$ (where f^* is the value of f in some equilibria of the system). However, this function is unable to give Lyapunov conditions outside the set C . Then, the function $V(x, \epsilon) = f(x) - f^* + \frac{1}{\epsilon} \sum_{i=1}^m [g_i(x)] + \frac{1}{\epsilon} \sum_{j=1}^k |h_j(x)|$ is introduced. Thanks to the penalty functions literature, this function allows us to work outside the constraints set C . Finally, the function $W_\alpha(x)$ is introduced in [14, Section V.C].

Readers familiar with the Safe Gradient Flow might think that in order to prove set input to state stability of C we would want to use $W_\alpha(x)$, as it is the function used in the proof of 1.1.34. However, it is more convenient to use $V(x; \epsilon)$. This happens because $W_\alpha(x)$ is used in 1.1.34 together with [14, Lemma 5.20], which characterizes the positive limit set of the trajectories of the Safe Gradient Flow. $W_\alpha(x)$ is only used "locally", but cannot be used as a set size function for $(C_c, \tilde{\mathcal{A}})$, as it is not zero for every $x \in \tilde{\mathcal{A}}$. Thus, $W_\alpha(x)$ -just as $V(x, \epsilon)$ - only allows us to work with practical stability.

Lemma 4.1.3. *Given $\epsilon > 0 \in \mathbb{R}$, the function $\frac{1}{\epsilon}[g_i(x)]$ is regular and*

$$\partial \frac{1}{\epsilon}[g_i(x)] = \begin{cases} \vec{0}, & \text{if } g_i(x) < 0 \\ \frac{1}{\epsilon} \cdot \text{co}(\{\vec{0}, \nabla g_i(x)\}), & \text{if } g_i(x) = 0 \\ \frac{1}{\epsilon} \nabla g_i(x), & \text{if } g_i(x) > 0 \end{cases} \quad (4.6)$$

$\forall i \in \{1, \dots, m\}$

Proof. $\partial \frac{1}{\epsilon}[g_i(x)] = \frac{1}{\epsilon} \partial(\max\{0, g_i(x)\})$

[30, Proposition 7] ensures that the function $\max\{0, g_i(x)\}$ is regular because both functions we are evaluating are regular, and also that

$$\partial(\max\{0, g_i(x)\}) = \text{co}(\{\emptyset \text{ if } 0 < g_i(x) ; \partial 0 \text{ otherwise}\} \cup \{\emptyset \text{ if } g_i(x) < 0 ; \partial g_i(x) \text{ otherwise}\}) \quad (4.7)$$

As both 0 and $g_i(x)$ are smooth, their generalized gradients are single-valued and respectively equal to $\vec{0}$ and $\nabla g_i(x)$. Substituting in 4.7, we get the proposed expression 4.6. $\square$

Lemma 4.1.4. *Given $\epsilon > 0 \in \mathbb{R}$, the function $\frac{1}{\epsilon}|h_j(x)|$ is regular and*

$$\partial \frac{1}{\epsilon}[h_j(x)] = \begin{cases} -\frac{1}{\epsilon} \nabla h_j(x), & \text{if } h_j(x) < 0 \\ \frac{1}{\epsilon} \cdot \text{co}\{-\nabla h_j(x), \nabla h_j(x)\}, & \text{if } h_j(x) = 0 \\ \frac{1}{\epsilon} \nabla h_j(x), & \text{if } h_j(x) > 0 \end{cases} \quad (4.8)$$

$\forall j \in \{1, \dots, k\}$

Proof. We reason just as in the previous lemma, only now taking into account that $|h_j(x)| = \max\{-h_j(x), h_j(x)\}$, therefore it is regular and we have that

$$\begin{aligned} \partial(\max\{-h_j(x), h_j(x)\}) = \\ \text{co}(\{\emptyset \text{ if } -h_j(x) < h_j(x) ; \partial(-h_j(x)) \text{ otherwise}\} \cup \{\emptyset \text{ if } h_j(x) < -h_j(x) ; \partial h_j(x) \text{ otherwise}\}) \end{aligned}$$

where $\partial(-h_j(x)) = -\nabla h_j(x)$ and $\partial(h_j(x)) = \nabla h_j(x)$ as $h_j(x)$ is a smooth function. $\square$

Definition 21. We define the function $V(x; \epsilon) : \mathbb{R}^n \times \mathbb{R}_{>0} \rightarrow \mathbb{R}$ as:

$$V(x; \epsilon) = f(x) - f^* + \frac{1}{\epsilon} \sum_{i=1}^m [g_i(x)] + \frac{1}{\epsilon} \sum_{j=1}^k |h_j(x)| \quad (4.9)$$

Lemma 4.1.5. If C is compact and MFCQ conditions hold $\forall x \in C$, then for all B compact set such that $C \subset B \subset \mathcal{X}$ there exists ϵ_1 such that $\forall \epsilon < \epsilon_1$, $V^{\text{ext}}(x; \epsilon)$ is a set size function for $(B, \mathcal{A})$

Proof. If MFCQ conditions hold $\forall x \in C$, then they hold for the restricted global minima. [38, Theorem 4(a)] states that, under these conditions, $V(x; \epsilon)$ is a weak penalty function. We take a compact Ω such that $B \subset \Omega$ and by the previous remark, $\exists \epsilon_1$ st $\forall \epsilon < \epsilon_1$ all the zeros (global minima) of $f(x) - f^*$ in $\Omega \cap C = C$ and $V(x; \epsilon)$ in Ω coincide. Thus, all of them in B do too, and so $V(x; \epsilon)$, and $f(x) - f^*$ share zeros in B . This guarantees that given $x \in B$, $V(x; \epsilon) = 0 \iff x \in \mathcal{A}$ and also $V^{\text{ext}}(x; \epsilon) \geq 0 \forall x \in B$. As B is compact, conditions in Lemma 3.2.1 hold and thus $V(x; \epsilon)$ is a set size function for $(B, \mathcal{A})$. $\square$

Lemma 4.1.6. $\forall B$ closed set such that $C \subset B \subset \mathcal{X}$, if either:

- B is bounded
- $\forall (x_n)_{n \geq 0} \subset B$ st $\|x_n\| \rightarrow \infty$ when $n \rightarrow \infty$; $\|G_\alpha(x_n)\| \rightarrow \infty$ with n too

Then, $\|G_\alpha(x)\|$ is a weak set size function for $(B, \mathcal{A})$.

Proof. Given $x \in B$, $\|G_\alpha(x)\| = 0 \iff x \in \tilde{\mathcal{A}} = X_{KKT}$. Also, $\|G_\alpha(x)\|$ is continuous and positive or equal to zero $\forall x \in \mathbb{R}^n$. Thus, conditions in Lemma 3.2.1 hold for $(C, \tilde{\mathcal{A}})$. We conclude $\|G_\alpha(x)\|$ is a weak set size function for $(C, \mathcal{A})$. $\square$

Lemma 4.1.7. Given a compact set B st $C \subset B \subset \mathcal{X}$, if $d(x, t) : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ is such that $\|d\|_\infty < \infty$, then $\exists \epsilon_2 > 0$ st $\forall \epsilon < \epsilon_2$:

$$\max \tilde{L}_{\tilde{G}_\alpha(x, t)} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 + \gamma(\|d(x, t)\|) \quad \forall (x, t) \in B \times \mathbb{R}_{\geq 0}$$

Proof. As B is bounded, we know that $|\nabla g_i(x)|$ and $|\nabla h_j(x)|$ are bounded in it.

$$\begin{aligned} |\nabla g_i(x)| &< T_g^i \in \mathbb{R} \quad \forall x \in B, \quad \forall i \in \{1, \dots, m\} \\ |\nabla h_j(x)| &< T_h^j \in \mathbb{R} \quad \forall x \in B, \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

Take $T_g := \max_{i \in \{1, \dots, m\}} T_g^i$ and $T_h := \max_{j \in \{1, \dots, k\}} T_h^j$. Then, given $x \in B$, $|\nabla g_i(x)| < T_g \ \forall i \in \{1, \dots, m\}$ and also $|\nabla h_j(x)| < T_h \ \forall j \in \{1, \dots, k\}$.

Also, following the same reasoning:

$$\exists T_f > 0 \text{ st } |\nabla f(x)| < T_f \ \forall x \in B$$

Now moving on to computing the Lie derivative, by lemma 1.1.4, $\tilde{\mathcal{L}}_{G_\alpha(x)+d(x,t)}V(x; \epsilon) = \tilde{\mathcal{L}}_{G_\alpha(x)}V(x; \epsilon) + \tilde{\mathcal{L}}_{d(x,t)}V(x; \epsilon)$.

Now, by lemma 1.1.2,

$$\tilde{\mathcal{L}}_{G_\alpha(x)}V(x; \epsilon) = (\tilde{\mathcal{L}}_{G_\alpha(x)}f(x))_{(1)} + \left(\sum_{i=1}^m \tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)]\right)_{(2)} + \left(\sum_{j=1}^k \tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)|\right)_{(3)}$$

and also

$$\tilde{\mathcal{L}}_{d(x,t)}V(x; \epsilon) = (\tilde{\mathcal{L}}_{d(x,t)}f(x))_{(4)} + \left(\sum_{i=1}^m \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)]\right)_{(5)} + \left(\sum_{j=1}^k \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)|\right)_{(6)}$$

We study this term by term, using the definition of the generalized Lie derivative and the expressions derived above for the generalized gradients:

(i) $\tilde{\mathcal{L}}_{G_\alpha(x)}(x)f(x) = \mathcal{L}_{G_\alpha(x)} \leq -\|G_\alpha(x)\|^2 + \alpha u^t g(x) + \alpha v^t h(x)$

(ii) for each $i \in \{1, \dots, m\}$:

- if $i \in I_-(x)$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)] = 0$
- if $i \in I_+(x)$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)] = \frac{1}{\epsilon} \nabla g_i(x)^t G_\alpha(x) \leq -\frac{\alpha}{\epsilon} g_i(x)$
- if $i \in I_0(x)$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)] = \{a \in \mathbb{R} \mid \frac{t}{\epsilon} \nabla g_i(x)^t G_\alpha(x) = a \ \forall t \in [0, 1]\}$.

Note that $\frac{t}{\epsilon} \nabla g_i(x)^t G_\alpha(x) \leq -\frac{\alpha t}{\epsilon} g_i(x) = 0 \ \forall t$ as $i \in I_0(x)$, meaning $g_i(x) = 0$, so in this case $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)] = 0$.

Notice that, as $u \in \mathbb{R}_{\geq 0}^m$:

$$\begin{aligned} \alpha u^t g(x) + \sum_{i=1}^m \tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} [g_i(x)] &\leq \sum_{i \in I_-(x) \cup I_0(x)} \alpha u_i g_i(x) + \alpha \cdot \sum_{i \in I_+(x)} (u_i - \frac{1}{\epsilon}) g_i(x) \\ &\leq \alpha \cdot \sum_{i \in I_+(x)} (u_i - \frac{1}{\epsilon}) g_i(x) \end{aligned}$$

(iii) for each $j \in \{1, \dots, k\}$:

- if $h_j(x) < 0$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)| = -\frac{1}{\epsilon} \nabla h_j(x)^t G_\alpha(x) = \frac{\alpha}{\epsilon} h_j(x)$
- if $h_j(x) > 0$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)| = \frac{1}{\epsilon} \nabla h_j(x)^t G_\alpha(x) = -\frac{\alpha}{\epsilon} h_j(x)$
- if $h_j(x) = 0$; $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)| =$
 $\{a \in \mathbb{R} \mid (-\frac{(1-t)}{\epsilon} \nabla h_j(x) + \frac{t}{\epsilon} \nabla h_j(x))^t G_\alpha(x) = a \ \forall t \in [0, 1]\} =$
 $\{a \in \mathbb{R} \mid \frac{2t}{\epsilon} \nabla h_j(x)^t G_\alpha(x) - \frac{1}{\epsilon} \nabla h_j(x)^t G_\alpha(x) = a \ \forall t \in [0, 1]\} =$
 $\{a \in \mathbb{R} \mid -\frac{2t}{\epsilon} \alpha h_j(x) + \frac{1}{\epsilon} \alpha h_j(x) = a \ \forall t \in [0, 1]\} = 0$
because $h_j(x) = 0$.

We can summarize this results in $\tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)| = -\frac{\alpha}{\epsilon} |h_j(x)|$.

Note also that:

$$\sum_{j=1}^k \tilde{\mathcal{L}}_{G_\alpha(x)} \frac{1}{\epsilon} |h_j(x)| + \alpha v^t h(x) \leq \alpha \cdot \sum_{j=1}^k (|v_j| - \frac{1}{\epsilon}) |h_j(x)|$$

(iv) $\tilde{\mathcal{L}}_{d(x,t)} f(x) = \nabla f(x)^t d(x, t)$

(v) for each $i \in \{1, \dots, m\}$:

- if $i \in I_-(x)$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = 0$
- if $i \in I_+(x)$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = \frac{1}{\epsilon} \nabla g_i(x)^t d(x, t)$
- if $i \in I_0(x)$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = \{a \in \mathbb{R} \mid \frac{1}{\epsilon} \nabla g_i(x)^t d(x, t) = a \ \forall t \in [0, 1]\}$.
In this case, $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = 0$ if $\nabla g_i(x)^t d(x, t) = 0$, else $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = \emptyset$ (if this happens, $\max \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] = -\infty$).

We can conclude that

$$\max \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} [g_i(x)] \leq \frac{1}{\epsilon} \|\nabla g_i(x)\| \|d(x, t)\|$$

(vi) for each $j \in \{1, \dots, k\}$:

- if $h_j(x) < 0$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = -\frac{1}{\epsilon} \nabla h_j(x)^t d(x, t)$
- if $h_j(x) > 0$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = \frac{1}{\epsilon} \nabla h_j(x)^t d(x, t)$
- if $h_j(x) = 0$; $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = \{a \in \mathbb{R} \mid (\frac{2t}{\epsilon} \nabla h_j(x) - \frac{1}{\epsilon} \nabla h_j(x))^t d(x, t) = a \ \forall t \in [0, 1]\} = \{a \in \mathbb{R} \mid \frac{1}{\epsilon} (2t - 1) \nabla h_j(x)^t d(x, t) = a \ \forall t \in [0, 1]\}$.
In this case, $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = 0$ if $\nabla h_j(x)^t d(x, t) = 0$, else $\tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = \emptyset$ (if this happens, $\max \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| = -\infty$).

We can conclude that

$$\max \tilde{\mathcal{L}}_{d(x,t)} \frac{1}{\epsilon} |h_j(x)| \leq \frac{1}{\epsilon} \|\nabla h_j(x)\| \|d(x, t)\|$$

Putting (1),(2) and (3) together we get that

$$\tilde{\mathcal{L}}_{G_\alpha(x)} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 + \alpha \cdot \sum_{i \in I_+(x)} (u_i - \frac{1}{\epsilon}) g_i(x) + \alpha \cdot \sum_{j=1}^k (|v_j| - \frac{1}{\epsilon}) |h_j(x)|$$

Choosing $\epsilon < \epsilon_2 < \frac{1}{B}$ as given by Lemma C.1 in [8], we have:

$$\alpha \cdot \sum_{i \in I_+(x)} (u_i - \frac{1}{\epsilon}) g_i(x) + \alpha \cdot \sum_{j=1}^k (|v_j| - \frac{1}{\epsilon}) |h_j(x)| < 0$$

Therefore, $\forall \epsilon < \epsilon_2$:

$$\max \tilde{\mathcal{L}}_{G_\alpha(x)} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 \quad \forall x \in \mathcal{X}$$

Now, we put (4),(5),(6) together with the bounds derived in the beginning of this proof, getting:

$$\max \tilde{\mathcal{L}}_{d(x,t)} V(x; \epsilon) \leq (T_f + \frac{m}{\epsilon} T_g + \frac{k}{\epsilon} T_h) \cdot \|d(x, t)\| \quad \forall \epsilon < \epsilon_2, \quad \forall (x, t) \in B \times \mathbb{R}_{\geq 0}$$

We define $\gamma(r) := (T_f + \frac{m}{\epsilon} T_g + \frac{k}{\epsilon} T_h) \cdot r$. Clearly, $\gamma \in \mathcal{K}_\infty$.

We can finally conclude that, $\forall \epsilon < \epsilon_2$,

$$\max \tilde{\mathcal{L}}_{G_\alpha(x)+d(x,t)} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 + \gamma(\|d(x, t)\|) \quad \forall (x, t) \in B \times \mathbb{R}_{\geq 0}$$

□

Proposition 4.1.8. *If the safe set C is bounded and the MFCQ conditions hold $\forall x \in C$, then $\forall \bar{\epsilon} > 0$:*

- $\exists c > 0$ st C is ISSf with constant c under the perturbed SGF dynamics $\dot{x} = G_\alpha(x) + d(x, t)$ and C_c is still bounded.
- $\exists \epsilon^*$ st $V(x; \epsilon)$ is a generalized SISS-Lyapunov function for $(C_c, \mathcal{A})$ $\forall \epsilon < \epsilon^*$

Proof. Given $\bar{\epsilon}$, as C is bounded the conditions in proposition 4.1.1 hold and therefore $\exists c > 0$ such that $C_c \subset D = D_{\bar{\epsilon}} \cap \mathcal{X}$ (built as described in the proposition) is forward invariant $\forall d \in \mathcal{M}_c$.

Take c small enough so that C_c is still bounded. This can be done because both g and h are continuously differentiable. Take a compact set B such that $C \subset C_c \subset B \subset X$. This can be done in the following way: for each $x \in C_c$, $\exists \epsilon_x$ such that $\bar{B}(x; \epsilon_x) \subset \mathcal{X}$. Clearly, $C_c \subset \bigcup_{x \in C_c} \bar{B}(x; \epsilon_x)$. Therefore as C_c is compact there exists a finite covering $\{\bar{B}(x_1; \epsilon_{x_1}), \dots, \bar{B}(x_f; \epsilon_{x_f})\}$ (with $0 < f < \infty$) such that $C_c \subset \bigcup_{i=1}^f \bar{B}(x_i; \epsilon_{x_i})$.

$\bigcup_{i=1}^f \bar{B}(x_i; \epsilon_{x_i}) \subset \mathcal{X}$ is clearly compact (finite union of compact sets in a metric space) and fulfills our requirements for B , therefore we choose

$$B := \bigcup_{i=1}^f \bar{B}(x_i; \epsilon_{x_i})$$

Now, lemma 4.1.5 ensures that $\exists \epsilon_1$ such that $V(x; \epsilon)$ is a set size function for $(B, \mathcal{A})$ $\forall \epsilon < \epsilon_1$.

Clearly, $V(x; \epsilon)$ is a set size function for $(C_c, \mathcal{A})$ too and:

$$\max \tilde{\mathcal{L}}_{G_\alpha(x)+d(x,t)} V(x; \epsilon) = \max \tilde{\mathcal{L}}_{G_\alpha(x)} V(x; \epsilon) \quad \forall (x, t) \in C_c \times \mathbb{R}_{\geq 0}$$

Lemma 4.1.6 ensures that $\|G_\alpha(x)\|$ is a weak set size function for $(B, \mathcal{A})$ and therefore also for $(C_c, \mathcal{A})$. Finally, lemma 4.1.7 ensures that, given the set C_c :

$$\max \tilde{\mathcal{L}}_{G_\alpha(x)+d(x,t)} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 + \gamma(\|d(x, t)\|) \quad \forall (x, t) \in C_c \times \mathbb{R}_{\geq 0}, \quad \forall \epsilon < \epsilon_2$$

We conclude that taking $\epsilon^* \leq \min\{\epsilon_1, \epsilon_2\}$, we ensure that $\forall \epsilon < \epsilon^*$, $V(x; \epsilon)$ is a generalized SISS-Lyapunov function. □

Corollary 4.1.9. *If the safe set C is bounded and the MFCQ conditions hold $\forall x \in C$, then*

$\forall \bar{\epsilon} > 0 \exists c > 0$ and $C_c \subset \mathcal{X}$ such that the perturbed Safe Gradient Flow 4.1 is (c)small gain SISS in C_c

Proof. Under these conditions, the previous proposition holds and therefore we can apply Corollary 3.2.10, which tells us that the system is small gain weak SISS in C_c . $\square$

Corollary 4.1.10. *If the safe set C is bounded, the MFCQ conditions hold $\forall x \in C$ and the optimization problem is convex, then*

$\forall \bar{\epsilon} > 0 \exists c > 0$ and $C_c \subset \mathcal{X}$ such that the perturbed Safe Gradient Flow 4.1 is small gain SISS in C_c

Proof. If the optimization problem is convex, then $\mathcal{A} = X_{KKT}$ and therefore we can apply Corollary 3.2.10 together with remark 3.2.12 to get that the perturbed Safe Gradient Flow 4.1 is (c)small gain SISS in C_c . $\square$

To finish this section, we give a result on the Set Input to State Stability of the system on the sub-level sets of $V(x, \epsilon)$.

Proposition 4.1.11. *Let $\tilde{\mathcal{A}}_p$ be an isolated subset of $\tilde{\mathcal{A}}$ such that $f(x) = f^*$ for all $x \in \tilde{\mathcal{A}}_p$ and suppose the MFCQ conditions hold on $\tilde{\mathcal{A}}_p$. Then, for every compact set $\Omega \subset \mathcal{X}$ such that $\tilde{\mathcal{A}}_p \subset \Omega$, $\tilde{\mathcal{A}}_p \cap \partial\Omega = \emptyset$ and $(\tilde{\mathcal{A}} \setminus \tilde{\mathcal{A}}_p) \cap \Omega = \emptyset$ there exists $\epsilon^* > 0$ such that for all $\epsilon < \epsilon^*$ there exist $r > 0$ such that $S_r^V \cap \Omega$ is a (set of) path connected components of S_r^V , and for every such r there exists $c_r > 0$ such that the system is (c_r) small-gain SISS on $(S_r^V \cap \Omega, \tilde{\mathcal{A}}_p)$.*

What is more, if conditions in Proposition 4.1.1 hold, there also exist $c_r^ > 0$ such that the system is (c_r^*) small-gain SISS on $(S_r^V \cap \Omega \cap C_{c_r^*}, \tilde{\mathcal{A}}_p)$*

Proof. If the MFCQ conditions hold in $\tilde{\mathcal{A}}_p$, take Ω as stated in the Proposition. By [38, Theorem 4(a)], the function $V(x; \epsilon)$ defined as in (4.9) with f^* the value of f in $\tilde{\mathcal{A}}_p$ is a weak penalty function. Then, there exists ϵ^* such that for all $x \in \Omega$, $V(x, \epsilon) = 0 \iff x \in \tilde{\mathcal{A}}_p$ for all $\epsilon < \epsilon^*$. Now, take $r > 0$ so that $S_r^V \cap \Omega$ is a (set of) path connected components of S_r^V , which can always be done because $d(S_r^V, \tilde{\mathcal{A}}_p) \rightarrow 0$ as $r \rightarrow 0$. $S_r^V \cap \Omega$ is compact, so S_r^V is a set size function for $(S_r^V \cap \Omega, \tilde{\mathcal{A}}_p)$. Note that $\mathcal{G}_\alpha$ is now in fact also a set size function for $(S_r^V \cap \Omega, \tilde{\mathcal{A}}_p)$. Then, taking into account that $S_r^V \cap \Omega$ is compact we can proceed as in Lemma 4.1.7 to see that S_r^V is a SISS-Lyapunov function for $(S_r^V \cap \Omega, \tilde{\mathcal{A}}_p)$.

Apply Corollary 3.2.11 to get that there exists c_r such that the system is (c_r) small signal SISS on $((S_r^V \cap \Omega, \tilde{\mathcal{A}}_p))$.

If conditions in Proposition 4.1.1 hold, then for every $\epsilon < \epsilon^*$ there exists $c > 0$ such that $C_c \subset \mathcal{X}$ is forward invariant $\forall d \in \mathcal{M}_c$. Take $c_r^* := \min\{c_r, c\}$. Then, Proposition 3.2.9 grants the SGF is (c_r^*) small signal SISS on $(S_r^V \cap \Omega \cap C_{c_r^*}, \tilde{\mathcal{A}}_p)$.

Note that we have stability (and not practical stability) because V and G_α are now set size functions for the same set of equilibria, $\tilde{\mathcal{A}}_p$. $\square$

4.2 SISS property of the SGF under exclusively state-dependent additive disturbances

4.2.1 Errors in the objective function

In this section, we will assume that our knowledge of the objective function we want to minimize is faulty. This is, we have $\tilde{f} = f + \tilde{d}$, where we will assume that $\tilde{d} : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable and its derivative is LL.

If this happens, we will get $\tilde{K}_\alpha(x), \tilde{\Lambda}_\alpha(x)$ and $\tilde{\mathcal{X}}$. Note that the results in Section 1.1.11 for these new sets still hold, so we will get a controller $\tilde{u}(x), \tilde{v}(x)$ and a flow $\tilde{\mathcal{G}}_\alpha(x)$ that still renders C forward invariant.

As $\tilde{u}(x), \tilde{v}(x)$, the derivative of $\tilde{d}$ and the gradient $\nabla \tilde{f}(x)$ are all LL in $\tilde{\mathcal{X}}$, we can write

$$\tilde{\mathcal{G}}_\alpha(x) = \mathcal{G}_\alpha(x) - d(x) \quad (4.10)$$

where $d : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is LL and can be decomposed as $d = \nabla \tilde{d} + d_u + d_v$

Lemma 4.2.1. *If C is bounded, $(f(x) - f^*)$ is a generalized SISS-Lyapunov function for $(C, \mathcal{A})$ under the flow 4.10*

Proof. Clearly $f(x) - f^*$ is continuous. Also, as $f^* = \min_{x \in C} f(x)$, $f(x) - f^* \geq 0 \forall x \in C$. What is more, $\mathcal{A} \subset C$ and clearly $f(x) - f^* = 0 \iff f(x) = f^* \iff x \in \mathcal{A}$. Therefore, by Lemma 3.2.1 we conclude that $(f(x) - f^*)$ is a Set Size Function for $(C, \mathcal{A})$.

Now, we need to prove the SISS-Lyapunov inequality. We can work with $f - f^*$ instead of the extended version because they take the same values on C (and their derivatives too, as the extension is done on B). Also, we can work with the usual notion of Lie derivative as $f - f^*$ is continuous:

$$\mathcal{L}_{\mathcal{G}_\alpha(x) - d(x)}(f(x) - f^*) = (\mathcal{G}_\alpha(x) - d(x))^t \cdot \nabla f(x)$$

Substituting in 1.10, we get that $\nabla f(x) = -(\mathcal{G}_\alpha + \frac{\partial g(x)}{\partial x}^t \cdot u + \frac{\partial h(x)}{\partial x}^t \cdot v)$. Therefore, taking into account the third and fourth equalities in 1.10, we can write:

$$\begin{aligned} \mathcal{L}_{\mathcal{G}_\alpha(x) - d(x)}f(x) &= -\|\mathcal{G}_\alpha\|^2 + \alpha \cdot u^t \cdot g(x) + \alpha \cdot v^t \cdot h(x) + \\ &\quad d(x)^t \cdot \nabla f(x) \end{aligned}$$

Now take into account that $u \in \mathbf{R}_{\geq 0}^m$ and we are considering $x \in C \Rightarrow g(x) \leq 0$ and $h(x) = 0$. Take M the bound of $\|\nabla f(x)\|^2$ on C , and note that $d(x)^t \cdot \nabla f(x) \leq \|d(x)\| \cdot \|\nabla f(x)\|$. We are left with:

$$\mathcal{L}_{\mathcal{G}_\alpha(x)}f(x) \leq -\|\mathcal{G}_\alpha(x)\|^2 + M \cdot \|d(x)\| \quad \forall x \in C$$

Lemma 4.1.6 states that $\mathcal{G}_\alpha(x)$ is a weak set size function in $(C, \mathcal{A})$. Note also that $\alpha(r) := Mr$ is a class $\mathcal{K}_\infty$ function.

We can conclude $(f(x) - f^*)$ is a generalized SISS-Lyapunov function for $(C, \mathcal{A})$. $\square$

Proposition 4.2.2. *If C is bounded, then the safe gradient flow is SISpS on $(C, \mathcal{A})$ under errors on the objective function f .*

Proof. The above lemma gives us a generalized SISS-Lyapunov function for 4.10. The safe set C is still forward invariant under the new controllers $\tilde{u}, \tilde{v}$. Therefore, by Proposition 3.2.9 the system is weak SISS (we are only imposing $\|d\|_\infty < \infty$, therefore we do not need the small signal condition) $\square$

4.2.2 Errors in the constraints set

Assume now that there are errors on the functions that define our safe set C . This is, imagine we have $\tilde{g}(x), \tilde{h}(x)$ that determine $\tilde{C}$. We will again assume that the error is finite and LL; this is, $\tilde{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined as $\tilde{g}(x) = g(x) + d_g(x)$ where $d_g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuously differentiable, its derivative is LL and $\|d_g\|_\infty < \infty$; and $\tilde{h} : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is defined as $\tilde{h}(x) = h(x) + d_h(x)$ where $d_h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is continuously differentiable, its derivative is LL and $\|d_h\|_\infty < \infty$.

These errors give raise to $\tilde{K}_\alpha(x), \tilde{\Lambda}_\alpha(x)$ and $\tilde{C} \subset \tilde{\mathcal{X}}$. Note that the results in Section 1.1.11 for these new sets still hold, so we will get a controller $\tilde{u}(x), \tilde{v}(x)$ and a flow $\tilde{\mathcal{G}}_\alpha(x) = \mathcal{G}_\alpha(x) - d(x)$ that renders the erroneous safe set $\tilde{C}$ forward invariant.

In this case, we have not been able to prove a better SISS condition than in the general case. This seems reasonable, as when we work with sets with points outside C we need to guarantee they are still inside $\mathcal{X}$ if we want to be able to work with $V(x; \epsilon)$ (or, in general, any SISS-Lyapunov function on the set of equilibria $\mathcal{A}$ that takes advantage of the definition of the SGF).

We will prove that the system is small gain weak SISS on $(\tilde{C}, \mathcal{A})$ for disturbances under a more flexible and intuitive bound c than in the general case.

Lemma 4.2.3. *If $\tilde{C}$ is bounded, $\tilde{C} \subset \mathcal{X}$ and the MFCQ conditions hold $\forall x \in \tilde{C}$, then $\exists \epsilon^*$ such that $V(x; \epsilon)$ is a generalized SISS-Lyapunov function for $(\tilde{C}, \mathcal{A})$ $\forall \epsilon < \epsilon^*$.*

Proof. We are trying to determine Set Input to Stability with respect to a set of equilibria $\mathcal{A}$ that need not be contained in $\tilde{C}$. Therefore, our SISS-Lyapunov function must be greater or equal to 0 $\forall x \in \tilde{C}$ with equality iff $x \in \mathcal{A} \cap \tilde{C}$.

If $\tilde{C}$ is bounded, then it is also compact. We can always build a bigger set $\tilde{B}$ such that $\tilde{C} \subset \tilde{B} \subset \mathcal{X}$ and $\tilde{B}$ is still compact: $\forall x \in \tilde{C}$, take δ_x st $B(x; \delta_x) \subset \mathcal{X}$. Clearly, $\tilde{C} \subset \bigcup_{x \in \tilde{C}} B(x; \delta_x)$. As $\tilde{C}$ is compact, there exists a finite cover $\{1, \dots, s\}$ such that $x_i \in \tilde{C} \forall i \in \{1, \dots, s\}$ and $\tilde{C} \subset \bigcup_{i=1}^s \bar{B}(x_i; \delta_{x_i}) =: \tilde{B}$.

Now, we build $B_0 = (\tilde{B} \setminus \text{int}(C)) \cup (C \cap \tilde{C})$. If we subtract an open set ($\text{int}(C)$) from a closed set ($\tilde{B}$), the result is still closed. The intersection of closed sets is closed, and the finite union of closed sets is closed too. Therefore, B_0 is closed. As it is clearly bounded, we conclude B_0 is compact.

Now, $\forall y \in \partial C \cap \partial B_0$, take δ_y such that $B(y; \delta_y) \subset \mathcal{X}$. Clearly $\partial C \cap \partial B_0 \subset \bigcup_{y \in \partial C \cap \partial B_0} B(y; \delta_y)$. Notice that $\partial C \cap \partial B_0$ is compact, so $\exists$ finite cover $\{1, \dots, v\}$ such that $y_i \in \partial C \cap \partial B_0 \forall i \in \{1, \dots, v\}$ and $\partial C \cap \partial B_0 \subset \bigcup_{i=1}^v B(y_i; \delta_{y_i}) =: D$.

Now, we take $B := (B_0 \setminus D) \cup \tilde{C}$. Before, we had $B_0 \cap (C \setminus \tilde{C}) = \partial C \cap \partial B_0$, but with the new set, $B \cap (C \setminus \tilde{C}) = \emptyset$. Notice that B is a compact (it is still bounded and it is closed because the result of subtracting an open set (D) to a closed set (B_0) and adding another closed set ($\tilde{C}$)). We will use this set to define an exact penalty function. We proceed as follows:

Consider the optimization problem $\min_{B \cap C} f(x) - f^* \quad (\tilde{P})$. The MFCQ conditions are fulfilled $\forall x \in \mathcal{A} \cap \tilde{C}$, which are all the minima in problem $(\tilde{P})$. [38, Theorem 4(a)] states that under this conditions $V(x; \epsilon)$ is a weakly exact penalty function. This is, $\exists \epsilon_1$ such that the global minima of $V(x; \epsilon)$ in $\text{int}(B)$ and the global minima of $(\tilde{P})$ coincide $\forall \epsilon < \epsilon_1$. As the global minima of problem $(\tilde{P})$ are $\mathcal{A} \cap \tilde{C}$, $V(x; \epsilon)$ is a generalized SISS-Lyapunov function candidate for the flow $\tilde{\mathcal{G}}_\alpha(x)$ on $(\tilde{C}, \mathcal{A})$.

$V(x; \epsilon)$ is greater or equal to zero in B : its global minima are zeros, and they are reached iff $x \in \tilde{C} \cap \mathcal{A}$ (because its global minima are those of problem $(\tilde{P})$). Also, its sub-level sets are bounded. By Lemma 3.2.1 we conclude it is a set size function for $(\tilde{C}, \mathcal{A})$.

Now, note that we can replicate the results in Lemma 4.1.7: $\exists \epsilon_2 > 0$ st $\forall \epsilon < \epsilon_2$:

$$\max_{\tilde{L}_{\tilde{\mathcal{G}}_\alpha(x)}} V(x; \epsilon) \leq -\|G_\alpha(x)\|^2 + \gamma(\|d(x)\|) \quad \forall x \in B$$

To do this, take into account B is bounded and therefore $\|\nabla g_i(x)\|$, $\|\nabla h_j(x)\|$ and $\|\nabla f(x)\|$ are bounded on B . Take $M_g = \max_{i \in \{1, \dots, m\}} M_{g_i}$ (where $M_{g_i} = \max_{x \in B} \|\nabla g_i(x)\|$), $M_h = \max_{j \in \{1, \dots, k\}} M_{h_j}$ (where $M_{h_j} = \max_{x \in B} \|\nabla h_j(x)\|$) and $M_f = \max_{x \in B} \|\nabla f(x)\|$. Then, the class $\mathcal{K}_\infty$ function $\gamma(r) = (M_g + M_h + M_f) \cdot r$ together with the knowledge that $\tilde{C} \subset \mathcal{X}$ gives us the proposed bound. Note that we can reason as in Lemma 4.1.6 to get that $\|G_\alpha(x)\|$ is a weak set size function for $(\tilde{C}, \mathcal{A})$ (in fact, it is for the more general pair $(B, \mathcal{A})$).

We conclude that $V(x; \epsilon)$ is a generalized SISS-Lyapunov function for the flow $\tilde{\mathcal{G}}_\alpha(x)$ on $(\tilde{C}, \mathcal{A})$ $\forall \epsilon < \epsilon^* = \min\{\epsilon_1, \epsilon_2\}$ $\square$

Proposition 4.2.4. *If $\tilde{C}$ is bounded, $\tilde{C} \subset \mathcal{X}$ and MFCQ conditions hold $\forall x \in \tilde{C}$, then the SGF with the erroneous safe set $\tilde{C}$ is SISSp on $(\tilde{C}, \mathcal{A})$.*

Proof. The previous Lemma gives us a generalized SISS-Lyapunov function for this system under these conditions. As the safe gradient flow design renders $\tilde{C}$ forward invariant, Proposition 3.2.9 ensures the flow $\tilde{\mathcal{G}}_\alpha(x)$ is weak SISS on $(\tilde{C}, \mathcal{A})$.

Observe that we implicitly have a small gain condition, as we are asking $\tilde{C}$ to be included in $\mathcal{X}$. $\square$

Remark 4.2.5 (Errors in the controllers u, v). When there are errors in the controllers u and v , we can no longer guarantee the safety of the set C (or $\tilde{C}$ if there are errors in g and h too). Therefore, in the event of faulty controllers we need to rescind ourselves to the general case studied in Proposition $\bullet$

4.3 Discretization

In this section, we prove that under certain conditions the discretized safe gradient flow is weak small-gain SISS on C , C_c or an infinitesimally expanded version of those sets, depending on how restrictive our conditions are.

Proposition 4.3.1. *Given $\alpha \in (0, 1)$, $\xi \in C$, **either**:*

- $g(\xi) < 0$

- g linear

and **either**

- no equality constraints
- h linear

there exists a sequence $(h_t^\alpha)_{t \in \mathbb{N} \cup 0}$ such that $\forall (h_t)_{t \in \mathbb{N} \cup 0}$ st $h_t \leq h_t^\alpha \ \forall t$, the **Forward Euler Discretization of the SGF**, $x_{t+1} = x_t + h_t \cdot \mathcal{G}_\alpha(x_t)$ renders the safe set C forward invariant.

Proof. We will see that under these conditions the pair (g, h) is a DVCBF. Given $i \in \{1, \dots, m\}$, using Taylor's expansion:

$$g_i(x_1) = g_i(\xi + h \cdot \mathcal{G}_\alpha(\xi)) = g_i(\xi) + h \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2)$$

As $\xi \in \mathcal{X}$, we can use the relations in 1.10 to get that $h \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) \leq -h \cdot \alpha \cdot g_i(\xi)$. Therefore,

$$g_i(\xi) + h \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq g_i(\xi) - h \cdot \alpha \cdot g_i(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2)$$

and

$$\begin{aligned} g_i(\xi) - h \cdot \alpha \cdot g_i(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\leq (1 - \alpha)g_i(\xi) \\ \iff \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\leq -(1 - h) \cdot \alpha \cdot g_i(\xi) \end{aligned}$$

If $g(\xi) < 0$, taking $h \in (0, 1)$, $-(1 - h) \cdot \alpha \cdot g_i(\xi) > 0$ and also $-(1 - h) \cdot \alpha \cdot g_i(\xi) \rightarrow -\alpha \cdot g_i(\xi) > 0$ as $h \rightarrow 0$, while $\mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \rightarrow 0$ as $h \rightarrow 0$. Therefore, $\exists h_0^\alpha$ st $\forall h \leq h_0^\alpha$, $\mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq -(1 - h) \cdot \alpha \cdot g_i(\xi) \Rightarrow g_i(x_0) \leq (1 - \alpha)g_i(\xi)$.

If g is linear, then $\mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) = 0$ and we only need $h < 1$.

If there are no equality constraints, we can see x_1 is still in C .

If h is linear, via Taylor's expansion we will get

$$h_j(x_1) = h_j(\xi) + h \cdot \nabla h_j(\xi)^t \cdot \mathcal{G}_\alpha(x)$$

Using again the relations in 1.10 we see that $h \cdot \nabla h_j(\xi)^t \cdot \mathcal{G}_\alpha(x) = -h \cdot \alpha \cdot h_j(\xi)$, so:

$$\begin{aligned} h_j(x_1) &= h_j(\xi) - h \cdot \alpha \cdot h_j(\xi) = (1 - \alpha)h_j(\xi) \\ -h \cdot \alpha \cdot h_j(\xi) &= -\alpha h_j(\xi) \end{aligned}$$

which is true because $\xi \in C$. This ensures $x_1 \in C$ too.

We can repeat this same procedure recursively to find a sequence $(h_t^\alpha)_{t \in \mathbb{N} \cup 0}$ that renders the safe set C forward invariant by Proposition 1.1.24

□

Remark 4.3.2. If g is linear we can discretize with a constant step h that ensures forward invariance of C .

Corollary 4.3.3. *Given $\alpha \in (0, 1)$, $\xi \in C$, if **either** $g(\xi) < 0$ or g linear and **either** there are no equality constraints or h is linear; then $\exists h_\alpha$ such that $\forall h \leq h_\alpha$, the **finite Forward Euler Discretization of the SGF**, $x_{t+1} = x_t + h \cdot \mathcal{G}_\alpha(x_t)$ with $t \in \mathbb{N}_{\leq T}$, renders the safe set C forward invariant.*

Proof. We proceed as in the previous proposition and just choose $h_\alpha = \min_{t \in \{0, \dots, T\}} h_t^\alpha$. $\square$

Proposition 4.3.4. $\forall \epsilon > 0$ such that the (set of) maximal path connected components of the set

$$\{x \in \mathbb{R}^n \mid g(x) - \epsilon \cdot \mathbf{1} < 0; h(x) - \epsilon \cdot \mathbf{1} < 0; h(x) + \epsilon \cdot \mathbf{1} > 0\} \subset \mathcal{X},$$

that contain C - which we call C^ϵ - is included in $\mathcal{X}$, there exists $(h_t^\alpha)_{t \in \mathbb{N} \cup 0} < (1)_{t \in \mathbb{N} \cup 0}$ such that for all $(h_t)_{t \in \mathbb{N} \cup 0} \leq (h_t^\alpha)_{t \in \mathbb{N} \cup 0}$, C^ϵ is **forward invariant** under the **Forward Euler Discretization of the SGF**, $x_{t+1} = x_t + h_t \cdot \mathcal{G}_\alpha(x_t)$, when $\xi \in C^\epsilon$ and $\alpha \in (0, 1)$.

Proof. Given $\epsilon > 0$ and $\xi \in C^\epsilon$, we can write the DVCBF conditions as:

$$\begin{aligned} g_i(x_{t+1}) - g_i(x_t) + \alpha \cdot g_i(x_t) - \alpha \cdot \epsilon &\leq 0 \iff \\ g_i(x_{t+1}) &\leq (1 - \alpha)g_i(x_t) + \alpha \cdot \epsilon \quad \forall i \in \{1, \dots, m\} \end{aligned}$$

$$\begin{aligned} h_j(x_{t+1}) - h_j(x_t) + \alpha \cdot h_j(x_t) - \alpha \cdot \epsilon &\leq 0 \iff \\ h_j(x_{t+1}) &\leq (1 - \alpha)h_j(x_t) + \alpha \cdot \epsilon \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

$$\begin{aligned} -h_j(x_{t+1}) + h_j(x_t) - \alpha \cdot h_j(x_t) - \alpha \cdot \epsilon &\leq 0 \iff \\ h_j(x_{t+1}) &\geq (1 - \alpha)h_j(x_t) - \alpha \cdot \epsilon \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

Given $i \in \{1, \dots, m\}$, we perform Taylor's expansion as in the previous propositions, and as $\xi \in C^\epsilon \subset \mathcal{X}$:

$$g(x_1) = g_i(\xi) + h \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq g_i(\xi) - h \cdot \alpha \cdot g_i(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2)$$

and

$$\begin{aligned} g_i(\xi) - h \cdot \alpha \cdot g_i(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\leq (1 - \alpha)g_i(x_t) + \alpha \cdot \epsilon \\ \iff \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\leq -(1 - h) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \epsilon \end{aligned}$$

As $h^2 \|\mathcal{G}_\alpha(\xi)\|^2 \rightarrow 0$ as $h \rightarrow 0$ and $-(1 - h) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \epsilon \rightarrow -\alpha \cdot g_i(\xi) + \alpha \cdot \epsilon > 0$ as $h \rightarrow 0$ (h must be smaller than 1), $\exists h_0^g < 1$ such that $\forall h_0 \leq h_0^g$ the inequality is fulfilled. The smaller we want ϵ to be, the bigger h_0^g will have to be.

Also, given $j \in \{1, \dots, k\}$, we perform Taylor's expansion as in the previous propositions, and as $\xi \in C^\epsilon \subset \mathcal{X}$:

$$h_j(x_1) = h_j(\xi) + h \cdot \nabla h_j(\xi)^t \cdot \mathcal{G}_\alpha(x) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) = h_j(\xi) - h \cdot \alpha \cdot h_j(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2)$$

and

$$h_j(\xi) - h \cdot \alpha \cdot h_j(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq (1 - \alpha) \cdot h_j(x_t) + \alpha \cdot \epsilon$$

$$\iff \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq -(1-h) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \epsilon$$

As $h^2 \|\mathcal{G}_\alpha(\xi)\|^2 \rightarrow 0$ as $h \rightarrow 0$ and $-(1-h) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \epsilon \rightarrow -\alpha \cdot h_j(\xi) + \alpha \cdot \epsilon > 0$ as $h \rightarrow 0$ (h must be smaller than 1), $\exists h_0^+ < 1$ such that $\forall h_0 \leq h_0^+$ the inequality is fulfilled.

Finally we repeat this process for the other inequality involving h :

$$\begin{aligned} h_j(\xi) - h \cdot \alpha \cdot h_j(\xi) + \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\geq (1-h) \cdot h_j(\xi) - \alpha \cdot \epsilon \\ \iff \mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) &\geq -(1-h) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \epsilon \end{aligned}$$

As $h^2 \|\mathcal{G}_\alpha(\xi)\|^2 \rightarrow 0$ as $h \rightarrow 0$ and $-(1-h) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \epsilon \rightarrow -\alpha \cdot h_j(\xi) - \alpha \cdot \epsilon < 0$ as $h \rightarrow 0$ (h must be smaller than 1), $\exists h_0^- < 1$ such that $\forall h_0 \leq h_0^-$ the inequality is fulfilled.

If we take $h_0^\alpha := \min\{h_0^g, h_0^+, h_0^-, \}$, we are guaranteeing that $x_1 \in C^\epsilon$. Reasoning inductively just as in the previous propositions, we get a sequence of steps $(h_t^\alpha)_{t \in \mathbb{N} \cup 0}$ that renders C^ϵ forward invariant. □

Corollary 4.3.5. $\forall \epsilon > 0$ such that the set $C^\epsilon \subset \mathcal{X}$, $\exists h^\alpha < 1$ such that $\forall h \leq h^\alpha$, C^ϵ is **forward invariant** under the **finite Forward Euler Discretization of the SGF**, $x_{t+1} = x_t + h_t \cdot \mathcal{G}_\alpha(x_t)$ with $t \in \mathbb{N}_{\leq T}$, when $\xi \in C^\epsilon$ and $\alpha \in (0, 1)$

Proof. We reason as in the previous proposition and take $h^\alpha := \min_{t \in \{1, \dots, T\}} h_t^\alpha$. □

Remark 4.3.6. The results in Propositions 4.3.1 and 4.3.4 can also be applied to the perturbed Safe Gradient Flow:

- If there only are errors in the objective function, the discretized perturbed SGF still renders C forward invariant under the conditions in Proposition 4.3.1 and C^ϵ forward invariant under the conditions in Proposition 4.3.4.
- If there only are errors in the safe set (and the objective function), the discretized perturbed SGF still renders $\tilde{C}$ forward invariant under the conditions in Proposition 4.3.1 and $\tilde{C}^\epsilon$ forward invariant under the conditions in Proposition 4.3.4.

Remark 4.3.7. Using the Remainder's Theorem, we can get a lower bound for h_t^α at each time-step $t \in \{0, \dots, T\}$ •

Remark 4.3.8. This explains the observation 5.6 in [Ahmed & Jorge]: the bigger α , the smaller h^α will need to be to maintain (approximate) safety of the constraints set C . •

As $\mathcal{G}_\alpha(x)$ points towards C for all $x \in C$ by design (because $\mathcal{G}_\alpha$ is designed to fulfill the conditions of Nagumo's Theorem for Set Invariance, which demand the vector field to point towards the set C for every point of the set), for any convergent discretization scheme it is intuitive that at every time-step we should be able to find a small enough step-size such that the set C is still rendered forward invariant. Let us formalize that for one of the most popular discretization methods, the RK4, which will be our method of choice to perform simulations with the Safe Gradient Flow. Please note that both the procedure used here and the results obtained are in fact valid for any Runge-Kutta method.

Proposition 4.3.9 (Forward Invariance of C under RK45 method). *Given $\alpha \in (0, 1)$, $\xi \in C$ such that $g(\xi) < 0$ and h is linear (or there are no equality constraints), there exists a sequence $(h_t^\alpha)_{t \in \mathbb{N} \cup 0}$ such that $\forall (h_t)_{t \in \mathbb{N} \cup 0}$ st $h_t \leq h_t^\alpha \forall t$, the **RK45 Discretization of the SGF** renders the safe set C forward invariant.*

Proof. Recall that the RK45 discretization is the following: given x_t ,

$$x_{t+1} = x_t + \sum_{r=1}^6 \hat{c}_r \cdot k_r \quad (4.11)$$

where:

$$k_1 = h_t \cdot G_\alpha(x_t) \quad (4.12)$$

$$k_2 = h_t \cdot G_\alpha(x_t + \beta_{2,1} \cdot k_1) \quad (4.13)$$

$$k_3 = h_t \cdot G_\alpha(x_t + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2) \quad (4.14)$$

$$k_4 = h_t \cdot G_\alpha(x_t + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3} \cdot k_3) \quad (4.15)$$

$$k_5 = h_t \cdot G_\alpha(x_t + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4) \quad (4.16)$$

$$k_6 = h_t \cdot G_\alpha(x_t + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5) \quad (4.17)$$

where $\hat{c}_r$ and $\beta_{k,l}$ are constants for $r, k \in \{1, \dots, 6\}$ and $l \in \{1, \dots, k-1\}$. At each time-step, h_t might need to be updated and the calculations repeated with the new step-size to ensure that the truncated error is smaller than a given tolerance.

Reasoning by induction, we see that under these conditions we can adequately adjust h_t at each time-step so that the pair (g, h) is a DVCBF.

Let $H_j := \{x \in \mathbb{R}^n \mid h_j(x) = 0\}$ for every $j \in \{1, \dots, k\}$. If h is linear, then for each $j \in \{1, \dots, k\}$ we have $\nabla h_j(x) = \nabla h_j \forall x \in \mathbb{R}^n$. We can see that for every $x \in H_j \cap \mathcal{X}$, the tangent cone to the set H_j at x , $T_x H_j$, is the hyperplane $\{v \in \mathbb{R}^n \mid (\nabla h_j)^t \cdot v = 0\}$. Notice $T_x H_j = T(H_j) \forall x \in H_j \cap \mathcal{X}$. We know that $\mathcal{G}_\alpha(x) \in T_x C \forall x \in C$. As $C \subseteq H_j$, $T_x C \subseteq T_x H_j$ and thus $\lambda \cdot \mathcal{G}_\alpha(x) \in T_x H_j = T(H_j) \forall x \in C$ and $\forall \lambda \in \mathbb{R}$. Notice $G := \{x \in \mathcal{X} \mid g(x) < 0\}$ is an open set.

Notice that in this step $k_1 = h_0 \cdot \mathcal{G}_\alpha(\xi)$. There exists $h_0^{\alpha,1}$ such that $\forall h_0 \leq h_0^\alpha$ we have $\xi + \beta_{l,1} \cdot k_1 \in G \forall l \in \{2, \dots, 6\}$ (this holds because G is an open set). Then, $\xi + \beta_{2,1} \cdot k_1 \in G \subseteq \mathcal{X}$ and thus $\mathcal{G}_\alpha(\xi + \beta_{2,1} \cdot k_1)$ is well defined. Also, note that for every $j \in \{1, \dots, k\}$ it holds $h_j(\xi + \beta_{l,1} \cdot k_1) = h_j(\xi) + \beta_{l,1} \cdot h_0 \cdot \nabla h_j^t \cdot \mathcal{G}_\alpha(\xi) = \beta_{l,1} \cdot h_0 \cdot \nabla h_j^t \cdot \mathcal{G}_\alpha(\xi) = 0 \forall l \in \{2, \dots, 6\}$ because $\mathcal{G}_\alpha(\xi) \in T(H_j)$. We conclude that $\xi + \beta_{l,1} \cdot k_1 \in H_j \forall l \in \{2, \dots, 6\}$ and every $j \in \{1, \dots, k\}$.

Again, note that in this step $k_2 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{2,1} \cdot k_1)$. Note that $h_0^{\alpha,1}$ guarantees $\xi + \beta_{l,1} \cdot k_1 \in G \forall l \in \{2, \dots, 6\}$, so there exists $h_0^{\alpha,2} \leq h_0^{\alpha,1}$ such that $\forall h_0 \leq h_0^{\alpha,2}$ $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 \in G \forall l \in \{3, \dots, 6\}$. In particular, this guarantees that $\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2 \in G$ and thus $\mathcal{G}_\alpha(\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2)$ is well defined. Note also that for any $j \in \{1, \dots, k\}$ it holds $\xi + \beta_{l,1} \cdot k_1 \in H_j$. As $h_j(\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2) = h_j(\xi + \beta_{l,1} \cdot k_1) + \beta_{l,2} \cdot \nabla h_j^t \cdot k_2 = \beta_{l,2} \cdot \nabla h_j^t \cdot k_2 = 0 \forall l \in \{3, \dots, 6\}$ because $k_2 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{l,1} \cdot k_1) \in T(H_j)$ and so $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 \in H_j \forall l \in \{3, \dots, 6\}$.

In this step, $k_3 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2)$. Note that $h_0^{\alpha,2}$ guarantees $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 \in G \forall l \in \{3, \dots, 6\}$, so there exists $h_0^{\alpha,3} \leq h_0^{\alpha,2}$ such that $\forall h_0 \leq h_0^{\alpha,3}$ $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in \mathcal{X} \forall l \in \{4, \dots, 6\}$. In particular, this guarantees $\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3} \cdot k_3 \in G$ and thus

$\mathcal{G}_\alpha(\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3})$ is well defined. What is more, for any $j \in \{1, \dots, k\}$ it holds $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 \in H_j \ \forall l \in \{3, \dots, 6\}$. We can write $h_j(\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3) = h_j(\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2) + \beta_{l,3} \cdot \nabla h_j^t \cdot k_3 = \beta_{l,3} \cdot \nabla h_j^t \cdot k_3 = 0$ because $k_3 \in T(H_j)$ and therefore $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in H_j \ \forall l \in \{4, 5, 6\}$ and $\forall j \in \{1, \dots, k\}$.

Following the same procedure, in this step $k_4 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3})$. As the bound $h_0^{\alpha,3}$ guarantees that $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in G \ \forall l \in \{4, \dots, 6\}$, there exists $h_0^{\alpha,4}$ such that $\forall h \leq h_0^{\alpha,4} \ \xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 + \beta_{l,4} \cdot k_4 \in G \ \forall l \in \{5, 6\}$. In particular, this guarantees $\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4 \in G$ and thus $\mathcal{G}_\alpha(\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4)$ is well defined. For any $j \in \{1, \dots, k\}$ it holds $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in H_j \ \forall l \in \{4, 5, 6\}$. We can write $h_j(\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 + \beta_{l,4} \cdot k_4) = h_j(\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3) + \beta_{l,4} \cdot \nabla h_j^t \cdot k_4 = \beta_{l,4} \cdot \nabla h_j^t \cdot k_4 = 0$ because $k_4 \in T(H_j)$. Thus, $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 + \beta_{l,4} \cdot k_4 \in H_j \ \forall l \in \{5, 6\}$ and $\forall j \in \{1, \dots, k\}$.

Finally, $k_5 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4)$. As the bound $h_0^{\alpha,4}$ guarantees that $\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 \in G$, there exists $h_0^{\alpha,5}$ such that $\forall h_0 \leq h_0^{\alpha,5}$ we have $\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5 \in G$. This guarantees $\mathcal{G}_\alpha(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5)$ is well defined, and so is $k_6 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5)$. For any $j \in \{1, \dots, k\}$ it holds $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 + \beta_{l,4} \cdot k_4 \in H_j$ for $l = \{5, 6\}$. We can write $h_j(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5) = h_j(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4) + \beta_{6,5} \cdot \nabla h_j^t \cdot k_5 = \beta_{6,5} \cdot \nabla h_j^t \cdot k_5 = 0$ because $k_5 \in T(H_j)$. We conclude $\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5 \in H_j \ \forall j \in \{1, \dots, k\}$.

Now, take $i \in \{1, \dots, m\}$ and use Taylor's expansion on $g_i(x_1)$:

$$g_i(x_1) = g_i(\xi + \sum_{r=1}^6 \hat{c}_r \cdot k_r) = g_i(\xi) + \nabla g_i(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2)$$

We know that $k_r \rightarrow 0$ as $h_0 \rightarrow 0 \ \forall r \in \{1, \dots, 6\}$, so Let us separate the term $\nabla g_i(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r)$ in $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1$ and $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r)$. Now, notice that $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1 = h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi)$. As $\xi \in \mathcal{X}$, we can use the relations in 1.10 to get that $h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) \leq -h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi)$.

Therefore, for the DVCBF condition to be satisfied on g_i it is enough to see that:

$$\begin{aligned} g_i(\xi) - h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi) + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) &\leq (1 - \alpha) g_i(\xi) \\ \iff \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) &\leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \end{aligned}$$

We know that $k_r \rightarrow 0$ as $h_0 \rightarrow 0 \ \forall r \in \{1, \dots, 6\}$, so $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \rightarrow 0$ as $h_0 \rightarrow 0$.

As $g(\xi) < 0$, if $h_0 \leq \frac{1}{\hat{c}_1}$ then $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \geq 0$ and also $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \rightarrow -\alpha \cdot g_i(\xi) > 0$ as $h \rightarrow 0$. Therefore, $\exists h_0^{\alpha,5} \leq \min\{h_0^{\alpha,5}, \frac{1}{\hat{c}_1}\}$ st $\forall h_0 \leq h_0^{\alpha,5}$, $\mathcal{O}(h^2 \|\mathcal{G}_\alpha(\xi)\|^2) \leq -(1 - h) \cdot \alpha \cdot g_i(\xi) \Rightarrow g_i(x_0) \leq (1 - \alpha) g_i(\xi)$. Note that $g_i(x_1) \leq (1 - \alpha) g_i(\xi) < 0$ guarantee that the DVCBF condition holds for g_i no matter the index i (and thus for g).

The DVCBF conditions on h are:

$$h_j(x_1) = (1 - \alpha) h_j(\xi) \iff h_j(x_1) = 0 \ \forall j \in \{1, \dots, k\}$$

Note that $h_j(x_1) = h_j(\xi) + \nabla h_j^t \cdot \sum_{r=1}^6 \hat{c}_r \cdot k_r = \nabla h_j^t \cdot \sum_{r=1}^6 \hat{c}_r \cdot k_r$. The term $\sum_{r=1}^6 \hat{c}_r \cdot k_r$ is a linear combination of elements of $T(H_j)$, and thus we get that $\sum_{r=1}^6 \hat{c}_r \cdot k_r \in T(H_j)$, which yields $h_j(x_1) = 0$ (and this holds for all $j \in \{1, \dots, k\}$).

Finally, we compute TE and see if it is smaller than the tolerance. Was it not, we choose a smaller step size and repeat the calculations. Note that if this happens we do not need to check again that the interpolating points belong to $\mathcal{X}$ nor that the DVCBF conditions hold, as we have guaranteed these two conditions are satisfied for every $h_0 < h_0^\alpha$. Once we got a TE smaller than the tolerance, we simply compute the next step size candidate as $h_1 = (\frac{\text{tol}}{2 \cdot \text{TE}})^{\frac{1}{4}}$ (where tol is a fixed tolerance and TE is the truncation error, see [26] for details).

We can now repeat this same procedure for every time step to inductively find a sequence $(h_t^\alpha)_{t \in \mathbb{N} \cup 0}$ that renders the safe set C forward invariant by Proposition 1.1.24. $\square$

Corollary 4.3.10. *Under the conditions of Proposition 4.3.9, there exists h_α such that $\forall h \leq h^\alpha$, the **finite RK45 Discretization of the SGF** with $t \in \mathbb{N}_{\leq T}$, renders the safe set C forward invariant.*

Proof. We proceed as in Proposition 4.3.9 and just choose $h^\alpha = \min_{t \in \{0, \dots, T\}} h_t^\alpha$. $\square$

Remark 4.3.11. We need to require $g(\xi) < 0$ even if g is linear. If we simply ask for $\xi \in C$, let $\xi \in C \subset \mathcal{X}$. Let $G_i := \{x \in \mathbb{R}^n \mid g_i(x) \leq 0\}$ for every $i \in \{1, \dots, m\}$. For each $\hat{i} \in \{1, \dots, m\}$ such that $g_{\hat{i}}$ is linear (this is, $\nabla g_{\hat{i}}(x) = \nabla g_{\hat{i}} \in \mathbb{R}^n \ \forall x \in \mathbb{R}^n$), we can see that for every $x \in \partial G_{\hat{i}} \cap \mathcal{X}$, the tangent cone to the set $G_{\hat{i}}$ at x , $T_x G_{\hat{i}}$, is the half-space $\{v \in \mathbb{R}^n \mid (\nabla g_{\hat{i}})^t \cdot v \leq 0\}$. We know that $\mathcal{G}_\alpha(x) \in T_x C \ \forall x \in C$. As $C \subseteq G_{\hat{i}}$, $T_x C \subseteq T_x G_{\hat{i}}$ and thus $\mathcal{G}_\alpha(x) \in T_x G_{\hat{i}} \ \forall x \in C$. In particular, this means that given $x \in C \cap \partial G_{\hat{i}}$ we have that $\lambda \cdot \mathcal{G}_\alpha(x) \in T_x G_{\hat{i}} \ \forall \lambda \geq 0$. However, the beta coefficients are not all positive, so the RK45 method might exit the set $G_{\hat{i}}$ (and thus C). •

Remark 4.3.12. We can implement the RK45 method to execute the Safe Gradient Flow in a computationally efficient way, as it is easy to check whether the chosen step size guarantees the interpolation points belong to G or not.

In more general settings we will not be able to do this, so it is more convenient to simply fix an arbitrarily small "maximum step size" and hope that for every time step t and every step size smaller than the minimum step size it holds $x_t + \sum_{l=1}^{r-1} \beta_{r,l} \cdot k_l \in \mathcal{X}$. •

Proposition 4.3.13 (Forward Invariance of C^ϵ under RK45 discretization). $\forall \epsilon > 0$ such that the set C^ϵ is contained in $\mathcal{X}$, there exists $(h_t^\alpha)_{t \in \mathbb{N} \cup 0} < (1)_{t \in \mathbb{N} \cup 0}$ such that for all $(h_t)_{t \in \mathbb{N} \cup 0} \leq (h_t^\alpha)_{t \in \mathbb{N} \cup 0}$, C^ϵ is **forward invariant** under the **RK45 Discretization of the SGF** when $\xi \in C^\epsilon$ and $\alpha \in (0, 1)$.

Proof. Let $\xi \in C^\epsilon \subset \mathcal{X}$. Notice that in this step $k_1 = h_0 \cdot \mathcal{G}_\alpha(\xi)$. There exists $h_0^{\alpha,1}$ such that $\forall h_0 \leq h_0^\alpha$ we have $\xi + \beta_{l,1} \cdot k_1 \in \mathcal{X} \ \forall l \in \{2, \dots, 6\}$ (this holds because $\mathcal{X}$ is an open set). Then, $\xi + \beta_{2,1} \cdot k_1 \in \mathcal{X}$ and thus $\mathcal{G}_\alpha(\xi + \beta_{2,1} \cdot k_1)$ is well defined.

Again, note that in this step $k_2 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{2,1} \cdot k_1)$. Note that $h_0^{\alpha,1}$ guarantees $\xi + \beta_{l,1} \cdot k_1 \in \mathcal{X} \ \forall l \in \{2, \dots, 6\}$, so there exists $h_0^{\alpha,2} \leq h_0^{\alpha,1}$ such that $\forall h_0 \leq h_0^{\alpha,2} \ \xi + \beta_{l,1} \cdot k_1 +$

$\beta_{l,2} \cdot k_2 \in \mathcal{X} \ \forall l \in \{3, \dots, 6\}$. In particular, this guarantees that $\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2 \in \mathcal{X}$ and thus $\mathcal{G}_\alpha(\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2)$ is well defined.

In this step, $k_3 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{3,1} \cdot k_1 + \beta_{3,2} \cdot k_2)$. Note that $h_0^{\alpha,2}$ guarantees $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 \in \mathcal{X} \ \forall l \in \{3, \dots, 6\}$, so there exists $h_0^{\alpha,3}$ such that $\forall h_0 \leq h_0^{\alpha,3} \ \xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in \mathcal{X} \ \forall l \in \{4, \dots, 6\}$. In particular, this guarantees $\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3} \cdot k_3 \in \mathcal{X}$ and thus $\mathcal{G}_\alpha(\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3} \cdot k_3)$ is well defined.

Following the same procedure, in this step $k_4 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{4,1} \cdot k_1 + \beta_{4,2} \cdot k_2 + \beta_{4,3} \cdot k_3)$. As the bound $h_0^{\alpha,3}$ guarantees that $\xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 \in \mathcal{X} \ \forall l \in \{4, \dots, 6\}$, there exists $h_0^{\alpha,4}$ such that $\forall h \leq h_0^{\alpha,4} \ \xi + \beta_{l,1} \cdot k_1 + \beta_{l,2} \cdot k_2 + \beta_{l,3} \cdot k_3 + \beta_{l,4} \cdot k_4 \in \mathcal{X} \ \forall l \in \{5, 6\}$. In particular, this guarantees $\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4 \in \mathcal{X}$ and thus $\mathcal{G}_\alpha(\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4)$ is well defined.

Finally, $k_5 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{5,1} \cdot k_1 + \beta_{5,2} \cdot k_2 + \beta_{5,3} \cdot k_3 + \beta_{5,4} \cdot k_4)$. As the bound $h_0^{\alpha,4}$ guarantees that $\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 \in \mathcal{X}$, there exists $h_0^{\alpha,5}$ such that $\forall h_0 \leq h_0^{\alpha,5}$ we have $\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5 \in \mathcal{X}$. This guarantees $\mathcal{G}_\alpha(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5)$ is well defined, and so is $k_6 = h_0 \cdot \mathcal{G}_\alpha(\xi + \beta_{6,1} \cdot k_1 + \beta_{6,2} \cdot k_2 + \beta_{6,3} \cdot k_3 + \beta_{6,4} \cdot k_4 + \beta_{6,5} \cdot k_5)$.

Now, take $i \in \{1, \dots, m\}$ and use Taylor's expansion on $g_i(x_1)$:

$$g_i(x_1) = g_i(\xi + \sum_{r=1}^6 \hat{c}_r \cdot k_r) = g_i(\xi) + \nabla g_i(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2)$$

We know that $k_r \rightarrow 0$ as $h_0 \rightarrow 0 \ \forall r \in \{1, \dots, 6\}$, so Let us separate the term $\nabla g_i(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r)$ in $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1$ and $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r)$. Now, notice that $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1 = h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi)$. As $\xi \in \mathcal{X}$, we can use the relations in 1.10 to get that $h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) \leq -h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi)$.

Therefore, for the DVCBF conditions derived in 4.3.4 to be satisfied on g_i it is enough to see that:

$$\begin{aligned} g_i(\xi) - h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi) + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) &\leq (1 - \alpha)g_i(\xi) + \alpha \cdot \epsilon \\ \iff \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) &\leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \epsilon \end{aligned}$$

We know that $k_r \rightarrow 0$ as $h_0 \rightarrow 0 \ \forall r \in \{1, \dots, 6\}$, so $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \rightarrow 0$ as $h_0 \rightarrow 0$.

If $h_0 \leq \frac{1}{\hat{c}_1}$ then $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \geq 0$ and also $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \rightarrow -\alpha \cdot g_i(\xi) \geq 0$ as $h \rightarrow 0$.

As $\alpha \cdot \epsilon > 0$, $\exists h_0^\alpha \leq \min\{h_0^{\alpha,5}, \frac{1}{\hat{c}_1}\}$ st $\forall h_0 \leq h_0^\alpha$, $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \epsilon \Rightarrow g_i(x_1) \leq (1 - \alpha)g_i(\xi) + \alpha \cdot \epsilon$ and thus the DVCBF condition holds for g_i , $\forall i \in \{1, \dots, m\}$.

Now, for any $j \in \{1, \dots, k\}$, we can reason similarly to get $h_j(x_1) = h_j(\xi) + \nabla h_j(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2)$. As $\xi \in \mathcal{X}$, we can use the relations in 1.10 to get that $h_0 \cdot \hat{c}_1 \cdot \nabla h_j(\xi)^t \cdot \mathcal{G}_\alpha(\xi) = -h_0 \cdot \hat{c}_1 \cdot \alpha \cdot h_j(\xi)$. Plugging these relationships in the DVCBF conditions for h_j and C^ϵ derived in 4.3.4, we get the inequalities:

$$(1 - \alpha) \cdot h_j(\xi) - \alpha \cdot \epsilon \leq h_j(\xi) - h_0 \cdot \hat{c}_1 \cdot \alpha \cdot h_j(\xi) + \nabla h_j(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \leq (1 - \alpha) \cdot h_j(\xi) + \alpha \cdot \epsilon$$

which can be simplified to:

$$-(1-h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \epsilon \leq \nabla h_j(\xi)^t \cdot \left(\sum_{r=2}^6 \hat{c}_r \cdot k_r \right) + \mathcal{O}(h_0^2 \left\| \sum_{r=1}^6 \hat{c}_r \cdot k_r \right\|^2) \leq -(1-h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \epsilon$$

Note that, as $h_0 \rightarrow 0$, $-(1-h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \epsilon \rightarrow -\alpha \cdot (h_j(\xi) + \epsilon) < 0$ and $-(1-h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \epsilon \rightarrow -\alpha \cdot (h_j(\xi) - \epsilon) > 0$. Therefore, we can update h_0^α so that these two inequalities hold for every $j \in \{1, \dots, k\}$.

Finally, we compute TE and see if it is smaller than the tolerance. Was it not, we choose a smaller step size and repeat the calculations. Note that if this happens we do not need to check again that the interpolating points belong to $\mathcal{X}$ nor that the DVCBF conditions hold, as we have guaranteed these two conditions are satisfied for every $h_0 \leq h_0^\alpha$ or $h_0 \leq \tilde{h}_0^\alpha$, accordingly. Once we got a TE smaller than the tolerance, we simply compute the next step size candidate as $h_1 = \left(\frac{\text{tol}}{2 \cdot \text{TE}} \right)^{\frac{1}{4}}$ (where tol is a fixed tolerance and TE is the truncation error, see [26] for details).

Inductively, we repeat the same reasoning for every time step. By Proposition 1.1.24, this concludes the proof. $\square$

Proposition 4.3.14 (Forward invariance of C_c under RK45). *Under the conditions of Proposition 4.1.1, for all $\epsilon > 0$ there exists $c > 0$ and a sequence $(h_t^\alpha)_{t \in \mathbb{N} \cup \{0\}}$ such that for every step size sequence $(h_t)_{t \in \mathbb{N} \cup \{0\}} \leq (h_t^\alpha)_{t \in \mathbb{N} \cup \{0\}}$ the inside of the ISSf set $\overset{\circ}{C}_c \subset \mathcal{X}$ is forward invariant under the **RK45** discretization of the **perturbed Safe Gradient Flow** with $\alpha \in (0, 1)$ and an additive disturbance $d \in \mathcal{M}_c$.*

What is more, for every $\tilde{\epsilon}$ such that $C_c^{\tilde{\epsilon}} \subseteq \mathcal{X}$ (built as in Proposition 4.3.13) there exists $(\tilde{h}_t^\alpha)_{t \in \mathbb{N} \cup \{0\}}$ such that for all $(h_t)_{t \in \mathbb{N} \cup \{0\}} \leq (\tilde{h}_t^\alpha)_{t \in \mathbb{N} \cup \{0\}}$ $C_c^{\tilde{\epsilon}}$ is forward invariant under the **RK45** discretization of the **perturbed Safe Gradient Flow** with $\alpha \in (0, 1)$ and an additive disturbance $d \in \mathcal{M}_c$.

Proof. Recall the definition of the ISSf set C_c (4.2) and notice that the DVCBF conditions require that for every $t \in \mathbb{N} \cup \{0\}$:

$$\begin{aligned} g_i(x_{t+1}) &\leq (1-\alpha) \cdot g_i(x_t) + \alpha \cdot \gamma(c) \quad \forall i \in \{1, \dots, m\} \\ (1-\alpha) \cdot h_j(x_t) - \alpha \cdot \gamma(c) &\leq h_j(x_{t+1}) \leq (1-\alpha) \cdot h_j(x_t) + \alpha \cdot \gamma(c) \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

Putting the previous results together with the DVCBF conditions for C^ϵ derived in 4.3.4, we can see that the DVCBF conditions for C_c^ϵ require that for every $t \in \mathbb{N} \cup \{0\}$:

$$\begin{aligned} g_i(x_{t+1}) &\leq (1-\alpha) \cdot g_i(x_t) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \quad \forall i \in \{1, \dots, m\} \\ (1-\alpha) \cdot h_j(x_t) - \alpha \cdot (\gamma(c) + \tilde{\epsilon}) &\leq h_j(x_{t+1}) \leq h_j(x_{t+1}) \leq (1-\alpha) \cdot h_j(x_t) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \quad \forall j \in \{1, \dots, k\} \end{aligned}$$

Let $d \in \mathcal{M}_c$. We will reason inductively.

Proceed as in Proposition 4.3.13 to find $h_0^{\alpha,5}$ such that for every $h_0, \tilde{h}_0 \leq h_0^{\alpha,5}$ the interpolation points of the RK45 discretization (4.11) belong to $\mathcal{X}$, and so the step is well defined.

Now, take $i \in \{1, \dots, m\}$ and use Taylor's expansion on $g_i(x_1)$:

$$g_i(x_1) = g_i(\xi) + \sum_{r=1}^6 \hat{c}_r \cdot k_r = g_i(\xi) + \nabla g_i(\xi)^t \cdot \left(\sum_{r=1}^6 \hat{c}_r \cdot k_r \right) + \mathcal{O}(h^2 \left\| \sum_{r=1}^6 \hat{c}_r \cdot k_r \right\|^2)$$

Let us separate the term $\nabla g_i(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r)$ in $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1$ and $\nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r)$. Now, notice that $\nabla g_i(\xi)^t \cdot \hat{c}_1 \cdot k_1 = h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(\xi)$ where $\tilde{\mathcal{G}}_\alpha = \mathcal{G}_\alpha + d$. As $\xi \in \mathcal{X}$ and the disturbance does not affect g nor ∇g , we can use the relations in 1.10 to get that $h_0 \cdot \hat{c}_1 \cdot \nabla g_i(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(\xi) \leq -h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi) + h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\|$.

Therefore, for the DVCBF conditions derived in 4.3.4 to be satisfied on g_i it is enough to see that:

$$\begin{aligned} g_i(\xi) - h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi) + h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \\ \leq (1 - \alpha)g_i(\xi) + \alpha \cdot \gamma(c) \\ \iff h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \\ \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \gamma(c) \end{aligned}$$

for $\overset{\circ}{C}_c$ and

$$\begin{aligned} g_i(\xi) - h_0 \cdot \hat{c}_1 \cdot \alpha \cdot g_i(\xi) + h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \\ \leq (1 - \alpha)g_i(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \\ \iff h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \\ \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \end{aligned}$$

for $C_c^{\tilde{\epsilon}}$.

We know that $k_r \rightarrow 0$ as $h_0 \rightarrow 0 \forall r \in \{1, \dots, 6\}$, so $h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \rightarrow 0$ as $h_0 \rightarrow 0$.

$-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) \rightarrow -\alpha \cdot g_i(\xi)$ as $h \rightarrow 0$, and $-\alpha \cdot (g_i(\xi) - \gamma(c)) > 0$ if $\xi \in \overset{\circ}{C}_c$ and $-\alpha \cdot (g_i(\xi) - \gamma(c) - \tilde{\epsilon}) > 0$ if $\xi \in C_c^{\tilde{\epsilon}}$.

As both $\alpha \cdot \gamma(c) > 0$ and $\alpha \cdot (\gamma(c) + \tilde{\epsilon}) > 0$, $\exists h_0^\alpha, \tilde{h}_0^\alpha \leq \min\{h_0^{\alpha,5}, \frac{1}{\tilde{c}_1}\}$ st $\forall h_0 \leq h_0^\alpha$, $h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot \gamma(c) \Rightarrow g_i(x_1) \leq (1 - \alpha)g_i(\xi) + \alpha \cdot \gamma(c)$ and $\forall h_0 \leq \tilde{h}_0^\alpha$, $h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla g_i(\xi)\| + \nabla g_i(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot g_i(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \Rightarrow g_i(x_1) \leq (1 - \alpha)g_i(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon})$ and thus the DVCBF conditions for $\overset{\circ}{C}_c$ and $C_c^{\tilde{\epsilon}}$ hold for g_i , $\forall i \in \{1, \dots, m\}$.

Now, for any $j \in \{1, \dots, k\}$, we can reason similarly to get $h_j(x_1) = h_j(\xi) + \nabla h_j(\xi)^t \cdot (\sum_{r=1}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2)$. As $\xi \in \mathcal{X}$, we can use the relations in 1.10 to get that $h_0 \cdot \hat{c}_1 \cdot \nabla h_j(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(\xi) \leq -h_0 \cdot \hat{c}_1 \cdot \alpha \cdot h_j(\xi) + h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla h_j(\xi)\|$. Plugging these relationships in the DVCBF conditions for h_j , we get the inequalities:

$$\begin{aligned} -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \gamma(c) \leq \\ h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla h_j(\xi)\| + \nabla h_j(\xi)^t \cdot (\sum_{r=2}^6 \hat{c}_r \cdot k_r) + \mathcal{O}(h_0^2 \|\sum_{r=1}^6 \hat{c}_r \cdot k_r\|^2) \\ \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \gamma(c) \end{aligned}$$

for $\overset{\circ}{C}_c$ and

$$\begin{aligned} & -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \leq \\ & h_0 \cdot \hat{c}_1 \cdot c \cdot \|\nabla h_j(\xi)\| + \nabla h_j(\xi)^t \cdot \left(\sum_{r=2}^6 \hat{c}_r \cdot k_r \right) + \mathcal{O}(h_0^2 \left\| \sum_{r=1}^6 \hat{c}_r \cdot k_r \right\|^2) \\ & \leq -(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \end{aligned}$$

for $C_c^\tilde{\epsilon}$.

Note that, as $h_0 \rightarrow 0$:

- If $\xi \in \overset{\circ}{C}_c$, $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot \gamma(c) \rightarrow -\alpha \cdot (h_j(\xi) + \gamma(c)) < 0$ and $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot \gamma(c) \rightarrow -\alpha \cdot (h_j(\xi) - \gamma(c)) > 0$
- If $\xi \in C_c^\tilde{\epsilon}$, $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) - \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \rightarrow -\alpha \cdot (h_j(\xi) + \gamma(c) + \tilde{\epsilon}) < 0$ and $-(1 - h_0 \cdot \hat{c}_1) \cdot \alpha \cdot h_j(\xi) + \alpha \cdot (\gamma(c) + \tilde{\epsilon}) \rightarrow -\alpha \cdot (h_j(\xi) - \gamma(c) - \tilde{\epsilon}) > 0$

Therefore, we can update h_0^α and $\tilde{h}_0^\alpha$ so that these two inequalities hold for every $h_0 \leq h_0^\alpha$ or $h_0 \leq \tilde{h}_0^\alpha$ and for all $j \in \{1, \dots, k\}$.

Finally, we compute TE and see if it is smaller than the tolerance. Was it not, we choose a smaller step size and repeat the calculations. Note that if this happens we do not need to check again that the interpolating points belong to $\mathcal{X}$ nor that the DVCBF conditions hold, as we have guaranteed these two conditions are satisfied for every $h_0 < h_0^\alpha$. Once we got a TE smaller than the tolerance, we simply compute the next step size candidate as $h_1 = (\frac{\text{tol}}{2 \cdot \text{TE}})^{\frac{1}{4}}$ (where tol is a fixed tolerance and TE is the truncation error, see [26] for details).

Inductively, we repeat the same reasoning for every time step. By Proposition 1.1.24, this concludes the proof. $\square$

Corollary 4.3.15. *Under the conditions of Proposition 4.3.14, there exists $h^\alpha, \tilde{h}^\alpha$ such that $\forall h \leq h^\alpha$ the **finite RK45 Discretization of the SGF** with $t \in \mathbb{N}_{\leq T}$ renders the $\overset{\circ}{C}_c$ forward invariant and $\forall h \leq \tilde{h}^\alpha$ it renders $C_c^\tilde{\epsilon}$ forward invariant.*

Proof. We proceed as in Proposition 4.3.9 and just choose $h^\alpha = \min_{t \in \{0, \dots, T\}} h_t^\alpha$ and $\tilde{h}^\alpha = \min_{t \in \{0, \dots, T\}} \tilde{h}_t^\alpha$ $\square$

Remark 4.3.16. The inequality $\nabla g_i(x)^t \cdot \tilde{\mathcal{G}}_\alpha(x) \leq -\alpha \cdot g_i(x) + c \cdot \|\nabla g_i(x)\|$ is true no matter if we consider the error to be only in the controllers and/or the safe set (in that case, we ideally have $\nabla g_i(\xi)^t \cdot \mathcal{G}_\alpha(\xi) \leq -\alpha \cdot g_i(\xi)$ but an erroneous $\tilde{\mathcal{G}}_\alpha$, so we have to bound $\nabla g_i(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(\xi)$), only on f (in that case, we have $\nabla g_i(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(\xi) \leq -\alpha \cdot g_i(\xi)$), or on both (mix of both previously described situations, the bound on the disturbance due to each error ends being additive on the right hand side anyway).

The equality relation with h and ∇h in 1.10 turns into the inequality: $\nabla h_j(\xi)^t \cdot \tilde{\mathcal{G}}_\alpha(x) \leq -\alpha \cdot h_j(x) + c \cdot \|\nabla h_j(x)\|$ due to the same causes. $\bullet$

Lemma 4.3.17. *Given some initial condition $\xi \in \mathcal{X}$, let $x(t; \xi) : \mathbb{R}_{\geq 0} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the solution to the SGF.*

Consider a sequence $\{t_0, \dots, t_N\}$ where $N \in \mathbb{N}$ and $x_{t_0} = x_0 = \xi$, $t_N = T$, $t_{i+1} - t_i = h > 0$ for $i \in \{0, \dots, N-1\}$ and choose a discretization scheme such that $\max_{i \in \{0, \dots, N\}} \|\text{step}_{i,i+1}(h, \mathcal{G}_\alpha, \xi)\| \rightarrow 0$ when $h \rightarrow 0$ and also

$\max_{i \in \{0, \dots, N\}} \|x_{t_i} - x(t_i, \xi)\| \rightarrow 0$ when $h \rightarrow 0$.

Consider also the curve created by uniting with a straight line each x_{t_i} with $x_{t_{i+1}}$ for $i \in \{0, \dots, N-1\}$. We will call it $\mu(t; \xi)$.

Then, given a LL function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$

$$|\alpha(\mu(s; \xi)) - \alpha(x(s; \xi))| \rightarrow 0 \text{ when } h \rightarrow 0 \forall s \in [0, T]$$

Proof. For each fixed ξ , $x(t; \xi)$ is a curve and it is LL as a function of t . Then, $x(t; \xi)|_{t \in [0, T]}$ is a compact.

The curve μ follows $\mu(t_i; \xi) = x_{t_i} \forall i \in \{0, \dots, N\}$. Observe $\mu(t; \xi)$ is only defined for $t \in [0, T]$.

There exists a compact set B that contains $x(t; \xi)|_{t \in [0, T]}$ and $\mu(t; \xi)$. $\alpha(x)$ is therefore L-Lipschitz in B , so given $s \in [0, T]$, $|\alpha(\mu(s; \xi)) - \alpha(x(s; \xi))| \leq L \cdot \|\mu(s; \xi) - x(s; \xi)\|$. Take the biggest t_i such that $t_i \leq s$. Then, $\|\mu(s; \xi) - x(s; \xi)\| \leq \|\mu(s; \xi) - \mu(t_i; \xi) + \mu(t_i; \xi) - x(t_i; \xi) + x(t_i; \xi) - x(s; \xi)\| \leq \|x_{t_{i+1}} - x_{t_i}\| + \|x_{t_i} - x(t_i; \xi)\| + \int_{t_i}^{t_{i+1}} \|\mathcal{G}_\alpha(x(t; \xi))\|^2 dt$. As $\|\mathcal{G}_\alpha(x(t; \xi))\|^2$ is bounded in B (we will call its bound M_G), we end up getting:

$$\begin{aligned} & |\alpha(\mu(s; \xi)) - \alpha(x(s; \xi))| \leq \\ & L \cdot \left(\max_{i \in \{0, \dots, T\}} \|\text{step}_{i,i+1}(h, \mathcal{G}_\alpha, \xi)\| + \max_{i \in \{0, \dots, T\}} \|x_{t_i} - x(t_i, \xi)\| + h \cdot M_G \right) \end{aligned}$$

Therefore, $|\alpha(\mu(s; \xi)) - \alpha(x(s; \xi))| \rightarrow 0$ when $h \rightarrow 0$. $\square$

Remark 4.3.18. By the previous lemma, given a LL size function w , we can write $w(x(t; \xi)) = w(\mu(t; \xi)) - d_\alpha(t; \xi) \forall t \in [0, T]$ with $d_\alpha(t; \xi) \rightarrow 0$ when $h \rightarrow 0$ $\bullet$.

Proposition 4.3.19. If the perturbed SGF is (c^*) small signal SIS(p)S on $(C_c, \mathcal{A})$ (with $c^* \leq c$) and there exists $0 < \epsilon^* \leq \infty$ such that $\overline{C_c^{\epsilon^*}} \subset \mathcal{X}$, then the finite RK45 discretization of the perturbed SGF with an additive disturbance is approximately (c^*) small signal SIS(p)S on $(C_c, \mathcal{A})$.

Proof. Take $0 < \tilde{\epsilon} < \epsilon^*$. If the system is (c^*) small signal SIS(p)S, then $\forall d \in \mathcal{M}_{c^*}$ there exists a $\mathcal{KL}$ -function β and $\mathcal{K}$ -function γ ($\mathcal{K}$ if stability is practical) such that:

$$w_{\mathcal{A}}(x(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|) \forall t \in [0, T] \text{ and } \forall \xi \in C_c$$

for some $w_{\mathcal{A}}$ a set size function for $(C_c, \mathcal{A})$. In particular, choose $w_{\mathcal{A}}$ to be a LL size function for $(\mathcal{X}, \mathcal{A})$. Remember we can always do this by Lemma 1.1.12. This ensures that $w_{\mathcal{A}}$ is a LL set size function for $(\overline{C_c^{\tilde{\epsilon}}}, \mathcal{A})$ for all $\tilde{\epsilon} \leq \epsilon^*$. By our definition of the expanded set $C_c^{\tilde{\epsilon}}$, it is clear that given $\tilde{\epsilon}_1 < \tilde{\epsilon}_2$, it holds that $C_c^{\tilde{\epsilon}_1} \subseteq C_c^{\tilde{\epsilon}_2}$ and also that $d(C_c, C_c^{\tilde{\epsilon}}) \rightarrow 0$ as $\tilde{\epsilon} \rightarrow 0$.

By remark 4.3.18, if $\mu(t, \xi, d)$ is the discretized trajectory for the perturbed SGF, we can rewrite:

$$w_{\mathcal{A}}(x(t, \xi, d)) = w_{\mathcal{A}}(\mu(t, \xi, d)) - d_{w_{\mathcal{A}}}(t; \xi) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|_\infty)$$

$$\begin{aligned} &\Longleftrightarrow w_{\mathcal{A}}(\mu(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|_{\infty}) + d_{w_{\mathcal{A}}}(t; \xi) \\ &\leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|_{\infty}) + \max_{t \in [0, T]} |d_{w_{\mathcal{A}}}(t; \xi)| \end{aligned}$$

Now, for our $\tilde{\epsilon} > 0$ there $\exists h_w$ st $\forall h < h_w$, $\max_{t \in [0, T]} |d_{w_{\mathcal{A}}}(t; \xi)| < \epsilon$ because the RK45 discretization scheme fulfills the conditions in Lemma 4.3.17. Thus:

$$w_{\mathcal{A}}(\mu(t, \xi, d)) \leq \beta(w_{\mathcal{A}}(\xi), t) + \gamma(\|d\|_{\infty}) + \tilde{\epsilon} \quad \forall t \in [0, T], \xi \in B$$

Define now $\tilde{\gamma} := \gamma(\cdot) + \epsilon \in \tilde{\mathcal{K}}$. We know that there exists $\tilde{h}^{\alpha}$ such that the finite RK45 discretization renders $C_c^{\tilde{\epsilon}}$ for all $h \leq \tilde{h}^{\alpha}$. Choose $h^* < \min\{h_B, h_w\}$. We can conclude that, given $\xi \in C_c^{\tilde{\epsilon}}$, $\forall h \leq h^*$:

- $x_{t_i} \in C_c^{\tilde{\epsilon}} \quad \forall i \in \{0, \dots, N\}$
- $w_{\mathcal{A}}(x_{t_i}) \leq \beta(w_{\mathcal{A}}(\xi), t) + \tilde{\gamma}(\|d\|) \quad \forall i \in \{0, \dots, N\}$

This concludes the proof that the discretized conditions in Definition 19 are fulfilled, where the parameter λ is here the step size h . Thus, the discretized system is approximately SISpS on $(C_c, \mathcal{A})$. $\square$

Corollary 4.3.20. *If conditions in Proposition 4.1.9 hold and there exists $0 < \epsilon^* \leq \infty$ such that $\overline{C_c^{\epsilon^*}} \subset \mathcal{X}$, then the finite RK45 discretization of the perturbed SGF with an additive disturbance is approximately (c^*) small-signal SISpS on $(C_c, \mathcal{A})$.*

Proof. Proposition 4.1.9 gives us the required conditions in 4.3.19. $\square$

Proposition 4.3.21. *If conditions in Proposition 4.2.2 hold and there exists $0 < \epsilon^* \leq \infty$ such that $\overline{C^{\epsilon^*}} \subset \mathcal{X}$, then the finite RK45 discretization of the perturbed SGF with an erroneous objective function $\tilde{f}$ is approximately (c^*) small-signal SISpS on $(C, \mathcal{A})$.*

Proof. The proof is equivalent to the one in Proposition 4.3.19, only in this case the SISS condition holds in C instead of C_c , and it is Proposition 4.3.4 that grants the existence of a step size that renders C^{ϵ} forward invariant. Discretized SISpS is again given by Proposition 4.3.17. $\square$

Corollary 4.3.22. *If conditions in Proposition 4.2.2 hold and g and h are linear then the finite Forward Euler discretization of the perturbed SGF with an erroneous objective function $\tilde{f}$ is (c^*) small-signal SISpS on $(C, \mathcal{A})$.*

Proof. The proof is equivalent to the one in 4.3.21, but here a step size that grants the forward invariance of the set C is given by 4.3.1. $\square$

Proposition 4.3.23. *If conditions in Corollary 4.2.4 hold and there exists $0 < \epsilon^* \leq \infty$ such that $\overline{C^{\epsilon^*}} \subset \mathcal{X}$, then the finite RK45 discretization of the perturbed SGF with an erroneous safe set $\tilde{C}$ is approximately (c^*) small-signal SISpS on $(\tilde{C}, \mathcal{A})$.*

Proof. Equivalent to the proof in Proposition 4.3.21, only in this case we apply Proposition 4.3.4 to $\tilde{C}$ instead of C . $\square$

Corollary 4.3.24. *If conditions in Corollary 4.2.4 hold and g and h are linear then the finite Forward Euler discretization of the perturbed SGF with an erroneous safe set $\tilde{C}$ is (c^*) small-signal SISpS on $(\tilde{C}, \mathcal{A})$.*

4.4 Simulations

In this section we showcase the results on the stability under disturbances and discretization of the Safe Gradient Flow which we have obtained along this Chapter (4). This section has 2 goals:

- (i) To showcase the forward invariance of the Safe Set C under the discretized SGF and the conditions in Proposition 4.3.9.
- (ii) To showcase the approximate input to state practical stability of the discretized SGF on $(C_c, \mathcal{A})$ given a bounded C , as stated in Corollary 4.3.20.

To this end, we use a simple quadratic program in $\mathbb{R}^2$ centered at $(x, y) = (0, 0)$ to illustrate our results. The Karush-Kuhn-Tucker points in this problem always coincide with the restricted minima, so we do not need to worry about *practical stability*. We do not include simulations for the erroneous f or C cases because the results are equivalent to the ones we are already showcasing:

- under an erroneous f we would still have forward invariance of C and set input to state stability (if C is bounded) which would result in convergence to the new minima, whose distance to the original minima is determined by the error introduced in f .
- under an erroneous C we would still have forward invariance of $\tilde{C}$ and set input to state stability (if C is bounded) which would result in convergence to the new restricted minima, which are determined by the differences between C and $\tilde{C}$.

In conclusion, in this section we are looking at the restricted optimization problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^2} & \frac{1}{2} \cdot x^t \cdot Q \cdot x \\ \text{s.t.} & \quad g(x) \leq 0 \\ & \quad h(x) = 0 \end{aligned} \tag{4.18}$$

where Q is a symmetric 2x2 matrix.

In deference to Ahmed Allibhoy and Jorge Cortés, who introduced the Safe Gradient Flow algorithm, we use the Q matrix they used to test out the Safe Gradient Flow,

$$Q = \begin{bmatrix} 0.148 & -0.033 \\ -0.033 & 0.101 \end{bmatrix}$$

We use $\alpha = 0.5$.

4.4.1 Quadratic Program

Here, we assume the errors in the computed controllers u, v of the Safe Gradient Flow are negligible. We use the standard solver "ECOS" to compute the global minima of the problem and then use this to illustrate the convergence of the Safe Gradient Flow

algorithm to the global minima while maintaining forward invariance of the constraints set C .

Let $g(x) = A \cdot x - b$, where A is the matrix:

$$A = \begin{bmatrix} 0.5 & 2 \\ 3 & -1 \end{bmatrix}$$

and b is the vector

$$b = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

and let $h(x) = c^t \cdot x$ where c is the vector

$$c = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

This choice of h means we are forcing the solutions of the system to fulfill the equality $y = x$, this is, the system must evolve along the line $y = x$.

Observe that the conditions in Proposition 4.3.9 hold. To check forward invariance of C , we have randomly chosen different initial points $x_0 \in C$ and verified the solutions remain within C for 50 time-steps using the RK45 method. We show the results for $x_0 = (1, 1)^t$ in Figure 4.1.

4.4.2 Perturbed Quadratic Program

Finally, let us force a disturbance d into the discretized Safe Gradient Flow. To guarantee that the conditions in Corollary 4.3.20 are fulfilled, let us modify the function g adding an inequality that forces the x-coordinate to be bigger than -0.5 at all times. To do this, simply consider:

$$A = \begin{bmatrix} 0.5 & 2 \\ 3 & -1 \\ -1 & 0 \end{bmatrix}$$

and b is the vector

$$b = \begin{bmatrix} 5 \\ 4 \\ 0.5 \end{bmatrix}$$

The introduced disturbance is added to the Safe Gradient Flow in each time step and is randomly generated using a uniform distribution in between -0.01 and 0.01 for each coordinate. We showcase the forward invariance of an expanded safe set characterized by $|h(x)| \leq \epsilon$ (with $\epsilon = 0.05$) instead of $h(x) = 0$ and convergence to a neighborhood of the restricted global minimum for both a starting point $x_0 = (1, 1)^t \in C$ and $x_0 = (1.01, 0.99)^t$, which belongs to the expanded set characterized by $|h(x)| \leq \epsilon$ (Figures 4.2 and 4.3, respectively)

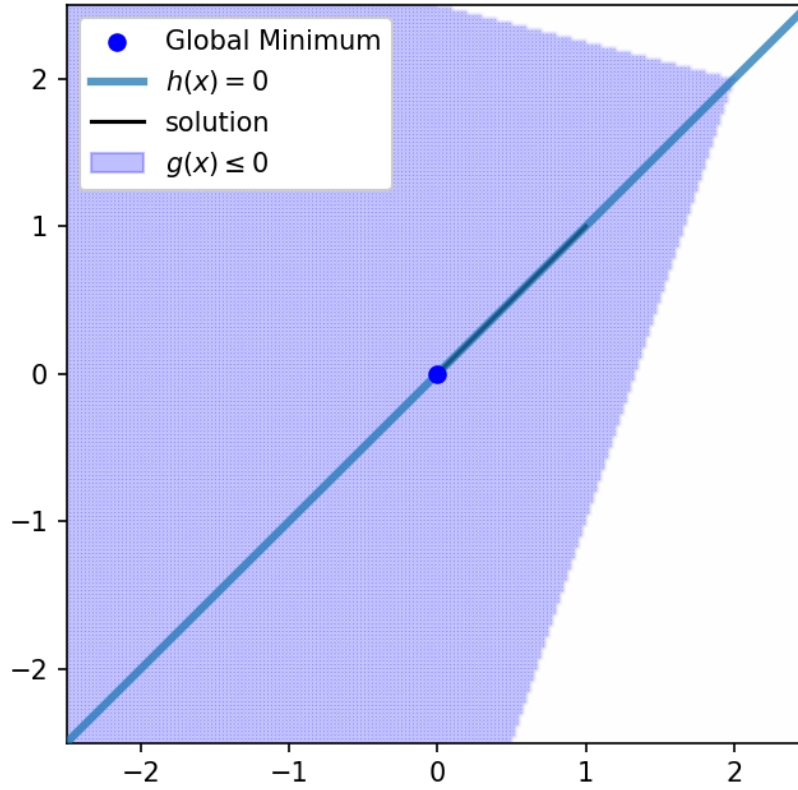


Figure 4.1: The figure displays the first 50 time steps of the trajectory of the discretized Safe Gradient Flow starting from $x_0 = (1, 1)^t$. We can see the set C is rendered forward invariant and there is convergence towards the restricted global minima.

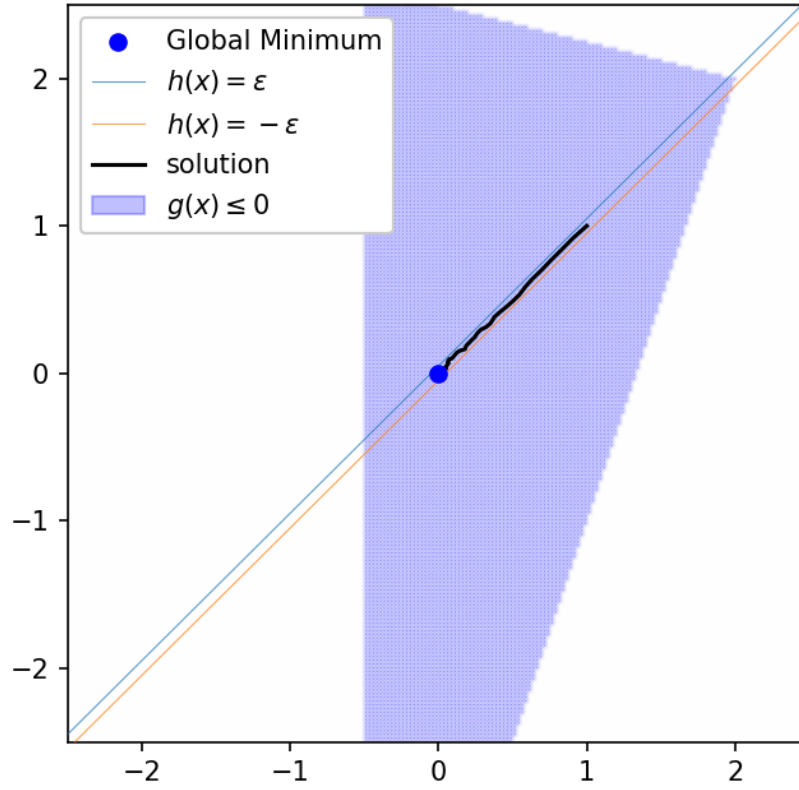


Figure 4.2: The figure displays the first 50 time steps of the trajectory of the perturbed discretized Safe Gradient Flow starting from $x_0 = (1, 1)^t$. We can see the expanded constraints set is rendered forward invariant and there is convergence towards a neighborhood of the restricted global minima.

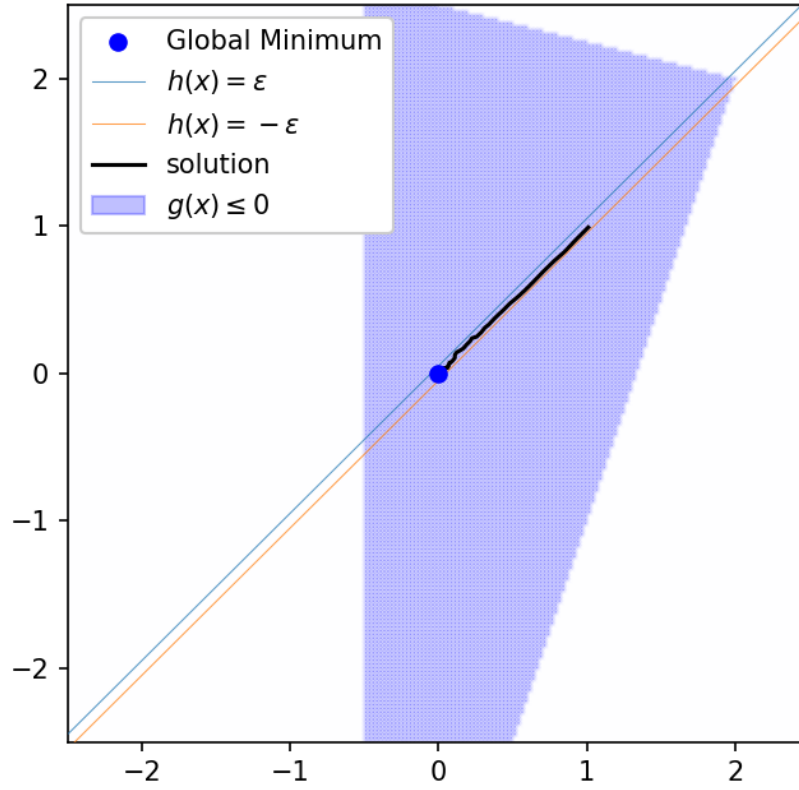


Figure 4.3: The figure displays the first 50 time steps of the trajectory of the discretized Safe Gradient Flow starting from $x_0 = (1.01, 0.99)^t$. We can see the expanded constraints set is rendered forward invariant and there is convergence towards a neighborhood of the global minima.

Chapter 5

Sustainability Report

This chapter consists of a sustainability report that evaluates the environmental, economic and social impact of the development of this document; as well as an analysis of its ethical implications and its relationship with the sustainable development goals.

Calculations in this chapter are made taking into account that I have worked for 8 hours a day, 5 days a week from September 12th 2023 until May 14th 2024. This accounts for approximately 1320 hours of work. This work has been carried on at UCSD, in La Jolla (San Diego), California, US.

5.1 Sustainability Matrix

Environmental, economic and social impacts are evaluated by means of a sustainability matrix that explores the implications of this document on each of those categories during its development and execution. What is more, it analyses future risks and limitations.

5.1.1 Environmental Impact

Development

A significant portion of my working hours has been spent in front of a paper sheet with just a pencil in hand. However, my computer has been in operation for most of the time, as I've utilized it to review existing literature, conduct inquiries, and draft this document along with the two papers that this document is expected to produce. Assuming that I have been using my computer for 80% of my working hours, that amounts to 1056 hours of computer usage. Assuming my computer (Dell XPS 15) consumes 130W while in use [39], the total electricity consumption has been $130 \cdot 1056 = 137280Wh = 137,28kWh$. The Carbon Density of energy in San Diego is $73gCO_2/kWh$ ([40]), so this amounts to approximately $10kgCO_2$. Nevertheless, I have not performed any heavy computations, so this amount should actually be considerably lower.

It is difficult to estimate the environmental impact of my stay at Jorge Cortés Lab at Franklin Antonio Hall at UCSD, as this is a collaborative laboratory which hosts many researchers. However, UCSD as an organization is deeply committed to sustainability ([41]).

The main environmental impact of the development of this work has been my trip to San Diego from Barcelona. The flights round trip amount for a total of $966,2kgCO_2$ ([42]).

Execution

This project can have deeply positive environmental implications. Now that the Safe Gradient Flow is guaranteed to be set input to state stable under disturbances, it can be implemented in real life settings.

Among its many applications lies the control of power networks to safely accomplish optimal power flow dispatch ([43]). Optimal Power Flow maximizes the efficiency of our energy networks, and doing so safely can avoid damages in installations such as power plants. This reduces the need for replacements and increases the life expectancy of the energy-producing installations. Therefore, Safe Optimal Power Flow has very positive environmental impacts ([44]).

Risks and Limitations

The abstract nature of this work, which is still in its initial stages, makes it difficult to predict its future risks and limitations.

5.1.2 Economic Impact

Development

I have received 3 scholarships to cover the trip to San Diego and the living expenses there, which amount to a total of approximately \$11700. Given that the minimum wage in California is \$14/h, the development of this project would have cost a minimum of \$18480.

The cost of the employed energy is negligible compared to these.

Execution

The implementation of the Safe Gradient Flow algorithm in real life settings can have enormous economic implications.

If used to achieve Safe Optimal Power Flow, the economic impact is given by the increased energy but also by the extended life expectancy of the energy-producing installations.

However, there are many other applications of the SGF. For example, it may be used to perform safe optimization in automatic systems such as driver-less cars or robots operating in delicate environments. These industries generate (or are expected to generate in the immediate future) enormous revenues every year ([45]).

Risks and Limitations

The abstract nature of this work, which is still in its initial stages, makes it difficult to predict its future risks and limitations.

5.1.3 Social Impact

Development

During the development of this project I have been fortunate to interact, interchange ideas and have shared experiences with people from a myriad of different backgrounds and nationalities. This has shaped me into a better person who is more conscious of the pressing social issues of our century.

UCSD is committed to generating positive social change, and has received extensive recognition for this effort ([46]).

Execution

The Safe Gradient Flow has been openly published ([14]), and our papers on Set Input to State Stability and the stability of the Safe Gradient Flow are expected to be published, too. This means anybody can implement and benefit from the Safe Gradient Flow and its properties without the need to pay for any kind of license to employ it.

Risks and Limitations

The abstract nature of this work, which is still in its initial stages, makes it difficult to predict its future risks and limitations.

5.1.4 Ethical Implications

Mathematics has been considered to be independent from social/philosophical considerations such as ethics, but the recent popularity of data science has made it increasingly difficult to deny the importance of ethics in mathematical practice. An example of this is the development of certain algorithms (in particular, machine learning algorithms) that discriminate people, as these algorithms themselves might include the biases and prejudices of either their creators or the information they are trained with.

The Safe Gradient Flow is an abstract algorithm in the form of a dynamical system that cannot possibly include any of its creator's biases. What is more, it does not need to be trained, so we do not need to worry about it being corrupted by biased data.

The first part of this document is an extension of the input to state stability theory, and this once again does not leave room for any social biases/prejudices.

However, both the Safe Gradient Flow and the theory introduced in this document might be used for unethical purposes. It is thus important to push for an ethical education that goes hand in hand with mathematical education and knowledge ([47]).

5.1.5 Relationship with the Sustainable Development Goals

The United Nations specify 17 sustainable development goals ([48]). Progress towards every one of them is evaluated yearly.

The Safe Gradient Flow is a transversal algorithm that can be useful in every system that employs optimization processes. Thus, it can indirectly have an impact in every sustainable development goal that may benefit from science & engineering solutions.

This document introduces a novel framework for the study of input to state stability, and is thus aligned with the sustainable development goal number 9.

Meanwhile, the Safe Gradient Flow is directly aligned with goal number 7 thanks to its applications in Safe Optimal Power Flow.

Chapter 6

Conclusions

We have introduced an extended framework which enables the study of Input to State Stability properties on closed sets inside an open set $\mathcal{X}$ in which the system is defined.

In doing this, we have eliminated the two main drawbacks of the Input to State Stability theory on open sets:

- Our framework can study systems that showcase input to state stability in a closed set but not in the whole space $\mathcal{X}$. The ISS property is very restrictive, but our extension can deal with a much broader class of systems.
- Our framework does not require to compute the Lie derivative of a function that explodes in the boundary of the set $\mathcal{X}$, easing calculations and allowing us to deal with systems with an unknown $\mathcal{X}$.

We have extended the notion of Local ISS to systems defined on open sets and found tight sufficient conditions and weak necessary conditions both directly related to the novel concept introduced in this paper: Set Input to State Stability. SISS is able to study the input to state stability of closed sets by decoupling the standard definition of Input to State Stability. In our framework, the forward invariance of the studied set is still a requirement, but now it can be studied separately (leveraging the extensive literature on closed set invariance) instead of demanding that the ISS inequality (1.2) directly guarantees it.

What is more, we have discussed the effects of dealing with 2 sets of equilibria such that one is contained inside the other and the SISS-Lyapunov inequality is formulated in terms of a set size function for the smaller set on the left side and a set size function for the bigger set on the right side. We have called this attribute weak SISS. This consideration is particularly useful when dealing with general gradient systems, in which the zeros of ∇V do not match the global minima of V . We have also showcased the broad applicability of our framework on gradient systems in which these two sets match.

We have also given a detailed characterization of the relationship between ISS, LISS and SISS - which is summarized in figure 1.1.

Using the Set Input to State Stability framework we have built, we have proved some stability properties of the Sage Gradient Flow algorithm. More explicitly,:

- We have proven Input to State Safety (ISSf) of the constraints set C of the SGF under very mild conditions.
- We have established conditions under which the SGF is small-signal SISpS under generic disturbances.
- We have proven the SGF satisfies improved SISS conditions under exclusively state-dependent disturbances such as errors in the objective function or in the definition of the safe set.
- We have found conditions for the SGF discretization to still render the safe set forward invariant, or approximately forward invariant under milder conditions. We have shown the finite discretization of the Safe Gradient Flow retains small-signal SIS(p)S if the continuous flow holds this property.

Future Work

We leave for future work the characterization of the relations between the ISpS and the SISpS properties. Future work will also study the discrete SISS property and the relationship ISS-LISS-SISS in the discrete case. What is more, future work will study the ISS, LISS and SISS properties on open sets with respect to closed sets of equilibria.

Regarding the implementation of the Safe Gradient Flow, future work will include deriving explicit bounds for the step sizes of the discretization algorithm used in order to guarantee that the properties of the SGF are maintained.

Another objective of future work will be to study the Set Input to State Stability of the Safe Monotone Flow and the Recursive Safe Monotone Flow, which is a generalization of the Safe Gradient Flow algorithm designed to solve variational inequalities while maintaining anytime safety of the constraints set.

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Appendix A

We present here an adapted version of Theorem B.1 in [35] for functions defined in a (closed) sub-level set of a size function for $(\mathcal{X}, \mathcal{A})$ (where $\mathcal{X}$ is an open set and $\mathcal{A}$ is a compact set of equilibria) from which we have removed the set of equilibria $\mathcal{A}$.

Theorem 6.0.1. *Let S_r be a sub-level set of a size function $w(x)$ for $(\mathcal{X}, \mathcal{A})$. Let $\mathcal{D} \subset \mathbb{R}^m$ compact. If there exist:*

- *a function $U : S_r \setminus \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ that is Lipschitz in every ring of $S_r \setminus \mathcal{A}$; this is, that given $0 < s_1 < s_2 \leq r$, U is Lipschitz on $K_{s_1, s_2} := \{x \in \mathcal{X} \mid s_1 \leq w(x) \leq s_2\}$*
- *a map $f : \mathbb{R}^n \times \mathcal{D} \rightarrow \mathbb{R}^n$ Locally Lipschitz on x uniformly on $\mathbf{d}$*
- *a continuous function $\alpha : S_r \setminus \mathcal{A} \rightarrow \mathbb{R}$ such that for every $\mathbf{d} \in \mathcal{D}$,*

$$\mathcal{L}_{f(\xi, \mathbf{d})}U(\xi) \leq \alpha(\xi)$$

almost everywhere on $S_r \setminus \mathcal{A}$

Then, for any pair of continuous functions $\mu, \nu : S_r \setminus \mathcal{A} \rightarrow \mathbb{R}_{>0}$ there exists a smooth function $V : S_r \setminus \mathcal{A} \rightarrow \mathbb{R}$ such that:

$$|U(\xi) - V(\xi)| < \mu(\xi) \quad \forall \xi \in S_r \setminus \mathcal{A}$$

for each $\mathbf{d} \in \mathcal{D}$

$$\mathcal{L}_{f(\xi, \mathbf{d})}V(\xi) \leq \alpha(\xi) + \nu(\xi) \quad \forall \xi \in S_r \setminus \mathcal{A}$$

Proof. The proof is analogous to the one in Theorem B.1 in [13]. We just need to make sure it holds for a set of the form $S_r \setminus \mathcal{A}$, which is not an open set (contrary to the assumptions in Theorem B.1).

Lemmas B.2, B.3 and B.4 are stated for compact sets of the open set $\mathcal{O}$ Theorem B.1 works with, in which U is Lipschitz as a consequence of the Local Lipschitzness of U in $\mathcal{O}$ assumption. As one of our assumptions is Lipschitzness of U in every ring $K_{s_1, s_2} \subset S_r \setminus \mathcal{A}$, Lemmas B.2, B.3 and B.4 hold in every ring K_{s_1, s_2} . These three lemmas combined give us the following result (Lemma B.5 in Theorem B.1):

Lemma 6.0.1. *Given K_{s_1, s_2} a ring of $S_r \setminus \mathcal{A}$, for every $\epsilon > 0$ exists a smooth function V_{s_1, s_2} defined on K_{s_1, s_2} such that:*

$$|V_{s_1, s_2}(\xi) - U(\xi)| < \epsilon \quad \text{and} \quad \mathcal{L}_{f(\xi, \mathbf{d})}V(\xi) \leq \alpha(\xi) + \epsilon$$

for all $\xi \in K_{s_1, s_2}$ and for all $\mathbf{d} \in \mathcal{D}$

Consider the series $(\frac{r}{2^k})_{k \in \mathbb{N} \cup \{0\}}$ and the series $(\frac{3r}{2^{k+2}})_{k \in \mathbb{N} \cup \{0\}}$ and let $\overset{\circ}{K}_i^m := \overset{\circ}{K}_{\frac{r}{2^{i+1}}, \frac{r}{2^i}}$, $\overset{\circ}{K}_j^s := \overset{\circ}{K}_{\frac{3r}{2^{j+3}}, \frac{2r}{2^{j+2}}}$. Consider the collection $\{\overset{\circ}{K}_i^m, \overset{\circ}{K}_i^s\}_{i \in \mathbb{N} \cup \{0\}}$. Clearly, this covers $\overset{\circ}{S}_r \setminus \mathcal{A}$.

Let $\mathcal{J}_\xi := \{i \mid \xi \in \overset{\circ}{K}_i^m\} \cup \{\tilde{j} = -j - 1 \mid \xi \in \overset{\circ}{K}_j^s\}$ defined for $\xi \in \overset{\circ}{S}_r \setminus \mathcal{A}$. Given $i \in \mathbb{N} \cup \{0\}$, note that if $w(\xi) = \frac{3r}{2^{i+2}}$ then ξ pertains only to $\overset{\circ}{K}_i^m$. Given $j \in \mathbb{N}$ ($j \neq 0$), if $w(\xi) = \frac{r}{2^j}$ then ξ pertains only to $\overset{\circ}{K}_{j-1}^s$. Finally, if $w(\xi) \in [\frac{3r}{4}, r)$ then ξ is only contained in $\overset{\circ}{K}_0^m$. If $\xi \in \overset{\circ}{S}_r \setminus \mathcal{A}$ does not follow any of the conditions stated here, then it belongs to two sets of the cover that are either $\overset{\circ}{K}_i^m$ and $\overset{\circ}{K}_i^s$ or $\overset{\circ}{K}_{i+1}^m$ and $\overset{\circ}{K}_i^s$, for some $i \in \mathbb{N} \cup \{0\}$. Therefore, the cardinal of $\mathcal{J}_\xi$ is:

$$\#\mathcal{J}_\xi = \begin{cases} 1 & \text{if } w(\xi) = \frac{r}{2^j} \text{ for some } j \in \mathbb{N}_{>0} \\ 1 & \text{if } w(\xi) = \frac{3r}{2^{i+2}} \text{ for some } i \in \mathbb{N} \cup \{0\} \\ 1 & \text{if } w(\xi) \geq \frac{3r}{4} \\ 2 & \text{if none of the above conditions hold} \end{cases}$$

Therefore, we have a countable, locally finite cover of $\overset{\circ}{S}_r \setminus \mathcal{A}$. As $\overset{\circ}{S}_r \setminus \mathcal{A} \subset \mathbb{R}^n$ is open, it is therefore a smooth manifold. This means there exists a smooth, locally finite partition of unity $\{\tilde{\beta}_i^m, \tilde{\beta}_i^s\}_{i \in \mathbb{N} \cup \{0\}}$ subordinate to the cover $\{\overset{\circ}{K}_i^m, \overset{\circ}{K}_i^s\}_{i \in \mathbb{N} \cup \{0\}}$.

Take $\xi \in \overset{\circ}{S}_r \setminus \mathcal{A}$ such that $w(\xi) \geq \frac{3r}{4}$. We have already seen that, in that case, ξ is only contained in $\overset{\circ}{K}_0^m$. As the partition is subordinate to the cover, for any ξ st $w(\xi) \geq \frac{3r}{4}$ it holds that $\tilde{\beta}_0^m(\xi) = 1$, $\tilde{\beta}_i^m(\xi) = 0 \ \forall i \in \mathbb{N}_{>0}$ and $\tilde{\beta}_j^s(\xi) = 0 \ \forall j \in \mathbb{N} \cup \{0\}$. Particularly, this means that $\forall \xi \in \overset{\circ}{S}_r \setminus \mathcal{A}$ st $w(\xi) \geq \frac{3r}{4}$: $\nabla \tilde{\beta}_i^m(\xi) = \nabla \tilde{\beta}_i^s = 0 \ \forall i \in \mathbb{N} \cup \{0\}$. Therefore, we can smoothly extend both $\tilde{\beta}_i^m, \tilde{\beta}_i^s$ to $S_r \setminus \mathcal{A}$ and, in fact, to $\mathcal{X} \setminus \mathcal{A}$: let

$$\beta_i^m(\xi) = \begin{cases} \tilde{\beta}_i^m(\xi) & \text{if } \xi \in \overset{\circ}{S}_r \setminus \mathcal{A} \\ 1 & \text{if } w(\xi) \geq r \text{ and } i = 0 \\ 0 & \text{if } w(\xi) \geq r \text{ and } i > 0 \end{cases}$$

for all $i \in \mathbb{N} \cup \{0\}$ and

$$\beta_j^s(\xi) = \begin{cases} \tilde{\beta}_j^s(\xi) & \text{if } \xi \in \overset{\circ}{S}_r \setminus \mathcal{A} \\ 0 & \text{if } w(\xi) \geq r \end{cases}$$

It is clear that the functions $\{\beta_i^m, \beta_i^s\}_{i \in \mathbb{N} \cup \{0\}}$ are smooth, and their restriction to $S_r \setminus \mathcal{A}$, $\{\beta_i^m|_{S_r \setminus \mathcal{A}}, \beta_i^s|_{S_r \setminus \mathcal{A}}\}_{i \in \mathbb{N} \cup \{0\}} =: P$ is a smooth, locally finite partition of unity subordinate to the cover $\{\tilde{K}_0^m, \{\overset{\circ}{K}_{i+1}^m, \overset{\circ}{K}_i^s\}_{i \in \mathbb{N} \cup \{0\}}\} =: C$ of $S_r \setminus \mathcal{A}$, where $\tilde{K}_0^m = \{x \in \mathcal{X} \mid \frac{r}{2} < w(x) \leq r\}$. Note that the closure of $\tilde{K}_0^m$ is $\overset{\circ}{K}_0^m$.

Now, for each $i \in \mathbb{N} \cup \{0\}$ consider $K_i^m = K_{\frac{r}{2^{i+1}}, \frac{r}{2^i}}$ and $K_i^s = K_{\frac{3r}{2^{i+3}}, \frac{3r}{2^{i+2}}}$ and let $\epsilon_i^m := \min \{\min_{\xi \in K_i^m} \mu(\xi), \min_{\xi \in K_i^m} \nu(\xi)\}$, $\epsilon_i^s := \min \{\min_{\xi \in K_i^s} \mu(\xi), \min_{\xi \in K_i^s} \nu(\xi)\}$, $\tau_i^m =$

$\max_{\xi \in K_i^m, \mathbf{d} \in \mathcal{D}} |\mathcal{L}_{f(x, \mathbf{d})} \beta_i^m(\xi)|$ and $\tau_i^s = \max_{\xi \in K_i^s, \mathbf{d} \in \mathcal{D}} |\mathcal{L}_{f(x, \mathbf{d})} \beta_i^s(\xi)|$. Redefine $\mathcal{J}_\xi$ as:

$$\mathcal{J}_\xi := \begin{cases} \{i \mid \xi \in \overset{\circ}{K}_i^m\} \cup \{j = -j - 1 \mid \xi \in \overset{\circ}{K}_j^s\} & \text{if } \xi \in \overset{\circ}{S}_r \\ \{0\} & \text{if } \xi \in \partial S_r \end{cases}$$

which captures the sets of the cover C in which each $\xi \in S_r \setminus \mathcal{A}$ is included because $\tilde{K}_0^m = \overset{\circ}{K}_0^m$ (if $\xi \in \overset{\circ}{S}_r \setminus \mathcal{A}$ and $\{0\} \in \mathcal{J}_\xi$, then this indicates $\xi \in \tilde{K}_0^m$) and if $\xi \in \tilde{K}_0^m \setminus \overset{\circ}{K}_0^m = \partial S_r$ then it is only included in $\overset{\circ}{K}_0^m$. Observe that the cardinal of the new $\mathcal{J}_\xi$ behaves exactly as the original.

Now, for each $i \in \mathbb{N} \cup \{0\}$ let

- V_i^m be the smooth function obtained by applying Lemma 6.0.1 to K_i^m with $\epsilon = \frac{\epsilon_i^m}{4(1+\tau_i^m)} (< \frac{\epsilon_i^s}{4})$
- V_i^s be the smooth function obtained by applying Lemma 6.0.1 to K_i^s with $\epsilon = \frac{\epsilon_i^s}{4(1+\tau_i^s)} (< \frac{\epsilon_i^s}{4})$

Define

$$V := \sum_{i=0}^{\infty} \beta_i^m V_i^m + \beta_i^s V_i^s = \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \beta_i^m V_i^m + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \beta_j^s V_j^s$$

where there exists at most one $i \geq 0$ st $i \in \mathcal{J}_\xi$ and there exists at most one $j \geq 0$ st $-j-1 \in \mathcal{J}_\xi \ \forall \xi \in S_r \setminus \mathcal{A}$.

Clearly, V is smooth in $S_r \setminus \mathcal{A}$. Note that $U(\xi) = \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \beta_i^m(\xi) U(\xi) + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \beta_j^s(\xi) U(\xi) \ \forall \xi \in S_r \setminus \mathcal{A}$. Then, $|V(\xi) - U(\xi)| \leq \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \beta_i^m(\xi) |V_i^m(\xi) - U(\xi)| + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \beta_j^s(\xi) |V_j^s(\xi) - U(\xi)| \leq \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} |V_i^m(\xi) - U(\xi)| + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} |V_j^s(\xi) - U(\xi)| < b^m(\xi) + b^s(\xi)$, where:

$$b^m(\xi) = \begin{cases} \frac{\epsilon_{i_\xi}^m}{4} & \text{if } \exists i_\xi \geq 0 \text{ st } i_\xi \in \mathcal{J}_\xi \\ 0 & \text{if not} \end{cases}$$

$$b^s(\xi) = \begin{cases} \frac{\epsilon_{j_\xi}^s}{4} & \text{if } \exists j_\xi \geq 0 \text{ st } -j_\xi - 1 \in \mathcal{J}_\xi \\ 0 & \text{if not} \end{cases}$$

Note that b^m and b^s are well defined for all $\xi \in S_r \setminus \mathcal{A}$ because both i_ξ and j_ξ , if they exist, are unique. Clearly $b^m(\xi) + b^s(\xi) \leq 2b^m(\xi) + 2b^s(\xi) \leq \min\{\mu(\xi), \nu(\xi)\}$. We conclude

$$|V(\xi) - U(\xi)| < \mu(\xi) \ \forall \xi \in S_r \setminus \mathcal{A}$$

Regarding the Lie Derivative of V along the trajectories of f (which exists for every $\xi \in S_r \setminus \mathcal{A}$ because V is smooth), note that given $i \in \mathbb{N} \cup \{0\}$, $\nabla \beta_i^m(\xi) = 0 \ \forall \xi \in S_r \setminus \overset{\circ}{K}_i^m \setminus \mathcal{A}$ and $\nabla \beta_i^s(\xi) = 0 \ \forall \xi \in S_r \setminus \overset{\circ}{K}_i^s \setminus \mathcal{A}$, as P is a smooth partition of unity subordinate to C .

Given $\xi \in S_r \setminus \mathcal{A}$ and $\mathbf{d} \in \mathcal{D}$, $\mathcal{L}_{f(\xi, \mathbf{d})} V(\xi) = \mathcal{L}_{f(\xi, \mathbf{d})} U(\xi) + \mathcal{L}_{f(\xi, \mathbf{d})} (\sum_{i=0}^{\infty} \beta_i^m (V_i^m - U) + \beta_i^s (V_i^s - U))(\xi) = \mathcal{L}_{f(\xi, \mathbf{d})} U(\xi) + \sum_{i=0}^{\infty} (V_i^m(\xi) - U(\xi)) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} \beta_i^m(\xi) +$

$$\sum_{i=0}^{\infty} (V_i^s(\xi) - U(\xi)) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} \beta_i^s(\xi) + \sum_{i=0}^{\infty} \beta_i^m(\xi) (\mathcal{L}_{f(\xi, \mathbf{d})} V_i^m(\xi) - \mathcal{L}_{f(\xi, \mathbf{d})} U(\xi)) + \sum_{i=0}^{\infty} \beta_i^s(\xi) (\mathcal{L}_{f(\xi, \mathbf{d})} V_i^s(\xi) - \mathcal{L}_{f(\xi, \mathbf{d})} U(\xi)).$$

Note that for any given $i \in \mathbb{N} \cup \{0\}$ $\mathcal{L}_{f(\xi, \mathbf{d})} \beta_i^m(\xi) = f(\xi, \mathbf{d})^t \cdot \nabla \beta_i^m(\xi) = 0 \ \forall \xi \in S_r \setminus \overset{\circ}{K}_i^m \setminus \mathcal{A}$; so given $\xi \in S_r \setminus \mathcal{A}$, $\mathcal{L}_{f(\xi, \mathbf{d})} \beta_i^m(\xi)$ can only be different than zero for $i \in \mathcal{J}_\xi$. (analogously for β_j^s and $\mathcal{J}_\xi$). Then,

$$\begin{aligned} \mathcal{L}_{f(\xi, \mathbf{d})} V(\xi) &= \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} (V_i^m(\xi) - U(\xi)) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} \beta_i^m(\xi) + \\ &\sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} (V_j^s(\xi) - U(\xi)) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} \beta_j^s(\xi) + \\ &\sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \beta_i^m(\xi) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} V_i^m(\xi) + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \beta_j^s(\xi) \cdot \mathcal{L}_{f(\xi, \mathbf{d})} V_j^s(\xi) < \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \tau_i^m \cdot \frac{\epsilon_i^m}{2(1+\tau_i^m)} + \\ &\sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \tau_j^s \cdot \frac{\epsilon_j^s}{2(1+\tau_j^s)} + \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \beta_i^m(\xi) \cdot (\alpha(\xi) + \frac{\epsilon_i^m}{2(1+\tau_i^m)}) + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \beta_j^s(\xi) \cdot (\alpha(\xi) + \frac{\epsilon_j^s}{2(1+\tau_j^s)}) \\ &\leq \alpha(\xi) + \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \frac{\epsilon_i^m}{2} + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \frac{\epsilon_j^s}{2} + \sum_{i \geq 0 \text{ and } i \in \mathcal{J}_\xi} \frac{\epsilon_i^m}{2} + \sum_{j \geq 0 \text{ and } -j-1 \in \mathcal{J}_\xi} \frac{\epsilon_j^s}{2} = \\ &\alpha(\xi) + 2b^m(\xi) + 2b^s(\xi). \end{aligned}$$

As $2b^m(\xi) + 2b^s(\xi) \leq \min\{\mu(\xi), \nu(\xi)\} \leq \nu(\xi)$, we conclude

$$\mathcal{L}_{f(\xi, \mathbf{d})} V(\xi) \leq \alpha(\xi) + \nu(\xi) \ \forall \xi \in S_r \setminus \mathcal{A}$$

where in fact we have seen strict inequality. This completes the proof. $\square$

Appendix B

SGF code

```
import cvxpy as cp
import numpy as np

from scipy.integrate import solve_ivp
from scipy.integrate import OdeSolver

class InfLocalModel:
    def __init__(self, n, f, df, gs=None, dgs=None, hs=None, dhs=None, solver=None):
        self.f = f
        self.df = df

        if gs is None:
            self.gs = list()
            self.dgs = list()
        else:
            self.gs = gs
            self.dgs = dgs

        if hs is None:
            self.hs = list()
            self.dhs = list()
        else:
            self.hs = hs
            self.dhs = dhs

        self.n = n
        self.solver = solver
        self.init_program()

    def init_params(self):
        self.x_param = cp.Parameter(shape=(self.n,))
        self.f_param = cp.Parameter()
        self.g_params = [cp.Parameter() for _ in self.gs]
```

```

self.h_params = [cp.Parameter() for _ in self.hs]

self.df_param = cp.Parameter(shape=(self.n,))
self.dg_params = [cp.Parameter(shape=(self.n,)) for _ in self.gs]
self.dh_params = [cp.Parameter(shape=(self.n,)) for _ in self.hs]

def init_vars(self):
    raise NotImplementedError

def init_constraints(self):
    self.ineq_constraints = None
    self.eq_constraints = None

def init_obj(self):
    self.obj = None

def init_program(self):
    self.init_params()
    self.init_vars()
    self.init_constraints()
    self.init_obj()
    self.problem = cp.Problem(cp.Minimize(self.obj),
        self.ineq_constraints + self.eq_constraints)

def do_solve(self, x, solver=None, verbose=False, **kwargs):
    self.set_params(x)
    self.problem.solve(solver=self.solver if solver is None else solver,
        verbose=verbose, **kwargs)

def dynamics(self, x, solver=None, verbose=False, **kwargs):
    self.do_solve(x, solver, verbose, **kwargs)
    return self.get_vector_field()

def simulate(self, t_range, x0, solver=None,
    verbose=False, s_args=dict(), **kwargs):
    return solve_ivp(lambda t,
        x: self.dynamics(x, solver, verbose, **s_args), t_range, x0, **kwargs)

def get_vector_field(self):
    raise NotImplementedError

def set_params(self, x):
    raise NotImplementedError

```

```

class SGFBase(InfLocalModel):

    def __init__(self, n, f, df, gs=None, dgs=None, hs=None,
dhs=None, solver=None,

        alpha=1.0):
self.alpha = alpha
super().__init__(n, f, df, gs, dgs, hs, dhs, solver=None)

    def update_alpha(self, alpha):
self.alpha = alpha
self.init_program()

    def init_vars(self):
self.xi_var = cp.Variable(shape=(self.n,))

    def init_obj(self):
self.obj = self.df_param @ self.xi_var + 0.5 * cp.sum_squares(self.xi_var)

    def set_params(self, x):
self.x_param.value = x
self.f_param.value = self.f(x)
self.df_param.value = self.df(x)
for (g, dg, g_param, dg_param)
    in zip(self.gs, self.dgs, self.g_params, self.dg_params):
g_param.value = g(x)
dg_param.value = dg(x)
for (h, dh, h_param, dh_param)
    in zip(self.hs, self.dhs, self.h_params, self.dh_params):
h_param.value = h(x)
dh_param.value = dh(x)

    def get_vector_field(self):
return self.xi_var.value

class SafeGradientFlow(SGFBase):

    def __init__(self, n, f, df, gs=None, dgs=None, hs=None, dhs=None,
        solver=cp.OSQP, alpha=1.0):
super().__init__(n, f, df, gs, dgs, hs, dhs, solver=solver, alpha=alpha)

    def init_constraints(self):
self.ineq_constraints = [self.alpha * g_param + dg_param @ self.xi_var <= 0
for (g_param, dg_param) in zip(self.g_params, self.dg_params)]

```

```

self.eq_constraints = [self.alpha * h_param + dh_param @ self.xi_var == 0
for (h_param, dh_param) in zip(self.h_params, self.dh_params)]

class DualSafeGradientFlow(InfLocalModel):

    def __init__(self, n, f, df, gs=None, dgs=None, hs=None,
        dhs=None, solver=None, alpha=1.0):
        self.alpha = alpha
        super().__init__(n, f, df, gs, dgs, hs, dhs, solver=None)

    def update_alpha(self, alpha):
        self.alpha = alpha
        self.init_program()

    def init_vars(self):
        if self.gs:
            self.u_var = cp.Variable(shape=(len(self.gs),))
            self.udg = sum([u * dg_param for (u, dg_param) in zip(self.u_var, self.dg_params)])

        if self.hs:
            self.v_var = cp.Variable(shape=(len(self.hs),))
            self.vdh = sum([v * dh_param for (v, dh_param) in zip(self.v_var, self.dh_params)])

        if self.gs and self.hs:
            self.flow = -self.df_param - self.udg - self.vdh
        elif self.gs:
            self.flow = -self.df_param - self.udg
        elif self.hs:
            self.flow = -self.df_param - self.vdh
        else:
            self.flow = -self.df_param

    def init_obj(self):
        if self.gs and self.hs:
            self.obj = 0.5 * cp.sum_squares(self.udg + self.vdh)
        elif self.gs:
            self.obj = 0.5 * cp.sum_squares(self.udg)
        elif self.hs:
            self.obj = 0.5 * cp.sum_squares(self.vdh)
        else:
            self.obj = 0.0

    def init_constraints(self):
        self.ineq_constraints = [dg_param @ self.flow + self.alpha * g_param <= 0

```

```

for (g_param, dg_param) in zip(self.g_params, self.dg_params)]
self.eq_constraints = [dh_param @ self.flow + self.alpha * h_param <= 0
for (h_param, dh_param) in zip(self.h_params, self.dh_params)]

def set_params(self, x):
self.x_param.value = x
self.f_param.value = self.f(x)
self.df_param.value = self.df(x)
for (g, dg, g_param, dg_param) in zip(self.gs, self.dgs,
self.g_params, self.dg_params):
g_param.value = g(x)
dg_param.value = dg(x)
for (h, dh, h_param, dh_param) in zip(self.hs, self.dhs,
self.h_params, self.dh_params):
h_param.value = h(x)
dh_param.value = dh(x)

def get_vector_field(self):
return self.flow.value

```

SGF application to solve a Quadratic Program

The code used to obtain the trajectory in Figure 4.1 (and the plot itself) was:

```

import sys

sys.path.append("../")

import cvxpy as cp
import numpy as np

from scipy.integrate import solve_ivp

#matplotlib inline
import matplotlib as mpl
import matplotlib.pyplot as plt

from opt_flows import SafeGradientFlow

n = 2
m = 2
k = 1

Q = np.array([
[0.148, -0.033],

```

```

[-0.033, 0.101],
])

A = np.array([
[0.5, 2],
[3, -1]]
)

b = np.array([5, 4])

c = np.array([-1, 1])
d = 0

# %%
x = cp.Variable(shape=(2,))
constraints = [A @ x - b <= 0, c @ x == d]
objective = cp.Minimize(cp.quad_form(x, Q))
prob = cp.Problem(objective, constraints)
prob.solve(solver=cp.ECOS, verbose=True)

x_star = x.value
u_star = constraints[0].dual_value
v_star = constraints[1].dual_value
# %%
sgf = SafeGradientFlow(n, f=lambda x: 0.5 * x @ (Q @ x), df=lambda x: Q @ x,
gs=[lambda x: A[i, :] @ x - b[i] for i in range(m)],
dgs=[lambda x: A[i, :] for i in range(m)],
hs=[lambda x: c[0, :] @ x - d], dhs=[lambda x: c[0, :]], alpha=0.5)
# %%
## Solve for a trajectory
l = np.random.randint(low = -2.5, high = 2.5, size=(50,1))
x0 = np.array([1, 1])
sol = sgf.simulate([0, 50], x0, method="RK45", max_step=0.1)

# %%%matplotlibinline

import matplotlib.patches as mpatches

fig, ax = plt.subplots(figsize=(5, 5), dpi=150)
ax.scatter(x_star[0], x_star[1], c="b", label="Global Minimum", zorder=5000)

x1_min = -2.5
x1_max = 2.5
x2_min = -2.5
x2_max = 2.5

```

```

x1_len = 200
x2_len = 200

x1_range = np.linspace(x1_min, x1_max, x1_len)
x2_range = np.linspace(x2_min, x2_max, x2_len)
X1, X2 = np.meshgrid(x1_range, x2_range)

G = np.zeros((x1_len, x2_len, m))
for i in range(X1.shape[0]):
    for j in range(X1.shape[1]):
        x = np.array([X1[i, j], X2[i, j]])
        G[i, j, :] = A @ x - b

if np.abs(c[0, 0]) > np.abs(c[0, 1]):
    hx2_range = np.arange(x2_min, x2_max, 0.01)
    hx1_range = 1.0 / c[0, 0] * (-c[0, 1] * hx2_range + d)
else:
    hx1_range = np.arange(x1_min, x1_max, 0.01)
    hx2_range = 1.0 / c[0, 1] * (-c[0, 0] * hx1_range + d)

ax.plot(hx1_range, hx2_range, zorder=9, alpha=0.75,
        label=r"$h(x) = 0$", linewidth=3.0)
ax.pcolormesh(X1, X2, np.all(G <= 0, axis=2),
              alpha=0.1, cmap=plt.get_cmap("bwr_r"), vmin=-1.0, vmax=1.0,
              shading="gouraud")
ax.plot(sol.y[0, :], sol.y[1, :], c="k", label="solution")
blue_patch = mpatches.Patch(color=plt.get_cmap("bwr")(-1.0), alpha=0.25,
                             label=r'$g(x) \leq 0$')
handles, _ = ax.get_legend_handles_labels()
ax.legend(handles=handles + [blue_patch], framealpha=1, loc=2).set_zorder(1000)

ax.set_xlim([x1_min, x1_max])
ax.set_ylim([x2_min, x2_max])

plt.show()
plt.close(fig)

```

Perturbed SGF to solve a Quadratic Program

The code used to obtain the trajectories in Figures 4.2 and 4.3 (and the plots themselves) was:

```
import sys
```

```

sys.path.append("../")

import cvxpy as cp
import numpy as np

from scipy.integrate import solve_ivp

#matplotlib inline
import matplotlib as mpl
import matplotlib.pyplot as plt

from opt_flows import SafeGradientFlow

n = 2
m = 3
k = 1

Q = np.array([
    [0.148, -0.033],
    [-0.033, 0.101],
])

A = np.array([
    [0.5, 2],
    [3, -1],
    [-1, 0]]
)

b = np.array([5, 4, 0.5])

c = np.array([-1, 1])
d = 0
epsilon = 0.05

# %%
x = cp.Variable(shape=(2,))
constraints = [A @ x - b <= 0, c @ x == d]
objective = cp.Minimize(cp.quad_form(x, Q))
prob = cp.Problem(objective, constraints)
prob.solve(solver=cp.ECOS, verbose=True)

x_star = x.value
u_star = constraints[0].dual_value
v_star = constraints[1].dual_value
# %%

```

```

sgf = SafeGradientFlow(n, f=lambda x: 0.5 * x @ (Q @ x), df=lambda x: Q @ x,
gs=[lambda x: A[i, :] @ x - b[i] for i in range(m)],
dgs=[lambda x: A[i, :] for i in range(m)],
hs=[lambda x: c[0, :] @ x - d], dhs=[lambda x: c[0, :]], alpha=0.5)
# %%
## Solve for a trajectory
l = np.random.randint(low = -2.5, high = 2.5,size=(50,1))
x0 = np.array([1.01, 0.99]).reshape(-1,1)
sol = x0

for i in range(50):
    aux = sgf.simulate([0,1], sol[:, -1].reshape(1,-1).flatten(), method="RK45",
    max_step=0.1)
    yaux = np.array([aux.y[0,-1],aux.y[1,-1]]).reshape(-1,1) +
    np.random.uniform(low=-0.01, high=0.01, size=(2,1))
    sol = np.concatenate((sol, yaux), axis=1)

# %%%matplotlibinline

import matplotlib.patches as mpatches

fig, ax = plt.subplots(figsize=(5, 5), dpi=150)
ax.scatter(x_star[0], x_star[1], c="b", label="Global Minimum", zorder=5000)

x1_min = -2.5
x1_max = 2.5
x2_min = -2.5
x2_max = 2.5

x1_len = 200
x2_len = 200

x1_range = np.linspace(x1_min, x1_max, x1_len)
x2_range = np.linspace(x2_min, x2_max, x2_len)
X1, X2 = np.meshgrid(x1_range, x2_range)

G = np.zeros((x1_len, x2_len, m))
for i in range(X1.shape[0]):
    for j in range(X1.shape[1]):
        x = np.array([X1[i, j], X2[i, j]])
        G[i, j, :] = A @ x - b

if np.abs(c[0, 0]) > np.abs(c[0, 1]):
    hx2_range1 = np.arange(x2_min, x2_max, 0.01)
    hx1_range1 = 1.0 / c[0, 0] * (-c[0, 1] * hx2_range1 + d + epsilon)

```

```

else:
    hx1_range1 = np.arange(x1_min, x1_max, 0.01)
    hx2_range1 = 1.0 / c[0, 1] * (-c[0, 0] * hx1_range1 + d + epsilon)

    if np.abs(c[0, 0]) > np.abs(c[0, 1]):
        hx2_range2 = np.arange(x2_min, x2_max, 0.01)
        hx1_range2 = 1.0 / c[0, 0] * (-c[0, 1] * hx2_range2 + d - epsilon)
    else:
        hx1_range2 = np.arange(x1_min, x1_max, 0.01)
        hx2_range2 = 1.0 / c[0, 1] * (-c[0, 0] * hx1_range2 + d - epsilon)

    ax.plot(hx1_range1, hx2_range1, zorder=9, alpha=0.75,
            label=r"$h(x) = \epsilon$", linewidth=0.5)
    ax.plot(hx1_range2, hx2_range2, zorder=9, alpha=0.75,
            label=r"$h(x) = -\epsilon$", linewidth=0.5)
    ax.pcolormesh(X1, X2, np.all(G <= 0, axis=2), alpha=0.1,
                  cmap=plt.get_cmap("bwr_r"), vmin=-1.0, vmax=1.0,
                  shading="gouraud")
    ax.plot(sol[0, :], sol[1, :], c="k", label="solution")
    blue_patch = mpatches.Patch(color=plt.get_cmap("bwr")(-1.0), alpha=0.25,
                                label=r'$g(x) \leq 0$')
    handles, _ = ax.get_legend_handles_labels()
    ax.legend(handles=handles + [blue_patch], framealpha=1, loc=2).set_zorder(1000)

    ax.set_xlim([x1_min, x1_max])
    ax.set_ylim([x2_min, x2_max])

    plt.savefig("Plot3.png")
    plt.show()
    plt.close(fig)

```